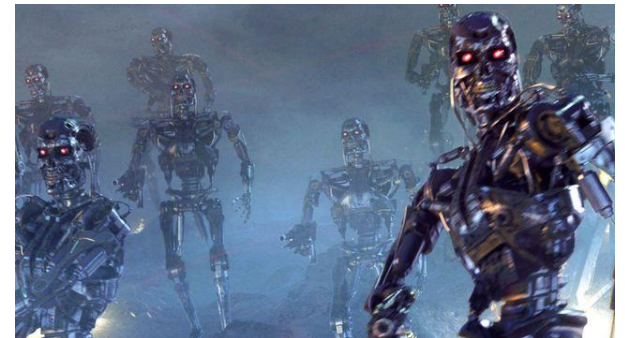
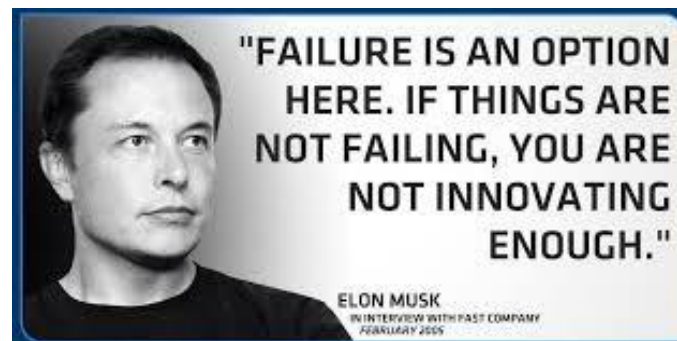




AGI et Finance : l'opportunité des réseaux neuronaux avancés

28/03/2017 //////////////

Olivier Croissant



Plan

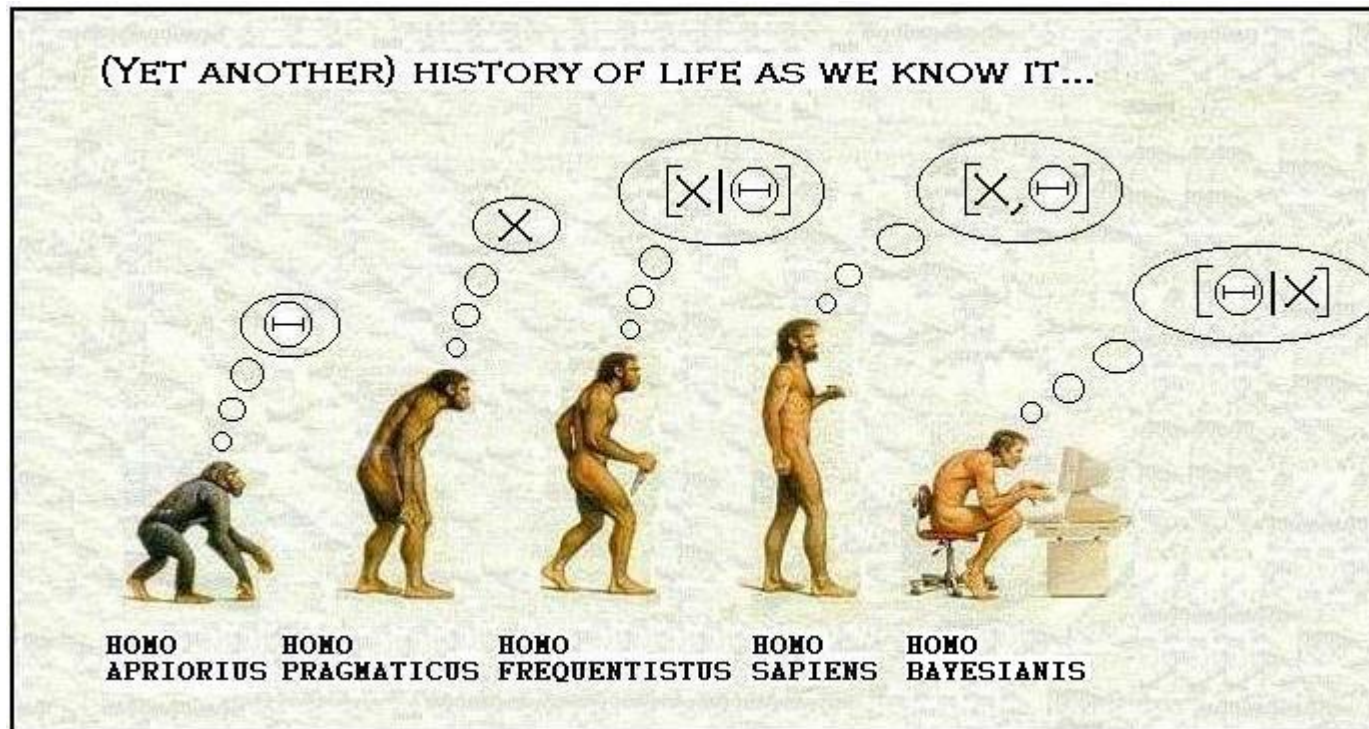


1. Introduction
 2. Concepts de base
 3. Inférence variationnelle
 4. HMM , RBM , Autoencodeur
 5. Autoencodeur variationnel
 6. LSTM et la mémoire longue
 7. DRAW et le génératif attentionnel
 8. DNC et le CPU différentiel
 9. GTMM ou l'apex neuronal
 10. DNI et les gradients synthétiques
 11. Le classificateur à sous-variété tangente
 12. Application a la prédiction financière
 13. Projet cible
 14. Conclusion
 15. Annexes
- Diagram illustrating the categorization of topics:
- Topics 4, 5, and 6 are grouped under the label **Monocouche**.
 - Topics 7, 8, and 9 are grouped under the label **Multicouche**.



$$P(\text{model} | \text{observations}) P(\text{observations}) \\ = P(\text{observations} | \text{model}) P(\text{model})$$

$$\begin{array}{c} \text{Posterior} \\ \downarrow \\ P(\theta | x) \end{array} = \frac{\begin{array}{c} \text{Likelihood} \\ \downarrow \\ P(x | \theta) \end{array} \begin{array}{c} \text{Prior} \\ \downarrow \\ P(\theta) \end{array}}{\begin{array}{c} \uparrow \\ \text{Evidence} \\ P(x) \end{array}}$$



↑
Illusion

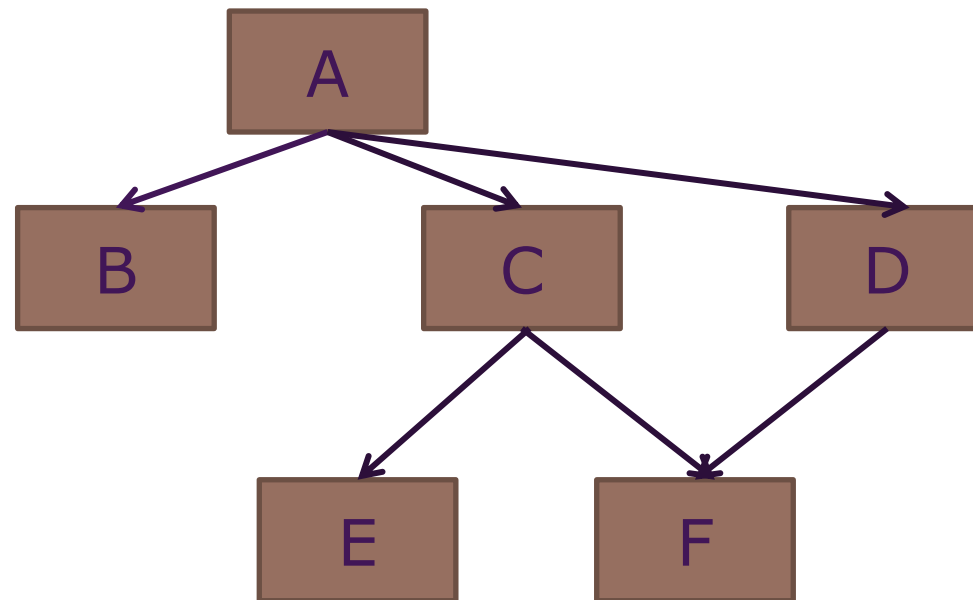
↑
Dogmatisme

↑
Démarche Scientifique

↑
Regard factuel

↑
Zenitude

Modèle Probabiliste Graphique



$$p(A = a; B = b; C = c; D = d; E = e; F = f) \\ = f_B(a, b)f_C(a, c)f_D(a, d)f_E(c, e)f_F(c, d, f)$$

Produit de facteurs

La backpropagation dans un réseau



Trouver θ tel que $distance(F(\theta), x)$ soit minimale

$dim(\theta)$ = jusqu'à un milliard de paramètres

x : base d'apprentissage

Backpropagation =

Newton

+ AD inverse

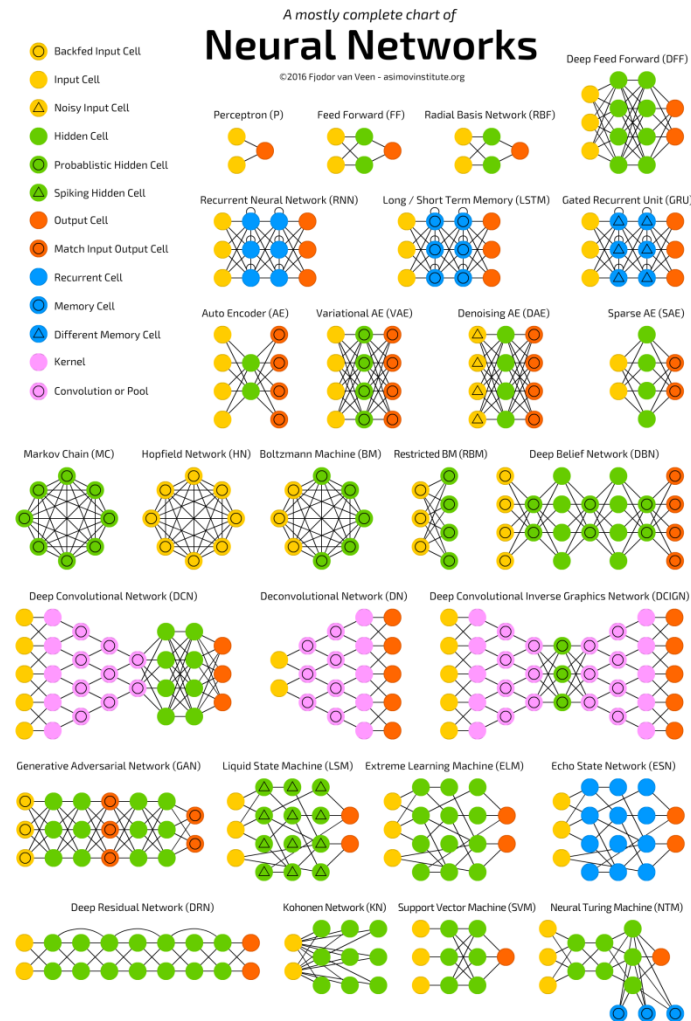
+ Hypothèse d'équipollence sur les paramètres

- On ne peut évaluer $F(\theta)$ que partiellement
- Il faut gérer les minima locaux

Descente de gradient stochastique (SGD) ...

Régularisation $\mathbb{L}^1$, $\mathbb{L}^2$, smart dropout ..

Une classification des réseaux neuronaux





Avantage du discriminatif

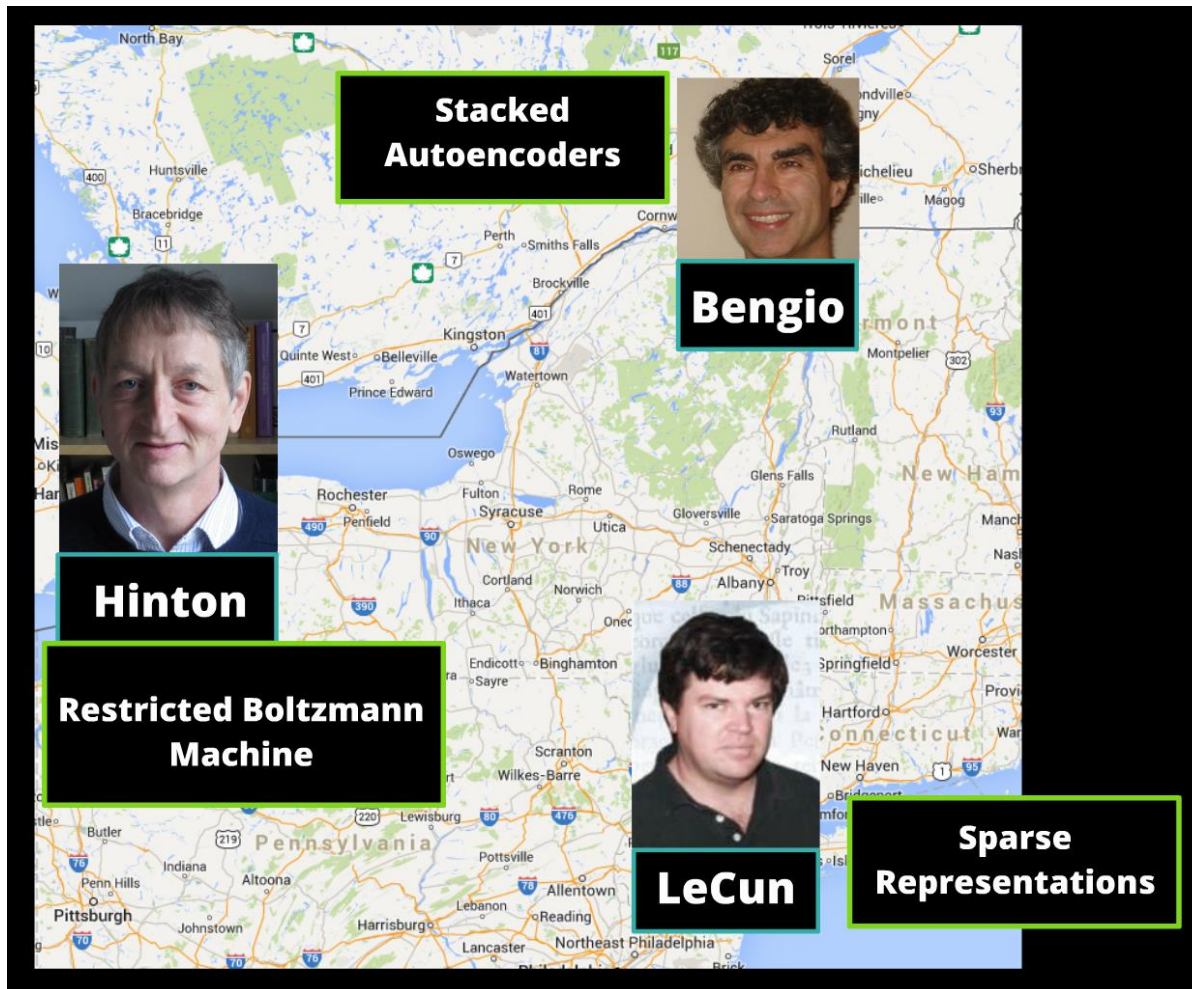
- Réseau est plus facile à mettre en œuvre. Il faut juste calculer $p(class|x)$

Avantage du génératif

- Efficience / données. Il est plus facile et robuste de calculer $p(class|z)$ que $p(class|x)$.
- On peut vérifier le modèle en regardant les exemples générés.
- Compréhension du modèle: on peut interpoler entre les variables latentes plus sûrement.



2006 l'année miracle pour le deep learning



Le Breakthrough

- Hinton, Osindero & Teh
« A Fast Learning Algorithm for Deep Belief Nets », *Neural Computation*, 2006
- Bengio, Lamblin, Popovici, Larochelle
« Greedy Layer-Wise Training of Deep Networks », *NIPS'2006*
- Ranzato, Poultney, Chopra, LeCun
« Efficient Learning of Sparse Representations with an Energy-Based Model », *NIPS'2006*

Deepmind



Solve intelligence. Use it to make the world a better place.



Google buys UK artificial intelligence startup Deepmind for £400m

fondée en 2010 par Demis Hassabis, Mustafa Suleyman et Shane Legg

Intelligence Artificielle ↔ Neurologie

Inventions :

- Machine de Turing neuronale
- Deep Reinforcement learning
- Learning learning
- Alphago

=> Sonnet



The Neural Processes Underpinning
Episodic Memory

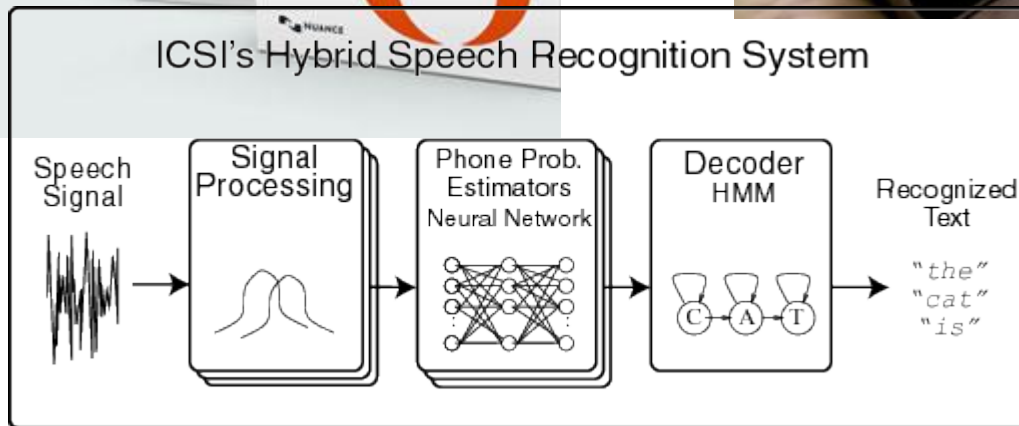
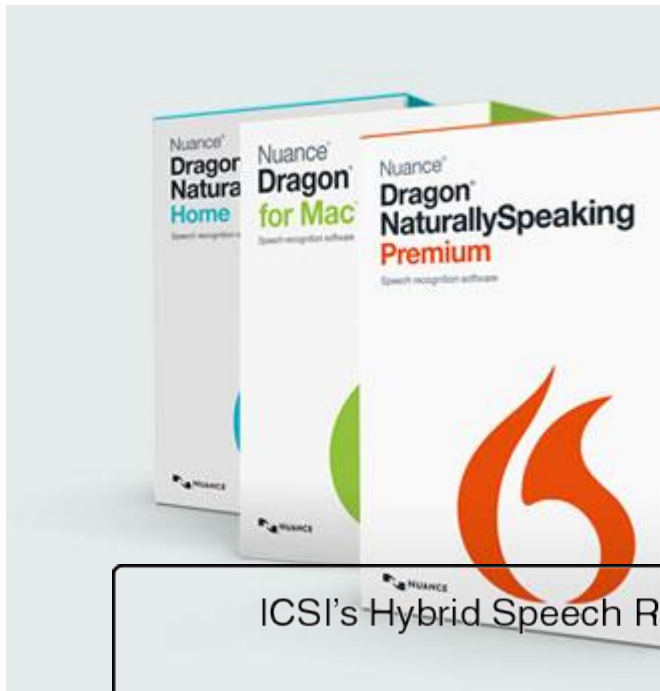
Demis Hassabis

Submitted for PhD in Cognitive Neuroscience

February 2009

, noté 2300 au classement Elo à 13 ans (en janvier 1990).

Reconnaissance de la parole



Shoebox, 1960 (IBM)

Etat de l'art en reconnaissance de la parole



IBM speech recognition is on the verge of super-human accuracy

Chris Weller · 9 Mar 2017, 16:58 · Tech Insider · 10,811



In the world of speech recognition software, 5.1% is kind of a magic number.

Companies that can create software with error rates falling in that ballpark are essentially matching the capabilities of humans, who miss roughly 5% of the words in a given conversation.



IBM's CEO, Ginny Rometty. Reuters

On March 7, IBM announced it had become the first to home in on that benchmark, having achieved a rate of 5.5%. The breakthrough signals a big win for artificial intelligence that could eventually live in smartphones and voice assistants like Siri, Alexa, and Google Assistant.

English Conversational Telephone Speech Recognition by Humans and Machines

George Saon, Gakuto Kurata*, Tom Sercu
Kartik Audhkhasi, Samuel Thomas, Dimitrios Dimitriadis
Xiaodong Cui, Bhuvana Ramabhadran, Michael Picheny

Lynn-Li Lim, Bergul Roomi, Phil Hall

IBM Watson, Yorktown Heights, USA
*IBM Research - Tokyo, Japan

Appen, Sydney, Australia

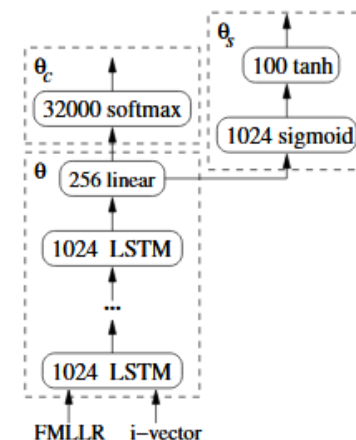
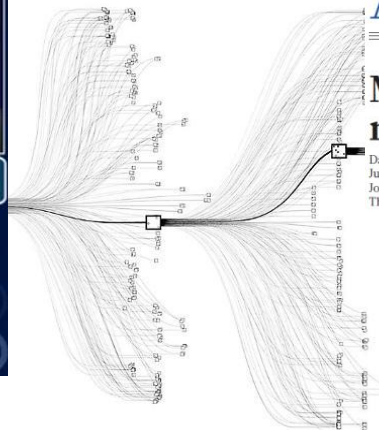


Figure 1: LSTM with speaker-adversarial MTL architecture.

	SWB	CH	RT'02	RT'03	RT'04	DEV'04f
n-gram	6.7	12.1	10.1	10.0	9.7	9.2
+ model-M	6.1	11.2	9.4	9.4	9.0	8.8
+ LSTM + DCC	5.5	10.3	8.3	8.3	8.0	8.0
'.' removal	5.5	10.3	8.3	8.0	7.7	7.1

AlphaGo (5 avril 2017)



ARTICLE

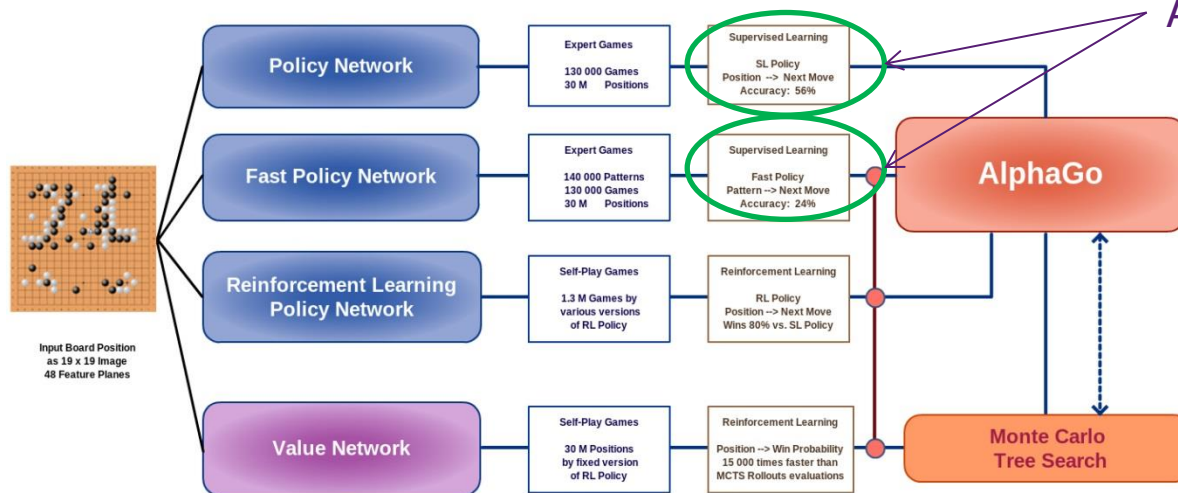
doi:10.1038/nature16961

Mastering the game of Go with deep neural networks and tree search

David Silver^{1*}, Aja Huang^{1*}, Chris J. Maddison¹, Arthur Guez¹, Laurent Sifre¹, George van den Driessche¹, Julian Schrittwieser¹, Ioannis Antonoglou¹, Veda Panneershelvam¹, Marc Lanctot¹, Sander Dieleman¹, Dominik Grewe¹, John Nham¹, Nal Kalchbrenner¹, Ilya Sutskever², Timothy Lillicrap¹, Madeleine Leach¹, Koray Kavukcuoglu¹, Thore Graepel¹ & Demis Hassabis¹



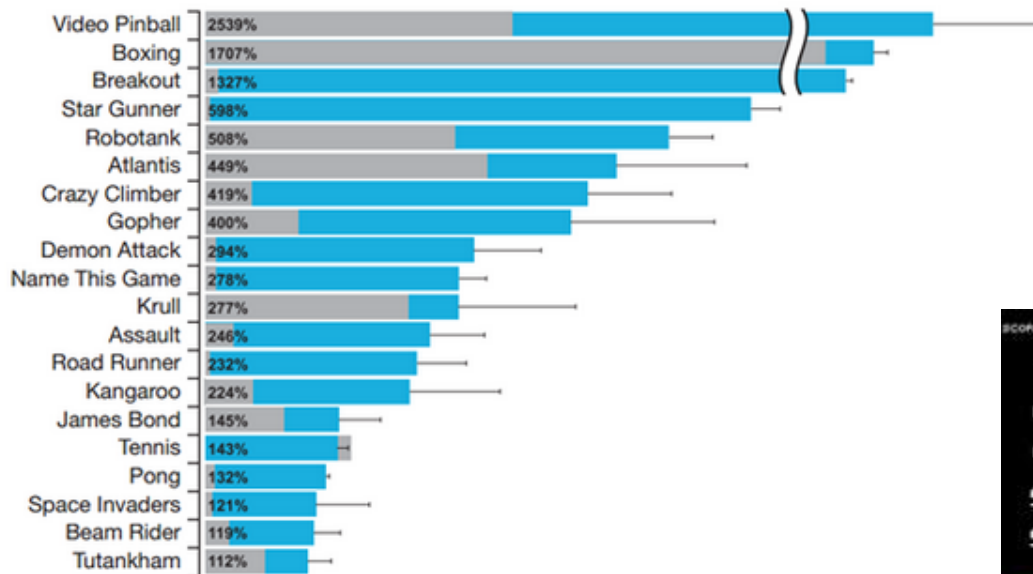
AlphaGo Overview



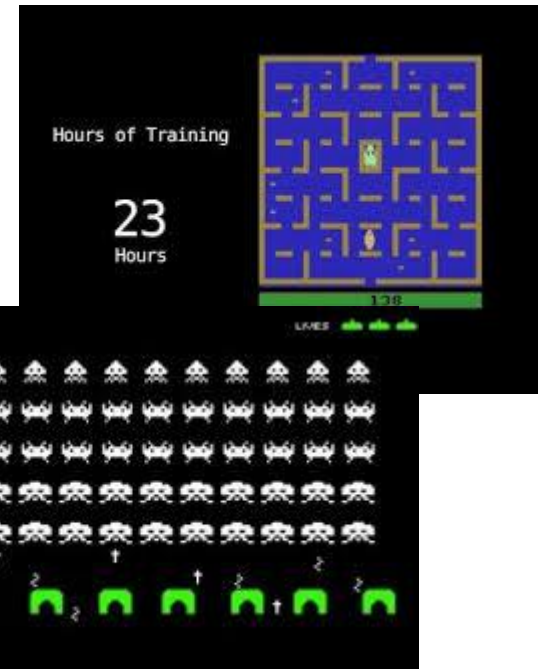
Apprentissage profond



L'IA dans les jeu pour arcade

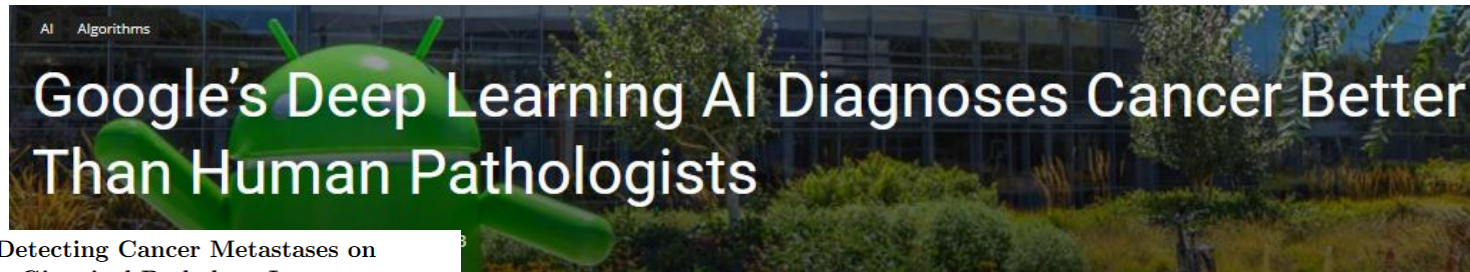


The games at which AIs in general perform better than humans (in grey) and DQN performs exceptionally (blue) tended to be more action oriented, lacking exploration or experimentation elements. The percentage is how much better the AI performed than a human. © Google DeepMind



DeepMind Achetée par Google le 26 janvier 2014 pour plus de 628 millions de \$

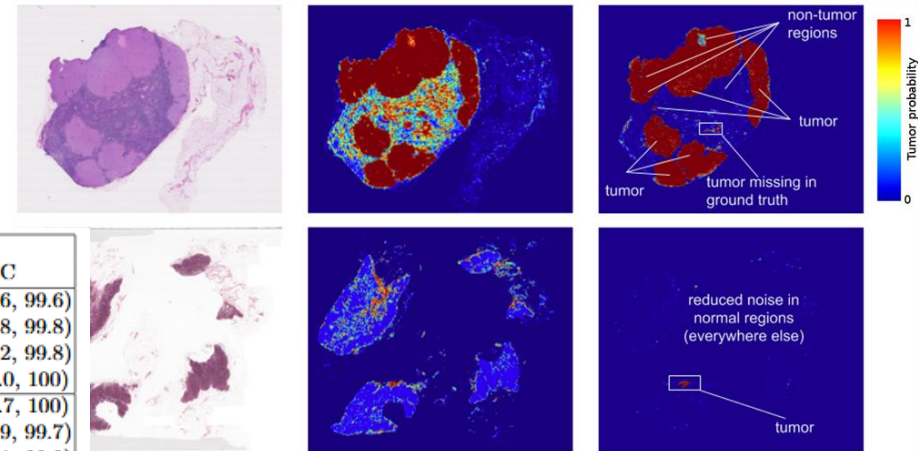
L'IA dans le diagnostique du cancer



Yun Liu^{1*}, Krishna Gadepalli¹, Mohammad Norouzi¹, George E. Dahl¹,
Timo Kohlberger¹, Aleksey Boyko¹, Subhashini Venugopalan^{2**},
Alekssei Timofeev², Philip Q. Nelson², Greg S. Corrado¹, Jason D. Hipp³,
Lily Peng¹, and Martin C. Stumpe¹

{liuyun,mnorouzi,gdahl,lhpeng,mstumpe}@google.com

¹Google Brain, ²Google Inc, ³Verily Life Sciences,
Mountain View, CA, USA



Input & model size	Validation			Test		
	FROC	@8FP	AUC	FROC	@8FP	AUC
40X	98.1	100	99.0	87.3 (83.2, 91.1)	91.1 (87.2, 94.5)	96.7 (92.6, 99.6)
40X-pretrained	99.3	100	100	85.5 (81.0, 89.5)	91.1 (86.8, 94.6)	97.5 (93.8, 99.8)
40X-small	99.3	100	100	86.4 (82.2, 90.4)	92.4 (88.8, 95.7)	97.1 (93.2, 99.8)
ensemble-of-3	-	-	-	88.5 (84.3, 92.2)	92.4 (88.7, 95.6)	97.7 (93.0, 100)
20X-small	94.7	100	99.6	85.5 (81.0, 89.7)	91.1 (86.9, 94.8)	98.6 (96.7, 100)
10X-small	88.7	97.2	97.7	79.3 (74.2, 84.1)	84.9 (80.0, 89.4)	96.5 (91.9, 99.7)
40X+20X-small	94.9	98.6	99.0	85.9 (81.6, 89.9)	92.9 (89.3, 96.1)	97.0 (93.1, 99.9)
40X+10X-small	93.8	98.6	100	82.2 (77.0, 86.7)	87.6 (83.2, 91.7)	98.6 (96.2, 99.9)
Pathologist [1]	-	-	-	73.3*	73.3*	96.6
Camelyon16 winner [1123]	-	-	-	80.7	82.7	99.4

Table 1. Results on Camelyon16 dataset (95% confidence intervals, CI). Bold indicates results within the CI of the best model. “Small” models contain 300K parameters per Inception tower instead of 20M. -: not reported. *A pathologist achieved this sensitivity (with no FP) using 30 hours.

99.4 % de précision !

Lecture des panneaux routiers



The challenge was won by a team based at the IDSIA institute (Lugano, Switzerland) with an *ensemble* of 25 ConvNets (named by the authors *multi-column* deep neural network, or MCDNN), which achieved a classification rate of 99.46% on the test data (Ciresan *et al.*, 2012 [6]). This result surpassed human classification accuracy, estimated to be 98.84% on the same data.

CVNAR Features

Road scene recognition and objects tracking

- ◆ Road boundaries
- ◆ Lane detection
- ◆ Vehicle recognition and tracking
- ◆ Distance and time to collision estimation
- ◆ Pedestrian detection and tracking
- ◆ Facade recognition and texture extraction
- ◆ Road signs recognition
- ◆ Parking slot recognition



Positioning

- ◆ Precise relative and absolute positioning
- ◆ Flexible data fusion and smooth Map Matching
- ◆ Automotive constrained SLAM

Integration and Fusion

1. Sensor data
2. External positioner data (optional)
3. External recognition engine integration
4. Pupil tracking
5. Telematics: V2I and V2V



Natural Augmented Reality

1. Basic vehicle data
2. Lanes and road boundaries
3. Road signs and cautions
4. Navigation data and hints
5. Facades highlights
6. Parking Places
7. Narrow street infographics
8. Street names and Complex Junctions
9. POI and OEM specific information

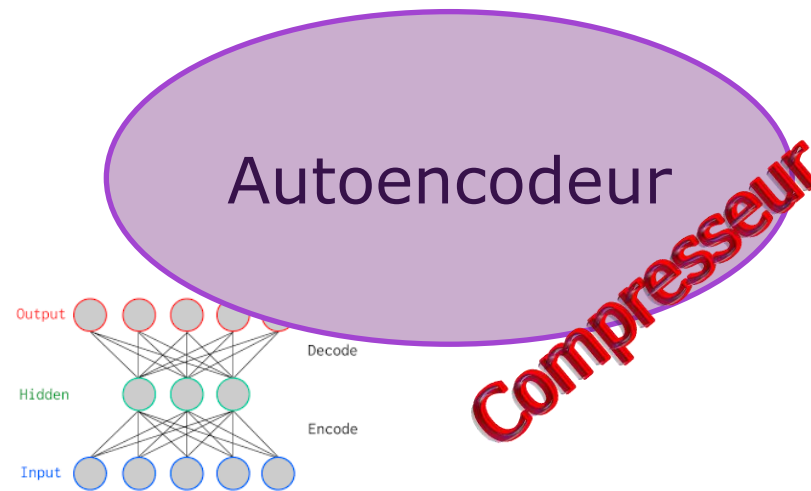
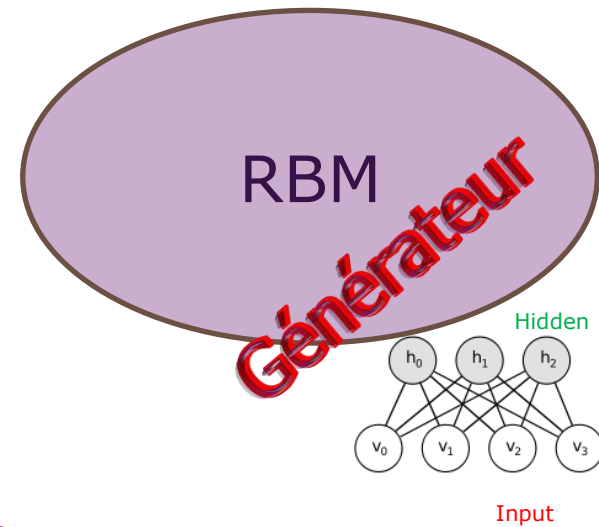
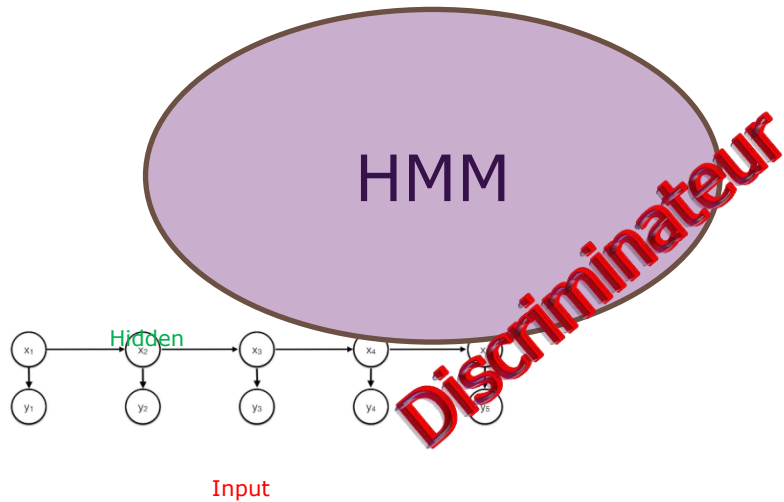
Highlights:

- ◆ CPU, GPU; OS: Linux, QNX, Android
- ◆ HW: Intel, nVidia, TI, Renesas
- ◆ Extrapolation engines for latency avoidance
- ◆ Neural networks: CNN, DNN, Deep Learning



www.kluge.com

Les « monocouches »

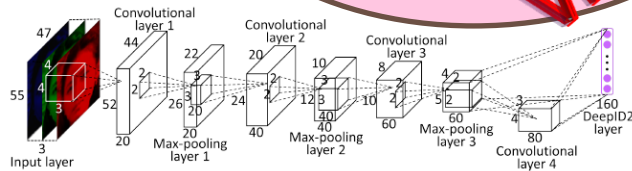


Plusieurs types de réseaux profonds



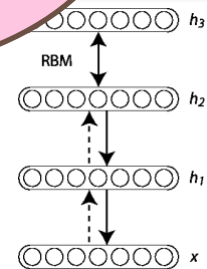
Convnet

Vision



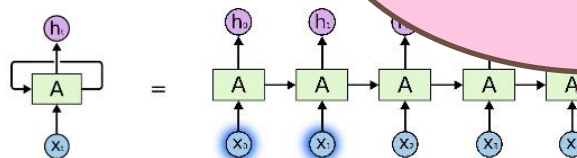
DBN

Training itératif



RNN

Langage naturel



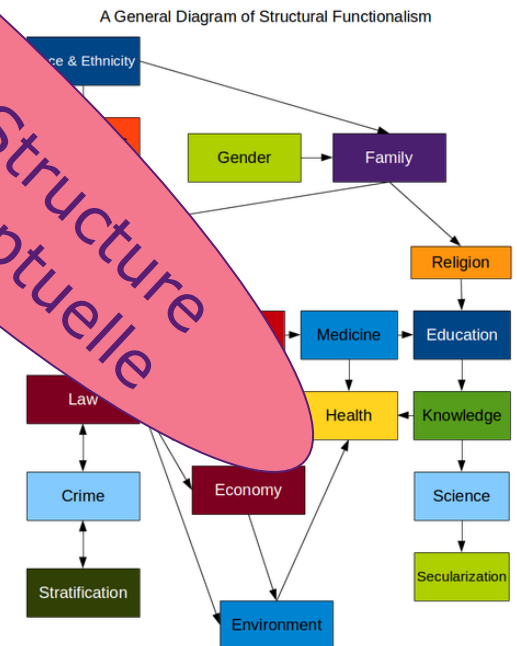
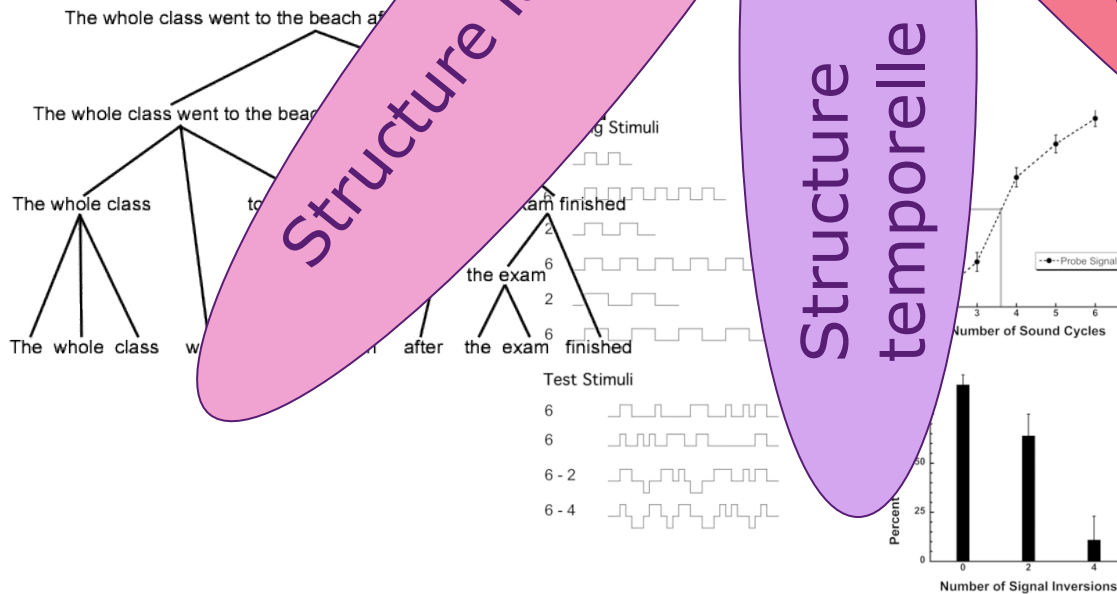
Les différentes représentation d'un RNN



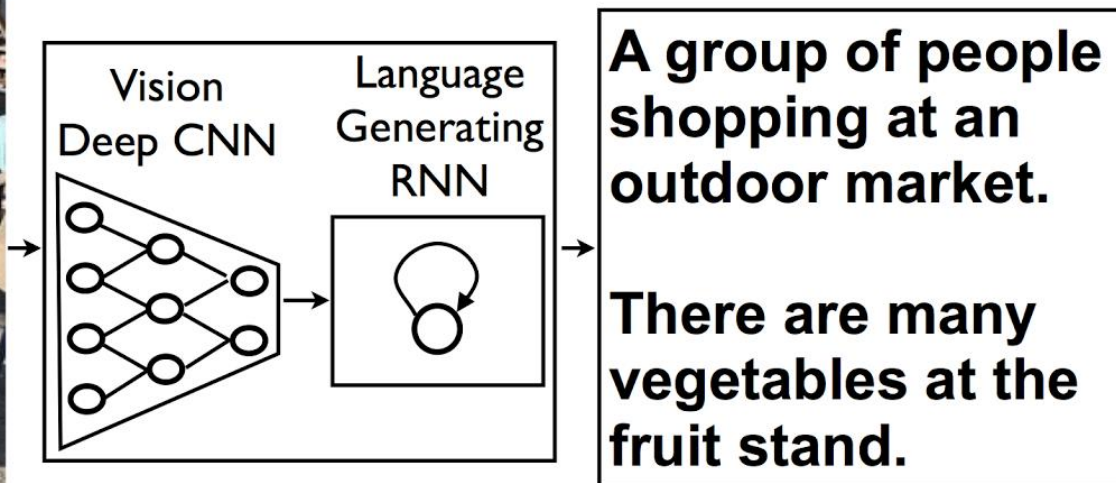
Structure langagière

Structure temporelle

Structure conceptuelle



Labélisation des images



Traduction automatique



FR: Nous sommes heureux que la FAA reconnaisse que « une expérience passager agréable » n'est pas incompatible avec la sécurité, a déclaré Roger Dow, Président et Directeur de la « US Travel association » .

Google Translate : `` We are pleased that the FAA recognizes that a pleasant passenger experience is not incompatible with safety '', said Roger Dow, president and director of the US Travel Association.

LSTM : `` We are pleased that the FAA recognizes that a pleasant passenger experience is not incompatible with security'', said Roger Dow, CEO of the US Travel Association.

Truth : `` We are pleased that the FAA recognizes that a pleasant passenger experience is not incompatible with safety and security'', said Roger Dow, CEO of the US Travel Association.

Générateur de texte (à la Shakespeare)



PANDARUS: Alas, I think he shall be come approached and the day
When little strain would be attain'd into being never fed, And who
is but a chain and subjects of his death, I should not sleep.

Second Senator: They are away this miseries, produced upon my
soul, Breaking and strongly should be buried, when I perish The
earth and thoughts of many states.

DUKE VINCENTIO: Well, your wit is in the care of side and that.

Second Lord: They would be ruled after this chamber, and my fair
nues begun out of the fact, to be conveyed, Whose noble souls I'll
have the heart of the wars. Clown: Come, sir, I will make did
behold your worship.

VIOLA: I'll drink it.

<http://colah.github.io/posts/2014-07-NLP-RNNs-Representations/>

Générateur de texte (à la Wikipedia)



Naturalism and decision for the majority of Arab countries' capitalide was grounded by the Irish language by [[John Clair]], [[An Imperial Japanese Revolt]], associated with Guangzham's sovereignty. His generals were the powerful ruler of the Portugal in the [[Protestant Immineners]], which could be said to be directly in Cantonese Communication, which followed a ceremony and set inspired prison, training. The emperor travelled back to [[Antioch, Perth, October 25|21]] to note, the Kingdom of Costa Rica, unsuccessful fashioned the [[Thrales]], [[Cynth's Dajoard]], known in western [[Scotland]], near Italy to the conquest of India with the conflict. Copyright was the succession of independence in the slop of Syrian influence that was a famous German movement based on a more popular servicious, non-doctrinal and sexual power post. Many governments recognize the military housing of the [[Civil Liberalization and Infantry Resolution 265 National Party in Hungary]], that is sympathetic to be to the [[Punjab Resolution]]

(PJS)[<http://www.humah.yahoo.com/guardian.cfm/7754800786d17551963s89.htm>

Official economics Adjoint for the Nazism, Montgomery was swear to advance to the resources for those Socialism's rule, was starting to signing a major tripad of aid exile.]]

Générateur de texte (à la XML)



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      <id>23</id>
    </contributor>
    <minor />
    <comment>Automated conversion</comment>
    <text xml:space="preserve">#REDIRECT
[[Christianity]]</text>
  </revision>
</page>
```

Générateur de texte (type Math)



For $\bigoplus_{n=1,\dots,m}$ where $\mathcal{L}_{m,\bullet} = 0$, hence we can find a closed subset $\mathcal{H}$ in $\mathcal{H}$ and any sets $\mathcal{F}$ on X , U is a closed immersion of S , then $U \rightarrow T$ is a separated algebraic space.

Proof. Proof of (1). It also start we get

$$S = \text{Spec}(R) = U \times_X U \times_X U$$

and the comparico in the fibre product covering we have to prove the lemma generated by $\coprod Z \times_U U \rightarrow V$. Consider the maps M along the set of points Sch_{fppf} and $U \rightarrow U$ is the fibre category of S in U in Section, ?? and the fact that any U affine, see Morphisms, Lemma ?? . Hence we obtain a scheme S and any open subset $W \subset U$ in $\text{Sh}(G)$ such that $\text{Spec}(R') \rightarrow S$ is smooth or an

$$U = \bigcup U_i \times_{S_i} U_i$$

which has a nonzero morphism we may assume that f_i is of finite presentation over S . We claim that $\mathcal{O}_{X,x}$ is a scheme where $x, x', s'' \in S'$ such that $\mathcal{O}_{X,x'} \rightarrow \mathcal{O}'_{X',x'}$ is separated. By Algebra, Lemma ?? we can define a map of complexes $\text{GL}_{S'}(x'/S'')$ and we win. $\square$

To prove study we see that $\mathcal{F}|_U$ is a covering of $\mathcal{X}'$, and $\mathcal{T}_i$ is an object of $\mathcal{F}_{X/S}$ for $i > 0$ and $\mathcal{F}_p$ exists and let $\mathcal{F}_i$ be a presheaf of $\mathcal{O}_X$ -modules on $\mathcal{C}$ as a $\mathcal{F}$ -module. In particular $\mathcal{F} = U/\mathcal{F}$ we have to show that

$$\widetilde{M}^\bullet = \mathcal{I}^\bullet \otimes_{\text{Spec}(k)} \mathcal{O}_{S,s} - i_X^{-1} \mathcal{F}$$

is a unique morphism of algebraic stacks. Note that

$$\text{Arrows} = (\text{Sch}/S)_{fppf}^{opp}, (\text{Sch}/S)_{fppf}$$

and

$$V = \Gamma(S, \mathcal{O}) \mapsto (U, \text{Spec}(A))$$

is an open subset of X . Thus U is affine. This is a continuous map of X is the inverse, the groupoid scheme S .

Proof. See discussion of sheaves of sets. $\square$

The result to prove any open covering follows from the less of Example ?? . It may replace S by $X_{spaces, \acute{e}tale}$ which gives an open subspace of X and T equal to S_{Zar} , see Descent, Lemma ?? . Namely, by Lemma ?? we see that R is geometrically regular over S .

Lemma 0.1. Assume (3) and (3) by the construction in the description.

Suppose $X = \lim |X|$ (by the formal open covering X and a single map $\text{Proj}_X(\mathcal{A}) = \text{Spec}(B)$ over U compatible with the complex

$$\text{Set}(\mathcal{A}) = \Gamma(X, \mathcal{O}_{X, \mathcal{O}_X}).$$

When in this case of to show that $\mathcal{Q} \rightarrow \mathcal{C}_{Z/X}$ is stable under the following result in the second conditions of (1), and (3). This finishes the proof. By Definition ?? (without element is when the closed subschemes are catenary. If T is surjective we may assume that T is connected with residue fields of S . Moreover there exists a closed subspace $Z \subset X$ of X where U in X' is proper (some defining as a closed subset of the uniqueness it suffices to check the fact that the following theorem

(1) f is locally of finite type. Since $S = \text{Spec}(R)$ and $Y = \text{Spec}(R)$.

Proof. This is form all sheaves of sheaves on X . But given a scheme U and a surjective étale morphism $U \rightarrow X$. Let $U \cap U = \coprod_{i=1,\dots,n} U_i$ be the scheme X over S at the schemes $X_i \rightarrow X$ and $U = \lim_i X_i$. $\square$

The following lemma surjective restrocomposes of this implies that $\mathcal{F}_{x_0} = \mathcal{F}_{x_0} = \mathcal{F}_{X,\dots,0}$.

Lemma 0.2. Let X be a locally Noetherian scheme over S , $E = \mathcal{F}_{X/S}$. Set $\mathcal{I} = \mathcal{I}_1 \subset \mathcal{I}'_n$. Since $\mathcal{I}^n \subset \mathcal{I}^n$ are nonzero over $i_0 \leq \mathfrak{p}$ is a subset of $\mathcal{I}_{n,0} \circ \overline{A}_2$ works.

Lemma 0.3. In Situation ?? . Hence we may assume $\mathfrak{q}' = 0$.

Proof. We will use the property we see that $\mathfrak{p}$ is the mex functor (??). On the other hand, by Lemma ?? we see that

$$D(\mathcal{O}_{X'}) = \mathcal{O}_X(D)$$

where K is an F -algebra where δ_{n+1} is a scheme over S . $\square$

Générateur de texte (type C-Code)



```
static int indicate_policy(void)
{
    int error;
    if (fd == MARN_EPT) {
        /*
         * The kernel blank will coeld it to userspace.
         */
        if (ss->segment < mem_total)
            unblock_graph_and_set_blocked();
        else
            ret = 1;
        goto bail;
    }
    segaddr = in_SB(in.addr);
    selector = seg / 16;
    setup_works = true;
    for (i = 0; i < blocks; i++) {
        seq = buf[i++];
        bpf = bd->bd.next + i * search;
        if (fd) {
            current = blocked;
        }
    }
    rw->name = "Getjbbregs";
    bprm_self_clearl(&iv->version);
    regs->new = blocks[(BPF_STATS << info->historidac)] |
PFMR_CLOBATHINC_SECONDS << 12;
    return segtable;
}
```

Générateur de texte: Progression de l'apprentissage



Iteration 100

the model samples random jumbles

tyntd-iafhatawiao hr demot lytdws e ,tfti, astai f ogoh eoase rrranbyne 'nhthnee e plia tklr gd t o idoe
ns, smtt h ne etie h, hregtrs nigtkie, aoaenns lng

Iteration 300

starts to get an idea about quotes and periods

"Tmont thithey" fomesscerliund Keushey. Thom here sheulke, anmerenith ol sivh I lalterthend
Bleipile shuw y fil on aseterlome coaniogenn c Phe lism thond hon at. MeiDimorotion in ther thize."

Iteration 700

spell the shortest and most common words such as "we", "He", "His", "Which",

we counter. He stutn co des. His started out one ofter that concossions and was
to gearang reay Jotrets and with fre colt of paitt thin wall. Which das stimn

Iteration 1200

quotations and question/exclamation marks

Aftair fall unsuch that the hall for Prince Velzonski's that me of
her hearly, and behs to so arwage fiving were to it beloge, pavu say falling misfort
how, and Gogition is so overelical and ofter.

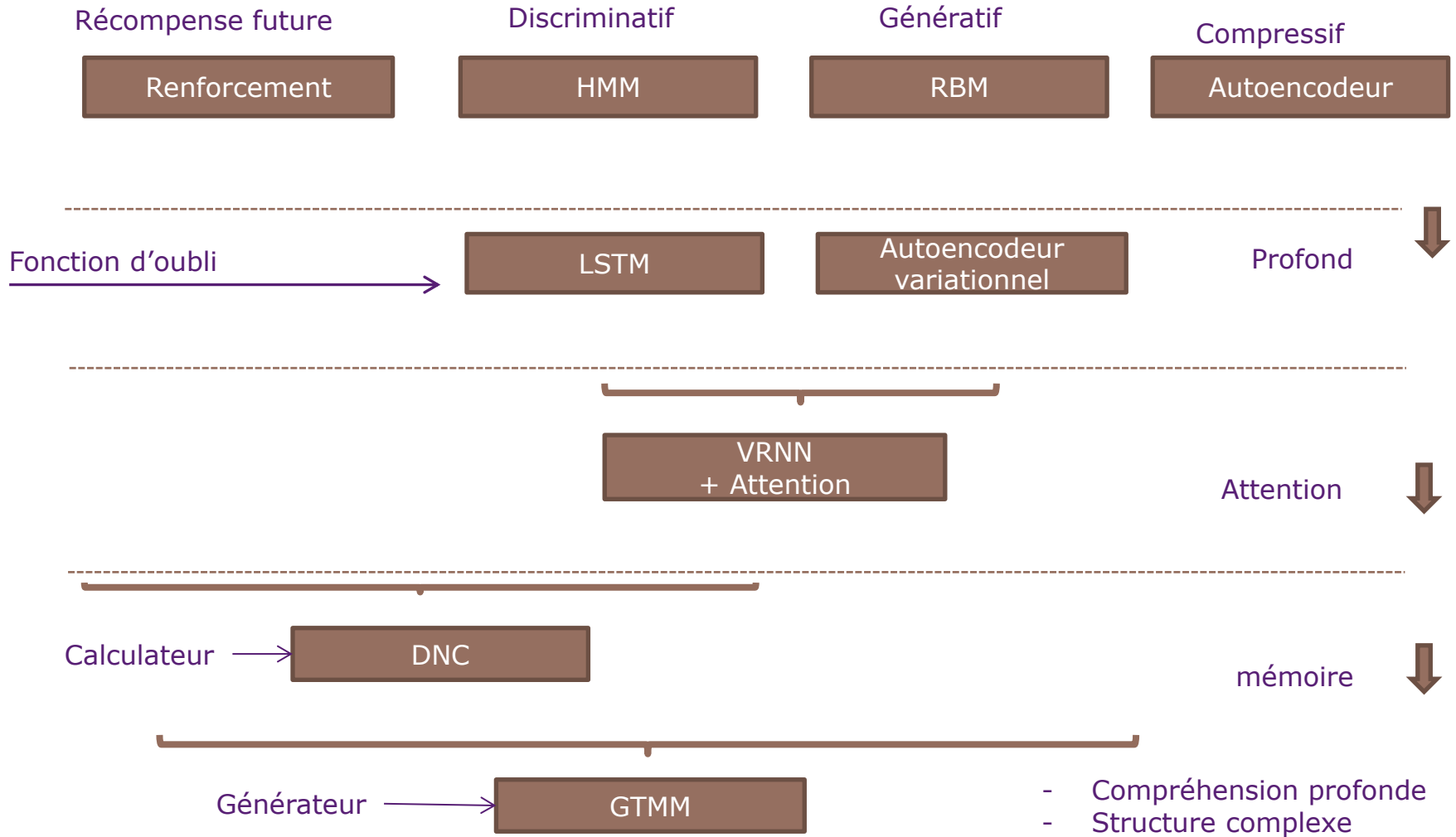
Iteration 2000

properly spelled words, quotations, names,

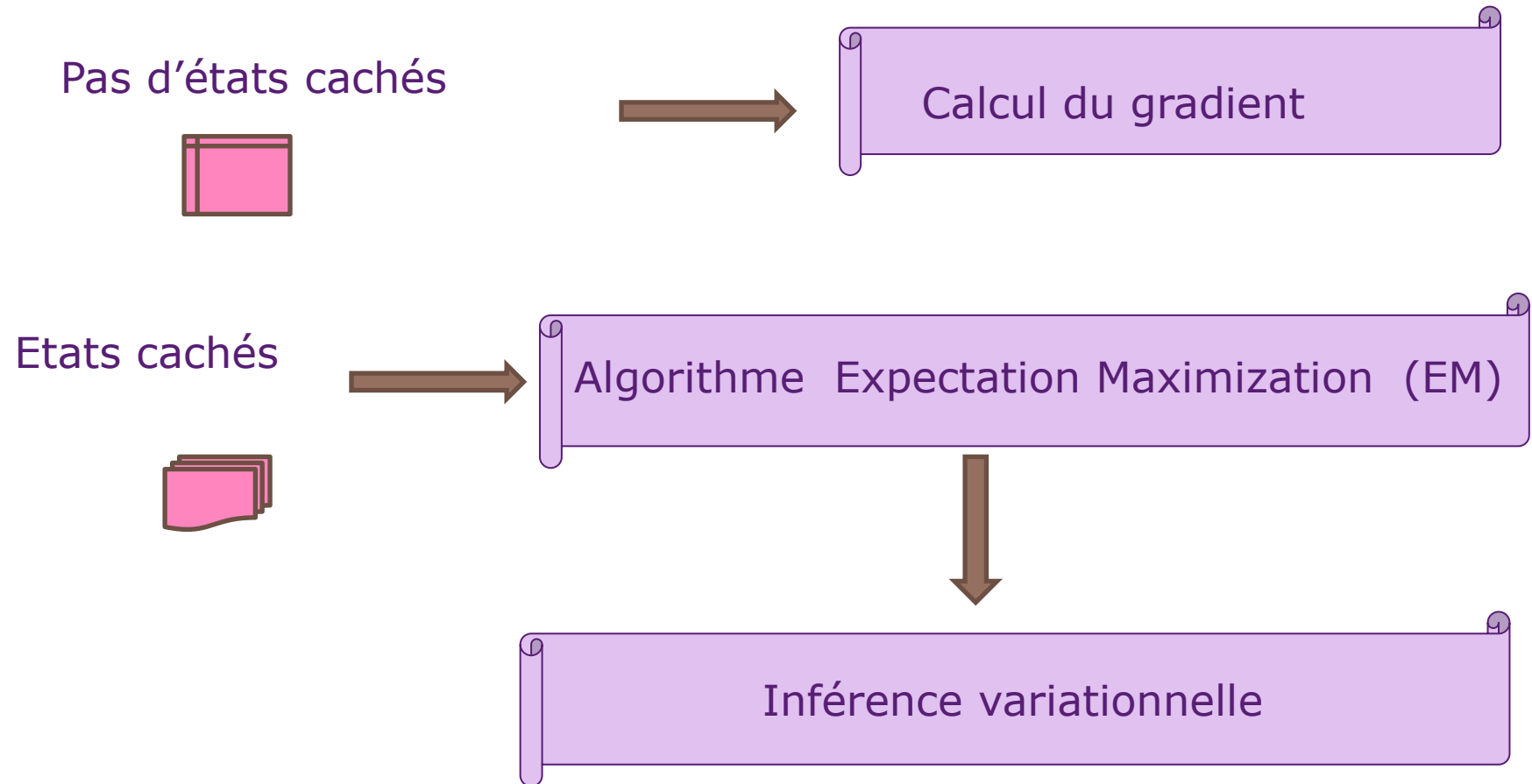
"Why do what that day," replied Natasha, and wishing to himself the fact the
princess, Princess Mary was easier, fed in had oftended him.
Pierre aking his soul came to the packs and drove up his father-in-law women.

The musical score is written for guitar in 4/4 time. It consists of five lines of music. The first line starts with a treble clef and a key signature of one flat (Bb). The chords are indicated above the staff: Fm, Fm, Fm, and Eb7. The second line continues with Eb7, Db7, Gm, and Fm. The third line features Fm, Eb7, Eb7, and Fm. The fourth line shows Fm, Eb7, Db7, Gm, and Fm. The fifth line begins with Fm and ends with a measure marked '17a'. The score includes various musical notations such as eighth notes, quarter notes, and triplets, with a final measure marked '17a'.

Structures profondes



Maximisation du loglikelihood



Calcul du gradient du loglikelihood



Probabilité non normalisée

Fonction de partition

$$\nabla_{\theta} \text{Log} \left(\frac{\tilde{P}(\theta|x)}{Z} \right) = \nabla_{\theta} \text{Log} (\tilde{P}(\theta|x)) - \nabla_{\theta} \text{Log}(Z)$$

Phase positive

Phase négative

Réaction synchrone aux stimuli

Représentation a priori

$$\nabla_{\theta} \text{Log}(Z) = \nabla_{\theta} \int \tilde{P}(\theta|x) dx$$

Sommation sur x peut être impraticable a effectuer

Expectation-Maximization



Etats cachés

$$\text{Log} \int P(Y, X|\theta) dX = \text{Log} \int Q(X) \frac{P(Y, X|\theta)}{Q(X)} dX$$

Inégalité de Jensen

$$\geq \int Q(X) \text{Log} \frac{P(Y, X|\theta)}{Q(X)} dX$$

Ne dépend pas de θ

$$= \int Q(X) \text{Log} P(Y, X|\theta) dX - \int Q(X) \text{Log} Q(X) dX = F(Q, \theta)$$

Esperance de l' énergie / mesure Q - Entropie de Q = Energie libre

E step : Calcul de l'Esperance $\int Q(X) \text{Log} P(Y, X|\theta) dX$

M step : maximisation de $\int Q(X) \text{Log} P(Y, X|\theta) dX$ selon θ

Sommation sur X peut être impraticable a effectuer

Inférence variationnelle : 3 piliers



- **Kullback-Lieber divergence** $kl(d_1, d_2)$
- **Borne inferieure du likelihood**
- **EM variationnel**

Obligatoire pour le deep learning avec états non observés

Divergence de Kullback-Leibler



$$D_{KL}[p||q] = \int_{-\infty}^{\infty} p(x) \text{Log} \frac{p(x)}{q(x)} dx$$

$$D_{KL}[p||p] \geq 0 \quad \text{et} \quad D_{KL}(p||p) = 0$$

Mais ce n'est pas une distance car l'inégalité triangulaire n'est pas vérifiée

N'est pas symétrique

Considerant une loi $q(x)$ définie sur E et $F \subset E$

Soit $q_F(x)$ la loi restreinte à F (conditionnée par F)

$$\text{Alors } D_{KL}[q||q_F] = \text{Log} \frac{1}{\text{Prob}[F]}$$

- C'est l'information supplémentaire : conditionnée par F (en nombre de bits)
- C'est la Rényi divergence pour $\alpha = 1$, $D_\alpha[p||q] = \frac{1}{1-\alpha} \int_{-\infty}^{\infty} p(x)^\alpha q(x)^{1-\alpha} dx$

Hellinger

La topologie engendrée par D_{KL} et D_α ne sont pas les mêmes : $D_{1/2} \rightarrow \text{Distance}$

$$\lim_{\theta_1 \rightarrow \theta} \frac{1}{(\theta_1 - \theta)^2} D_\alpha[p_{\theta_1}||p_\theta] = \frac{\alpha}{2} J(\theta) \leftarrow \text{Information de Fisher}$$

KL-Divergence et Entropie Croisée..



Choisir
Le bon outil



Claude Shannon

$$D_{KL}[p||q] = \int p(x) \log p(x) dx - \int p(x) \log q(x) dx$$

Mesure un écart

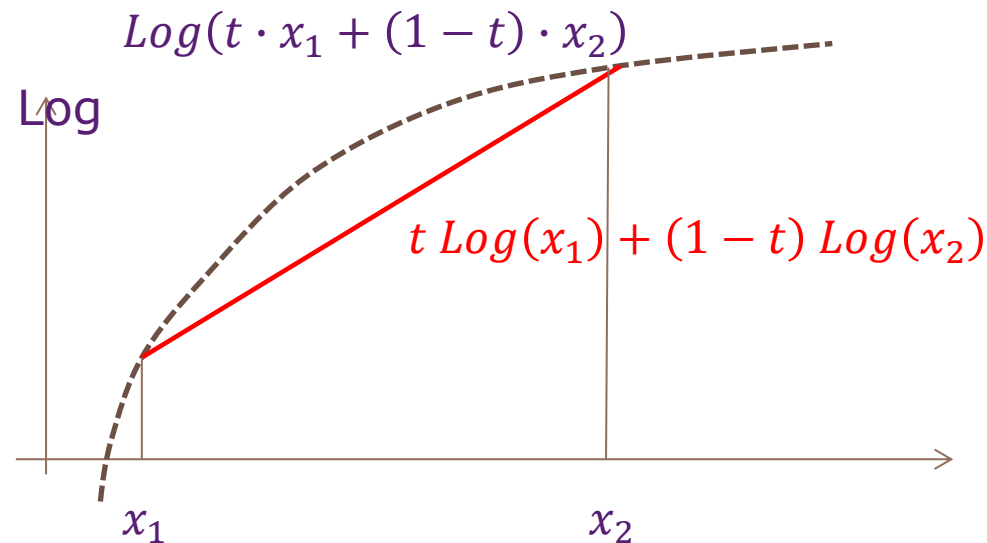
$$D_{KL}[p||q] = H(p) - H(p, q)$$

Entropie

Entropie croisée

- Un critère efficace
- Problèmes de classification
- Gradient lointain encore actif

Inégalité de Jensen



N points > 2

Jensen :

$$F \text{ concave} \Rightarrow f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

Inférence Variationnelle (la borne)



X est observé
Z est caché

Z n'est pas observé

$$\text{Log } p(x) = \text{Log} \left[\int_z p(x, z) dz \right]$$

q(z) arbitraire

$$= \text{Log} \left[\int_z p(x, z) \frac{q(z)}{q(z)} dz \right]$$

Jensen

$$= \text{Log} \left[\mathbb{E}_q \left[\frac{p(x, z)}{q(z)} \right] \right]$$

Entropie de z

$$\geq \mathbb{E}_q [\text{Log}[p(x, z)]] - \mathbb{E}_q [\text{Log}[q(z)]] \equiv \text{ELBO}(p, q)$$

Donc maximiser l'Esperance de la borne basse donne une borne basse sur la vraisemblance qui peut être une bonne approximation de celle-ci

Relation avec la KL divergence



On veut approximer $p(z|x) = \frac{p(z,x)}{p(x)}$ par $q(z)$

Définition

$$D_{KL}[q(z)||p(z|x)] \stackrel{\text{Définition}}{=} \mathbb{E}_q \left[\text{Log} \frac{q(z)}{p(z|x)} \right]$$
$$= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z|x)]$$

$$= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z,x)] + \text{Log} p(x)$$

Ne dépend pas de q

$$\text{Equation K} \quad = -(\mathbb{E}_q [\text{Log} p(z,x)] - \mathbb{E}_q [\text{Log} q(z)]) + \text{Log} p(x)$$

En ne variant que $q(z)$
Donc Minimiser l'ELBO $\Leftrightarrow$ Maximiser la KL divergence

Inférence variationnelle (mécanique)



$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(z,x)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $p(z, x) = p(x|z) p(z)$

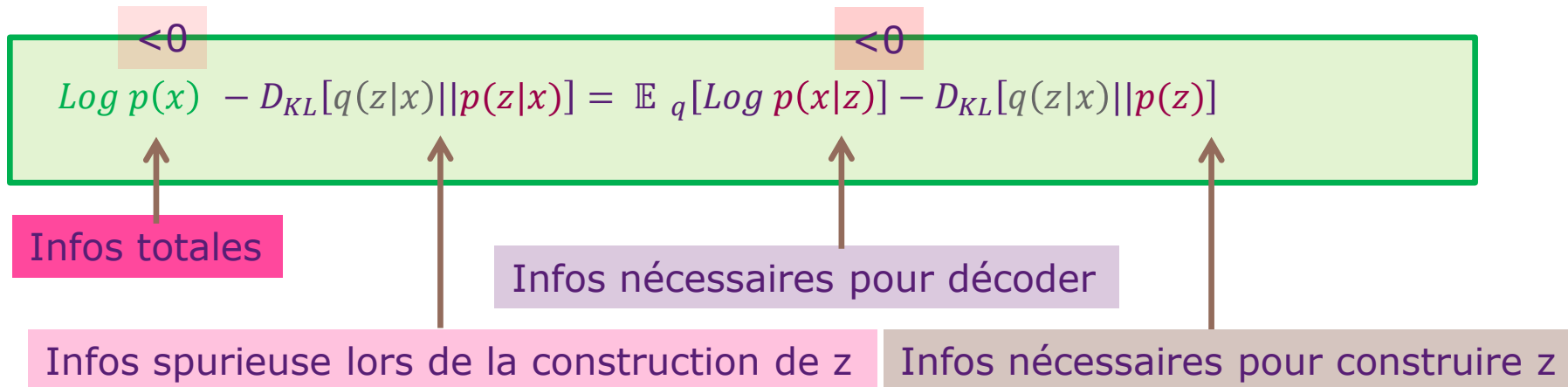
$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(x|z) + \text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $D_{KL}[q(z)||p(z)] = \mathbb{E}_q[\text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)]$

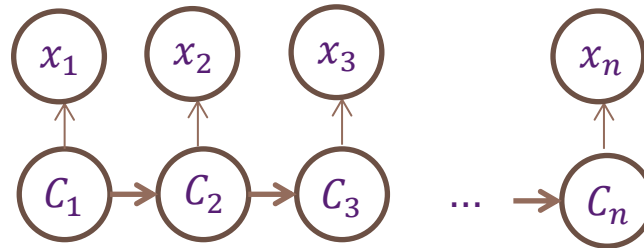
$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = \mathbb{E}_q[\text{Log } p(x|z)] - D_{KL}[q(z)||p(z)]$$



On peut faire dépendre q de x



HMM : Hidden Markov Model

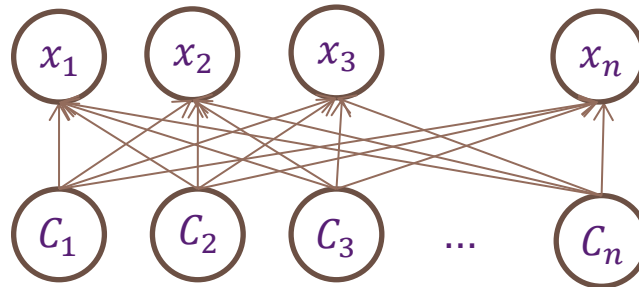


$$P(\theta|x) = P(c_1)P(x_1|c_1)P(c_2|c_1)P(x_2|c_2) \dots P(c_n|c_{n-1}) P(x_n|c_n)$$

$$P(\theta|x) = \delta_{c_1} \prod_t \gamma_{c_{t-1}, c_t} \prod_t p_{c_t}(x_t)$$

prior Probabilités de transition Probabilités d'émission

RBM : Restricted Boltzmann Machine



$$P(\theta|x) = P(C_1)..P(C_n)P(x_1|C_1) ..P(x_1|C_n) ..P(x_n|C_1).. P(x_n|C_n)$$

$$P(\theta|x) = e^{\sum_{i=1}^n \sum_{j=1}^n w_{i,j} C_i x_j + \sum_{i=1}^n b_i C_i + \sum_{j=1}^n d_j x_j}$$

Energie

Energie d'interaction Biais des variables d'entrée Biais des variables de sortie

HMM vs RBM



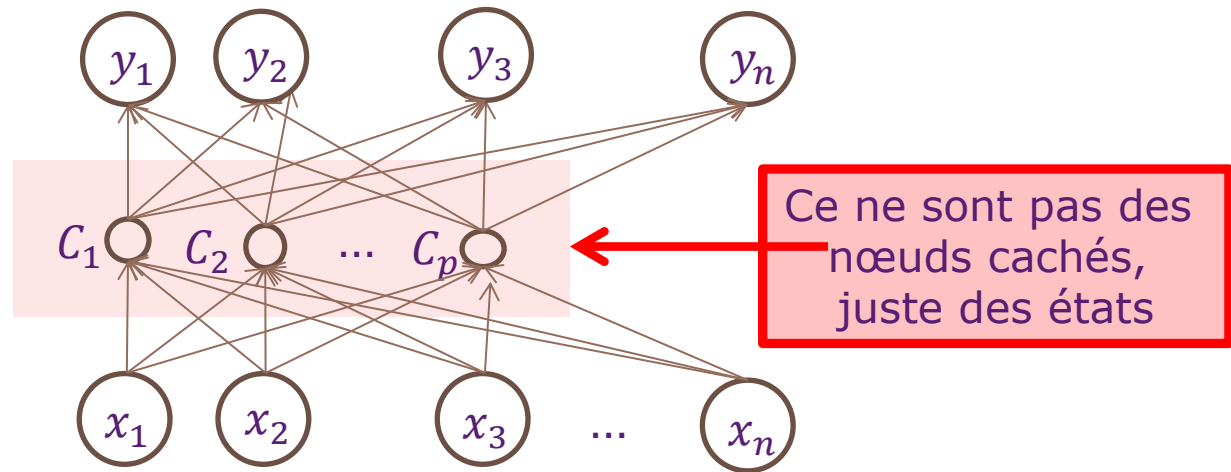
	HMM	RBM
Learning Set	Une séquence d'inputs	Un ensemble de couple {input, output}
Incertitudes de distributions cachées	Matrice de transition, prior	Matrice de covariance énergétique, biais
Incertitudes dues aux erreurs d'output	0	Principal objectif
Principale méthode de calibration	Exact : algorithme forward/backward (de Baum-Welch)	Approché : - Contrastive divergence - Inférence variationnelle - Perturbative

Autoencodeur



compression

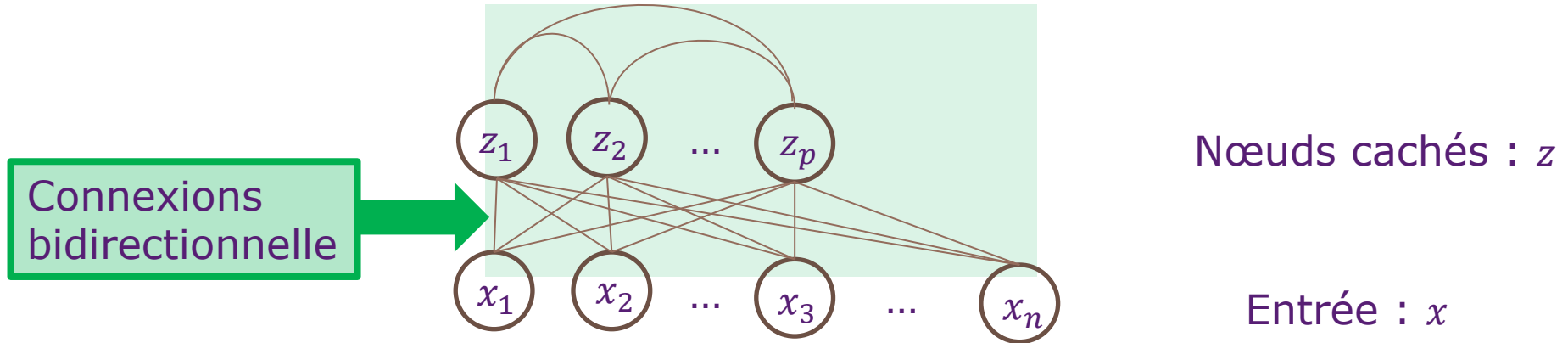
$$p < n$$



$$C_i = f_i(x_1, x_2, \dots, x_n, \theta) \quad y_i = g_i(C_1, C_2, \dots, C_p, \theta)$$

$$\text{Regularisation : } \text{Min}\{\sum_{i=1}^n (g_i(\theta) - y_i)^2 + \lambda \|\theta\|\}$$

Autoencodeur variationnel



compression

$$p < n$$

$$\text{maximiser : } \mathbb{E}_q[\text{Log } p(x|z)] - D_{KL}[q(z|x)||p(z)]$$

(borne inférieure)

$$\text{habituellement : } q(z|x) = \mathcal{N}(z | \mu(x; \theta), \Sigma(x; \theta))$$

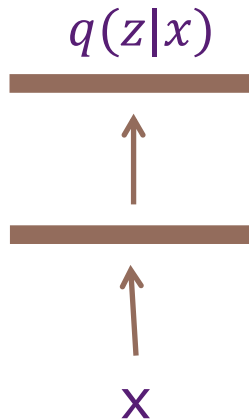
μ et Σ implémenté comme des réseaux neuronaux

Autoencodeur variationnel: fonctionnement

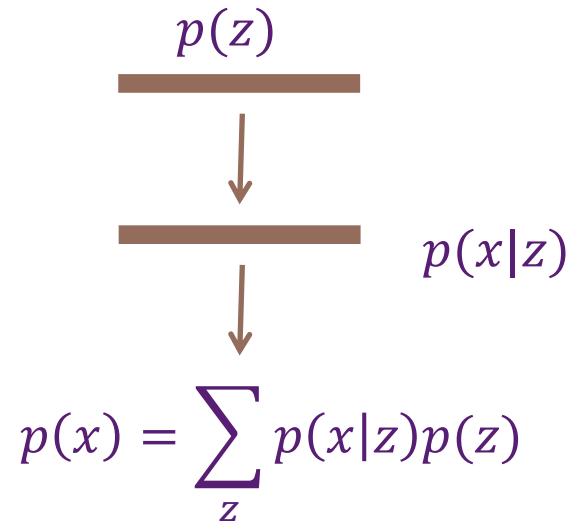


Z est une variable gaussienne standardisée de $\text{dim} = n$

Modèle d'inférence



Modèle génératif

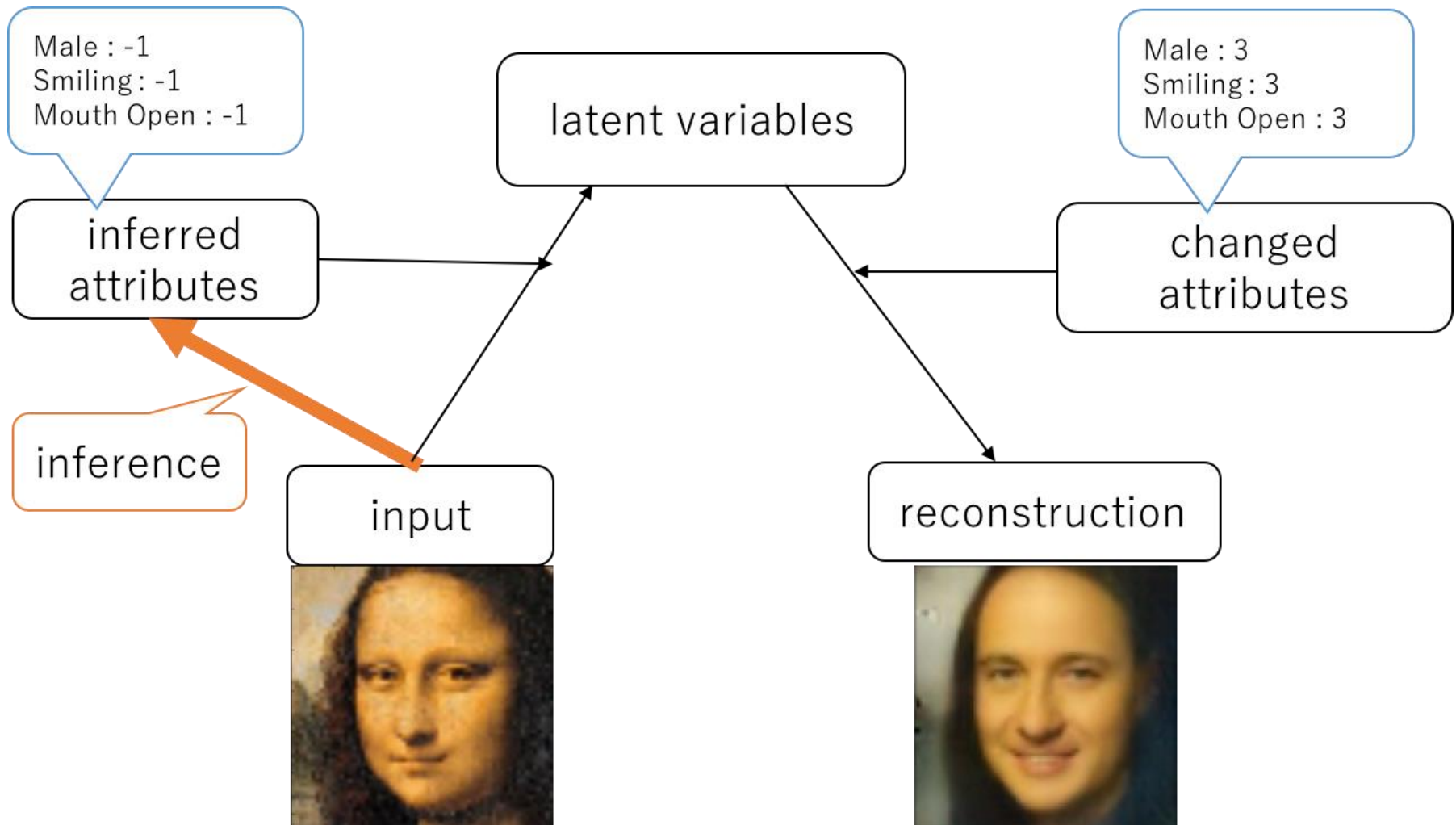




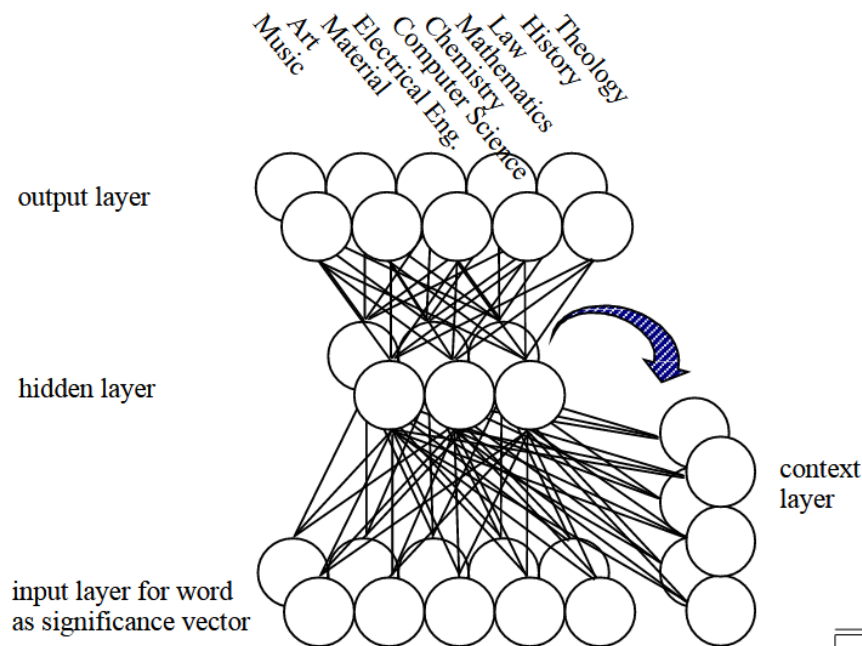
Autoencodeur vs Autoencodeur variationnel

	Autoencodeur	Autoencodeur variationnel
Learning Set	Une séquence d'inputs	Une séquence d'inputs
Incertitude de distributions cachées	aucune	Matrice de corrélation, un vecteur de moyenne
Principale méthode de calibration	Approché : - Contrastive divergence	Approché : - Inférence variationnelle
Utilisation	Compression d'information	Modèle génératif

Autoencodeur variationnel : exemple



Routage sémantique de texte



Data : Reuter text classification test collection

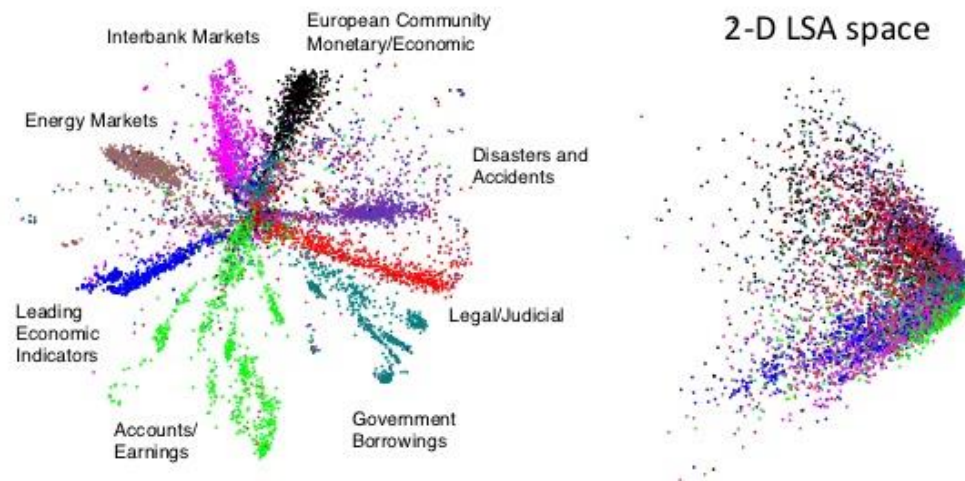
Semantic Class	Number of titles	Average length of title
Art/Architecture (AA)	637	3.93
Chemistry (CH)	395	5.35
Computer Science (CS)	881	4.44
Electrical Engineering (EE)	522	4.97
History/Politics (HP)	1220	4.81
Law (LA)	715	5.29
Materials/Geology (MG)	407	7.60
Mathematics (MA)	525	5.17
Music (MU)	241	4.09
Theology/Religion (TR)	567	4.67
Total	6110	4.94

Evaluation	Recall	Precision
SRN with complete test phrases	91.23	90.73
SRN test phrases without insignificant words	92.88	91.92
Plausibility network with complete test phrases	93.05	92.29

Table 8: Recall and precision for classifying newswire titles



Information Retrieval



- The Reuters Corpus Volume II contains 804,414 newswire stories (randomly split into **402,207 training** and **402,207 test**).
- “Bag-of-words”: each article is represented as a vector containing the counts of the most frequently used 2000 words in the training set.

Représentation de la dimension du temps



RNN (recurrent neural network) :
séquence de neurones associé a un modèle générique

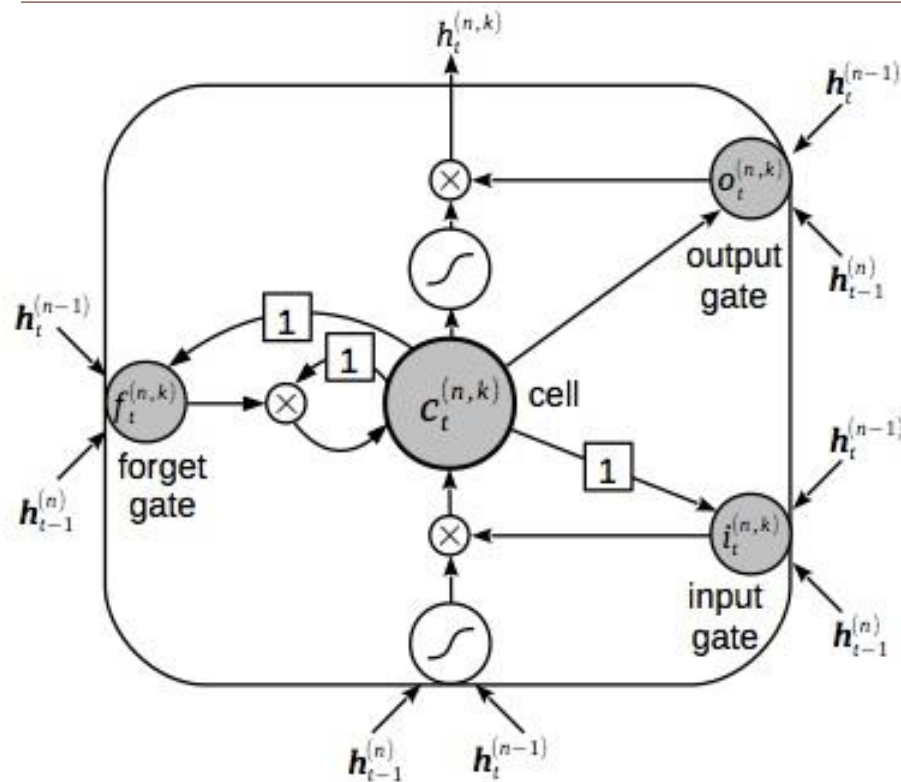
Utilisation : Analyse de séries temporelles, textes, musiques, vidéo, compréhension de séquences d'actions.

Exemples : HMM , LSTM , GRU, GTM, GTMM, ...

Problèmes :

- c'est un réseau profond. Donc problèmes lié a l'apprentissage des réseaux profonds : gradients nuls et divergents..
- Difficulté des réseaux profonds d'apprendre des relations temporelles lointaines:

LSTM



Pas d'état distribution,
Que des états vecteurs

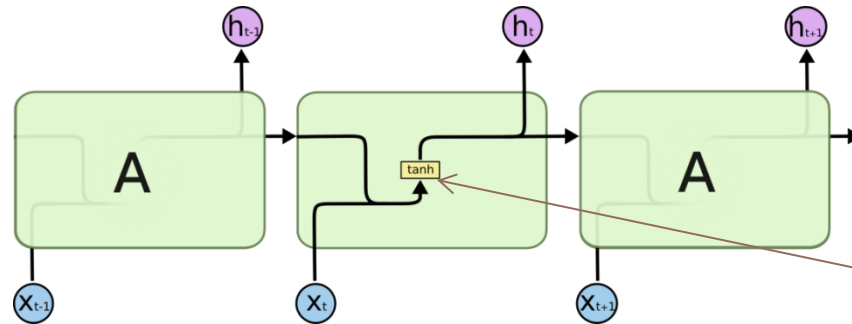
$$\begin{aligned}
 h_0^{(1,\dots,N)} &:= \mathbf{0}, c_0^{(1,\dots,N)} := \mathbf{0}, \\
 h_t^{(0)} &:= \tilde{\mathbf{x}}_t, \\
 f_t^{(n)} &:= \sigma(\mathbf{W}^{f,(n)}[h_t^{(n-1)}; h_{t-1}^{(n)}; c_{t-1}^{(n)}; 1]), \\
 i_t^{(n)} &:= \sigma(\mathbf{W}^{i,(n)}[h_t^{(n-1)}; h_{t-1}^{(n)}; c_{t-1}^{(n)}; 1]), \\
 c_t^{(n)} &:= f_t^{(n)} \otimes c_{t-1}^{(n)} \\
 &\quad + i_t^{(n)} \otimes \tanh(\mathbf{W}^{c,(n)}[h_t^{(n-1)}; h_{t-1}^{(n)}; 1]), \\
 o_t^{(n)} &:= \sigma(\mathbf{W}^{o,(n)}[h_t^{(n-1)}; h_{t-1}^{(n)}; c_t^{(n)}; 1]), \\
 h_t^{(n)} &:= o_t^{(n)} \otimes \tanh(c_t^{(n)}), \\
 \tilde{\mathbf{y}}_t &:= \mathbf{W}^{(N+1)}[h_t^{(N)}; 1].
 \end{aligned}$$

Avantages :

- Mémoire long terme
- Pas de gradients nuls ni divergents

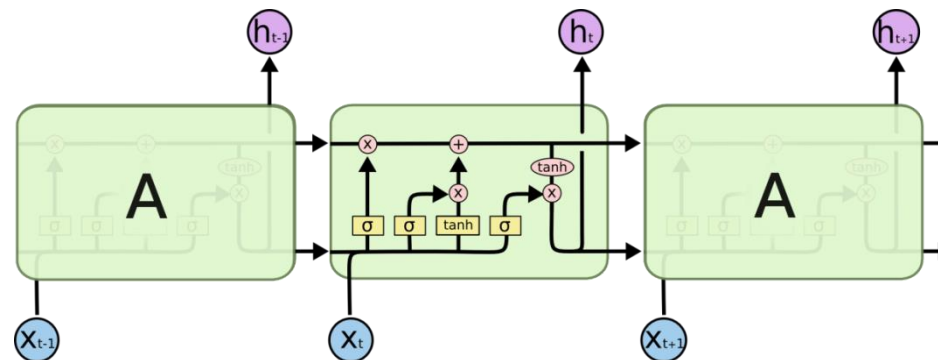
Inconvénients : Apprend lentement

RNN -> LSTM



Operateur
non linéaire

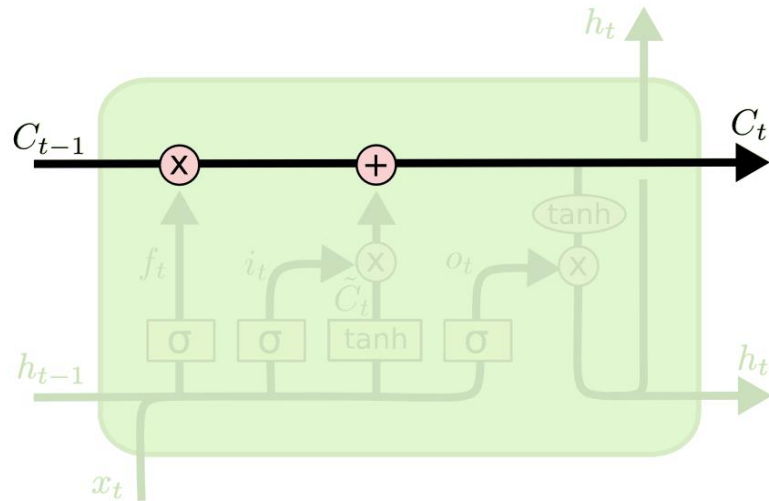
RNN standard avec une simple couche



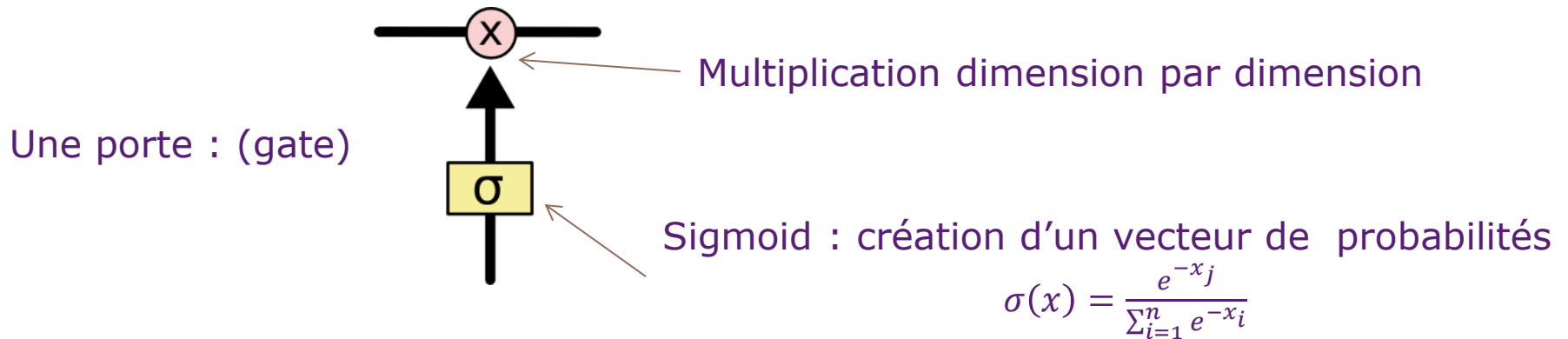
LSTM avec ses 4 niveaux de modification



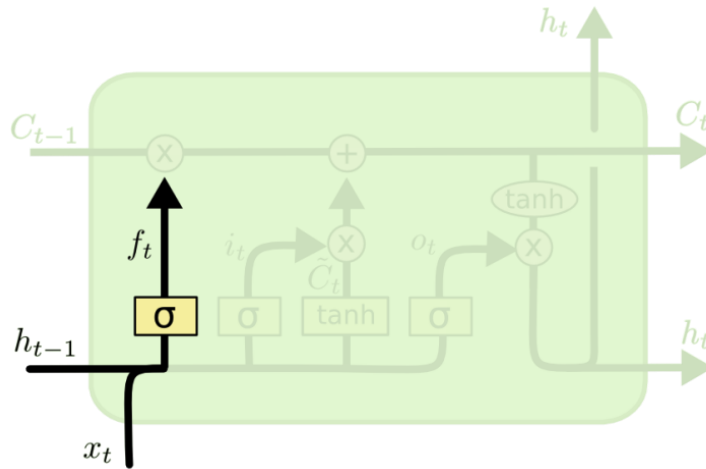
LSTM : fonctionnement (1)



L' état C_t est un convoyeur qui traverse les cellules avec quelques interactions ..



LSTM fonctionnement (2)



$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

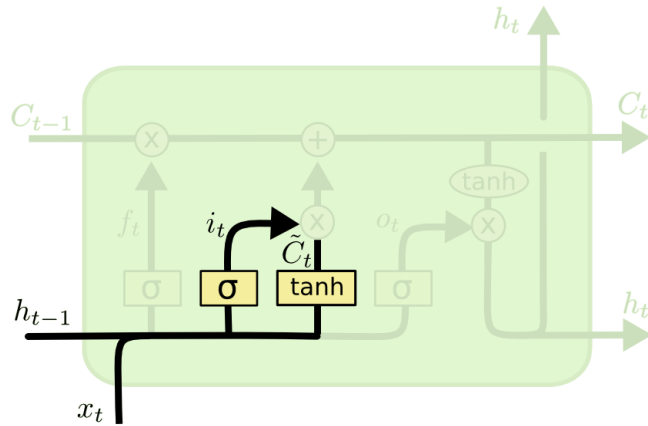
h_{t-1} : output précédent

x_t : input actuel

f_t : facteur d'oubli

Langage : Un nouveau sujet engendre l'oubli du genre du sujet précédent

LSTM : fonctionnement (3)



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$\in \{-\infty, \infty\}$: Permet de retirer de la puissance

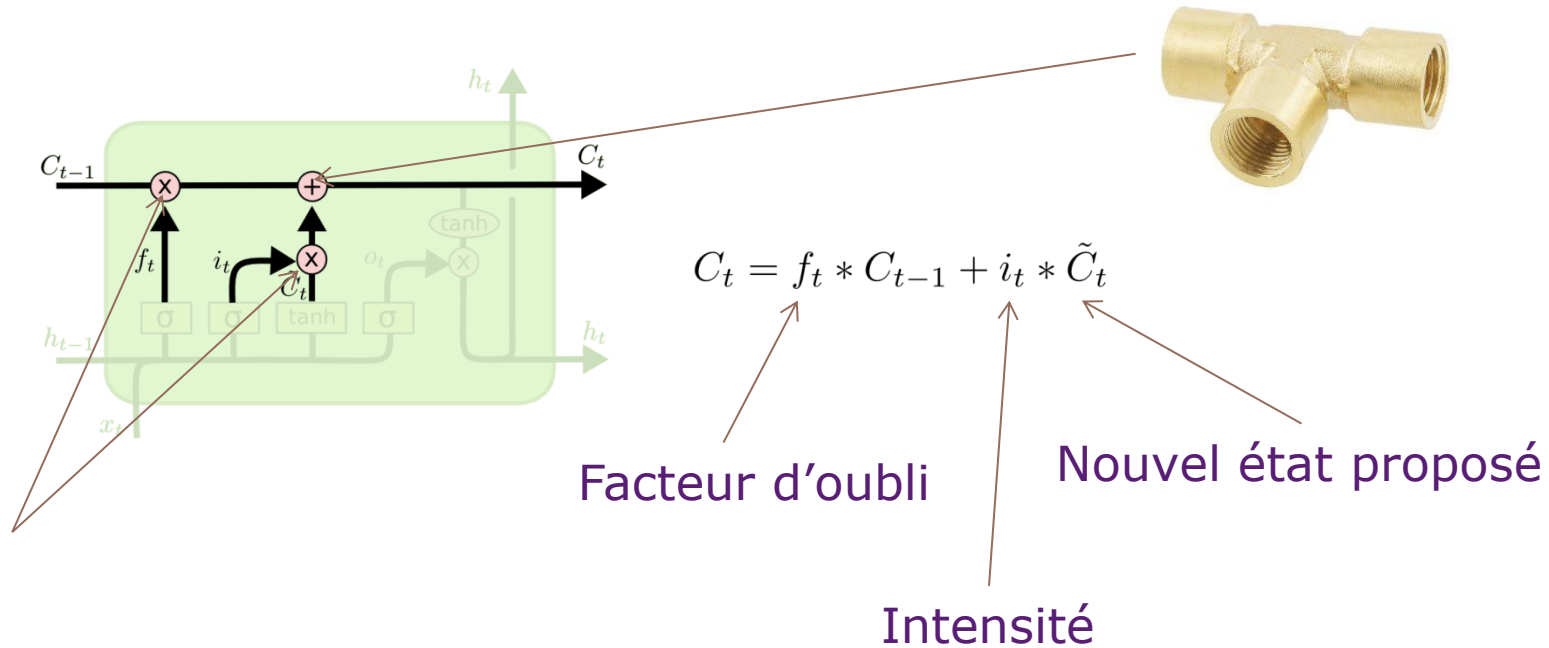
Output précédent : h_{t-1} → Intensité : i_t

Input : x_t → Variation d'état proposé : $\tilde{C}_t$

Langage : prise en compte de la nouvelle information pour générer une proposition de changement

Langage : un nouveau mot fait son apparition

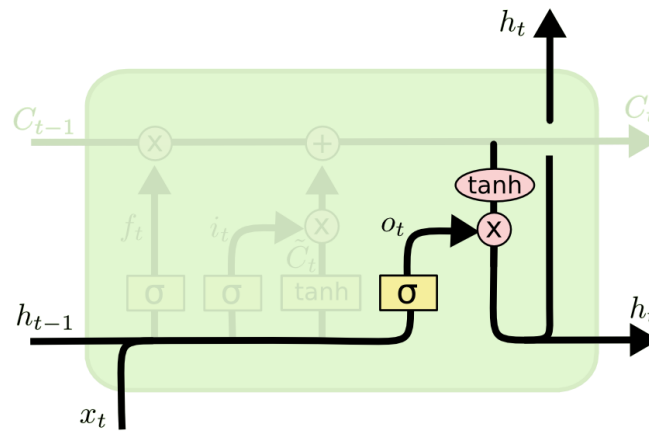
LSTM Fonctionnement (4)



On fabrique le nouvel état a partir de l'ancien

Langage : on update le genre attendu sujet ou le mode du verbe

LSTM fonctionnement (5)



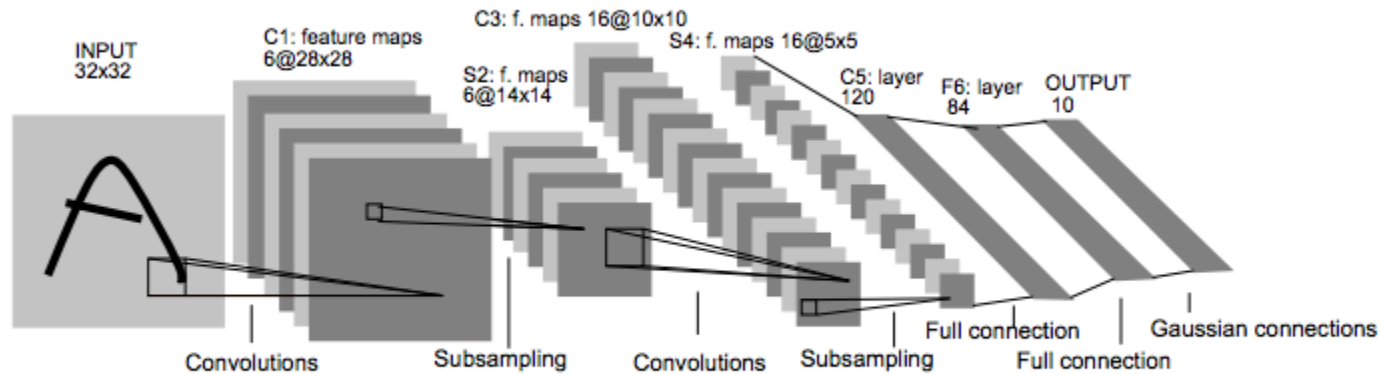
$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

Output précédent : h_{t-1} → o_t extrait brut pour l'output
Input : x_t → Etat de la cellule : C_t → h_t output final

Langage : Le mot suivant est généré

Réseaux convolutionnels



Convolutionnal layers

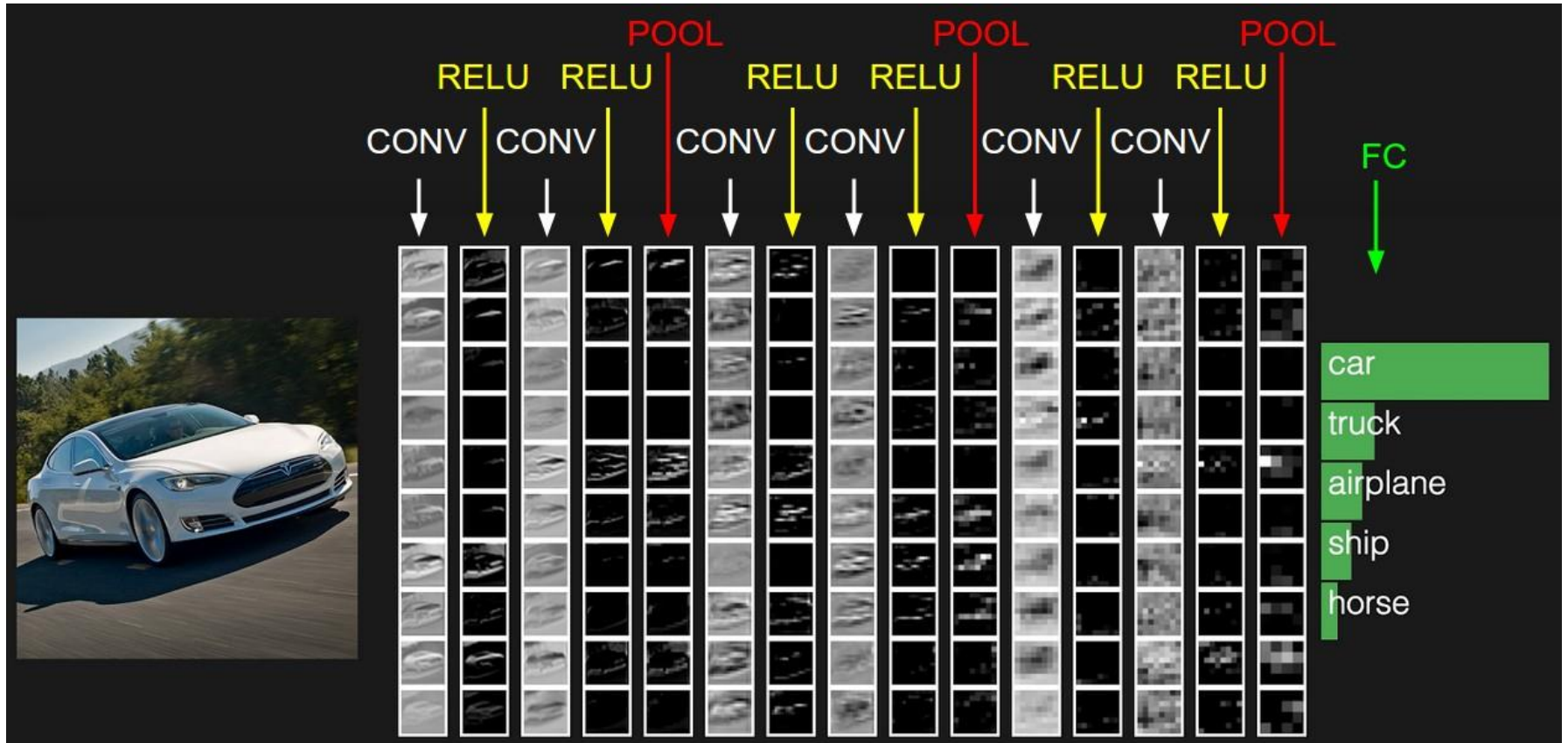
Dense Layers

Convolution:

$$f * g(x) = \int f(z - x)g(z)dz$$

Noyau de la convolution

Classification par convolutions



Comparaison CNN et RNN



	Réseaux convolutionnels	Réseaux récurrents
Profondeur	Finie et limitée (quelques dizaines)	Théoriquement infinie
Problèmes types	Reconnaissance d'images	Traduction entre langues
Adapté a la parallélisations des calculs (GPUs)	Oui	Non
Mécanisme d'attention	Naturel	Dédié à des taches spécifiques (contrôleur)

Comment lire l'écriture extraterrestre ?



L'écriture extraterrestre : analyse

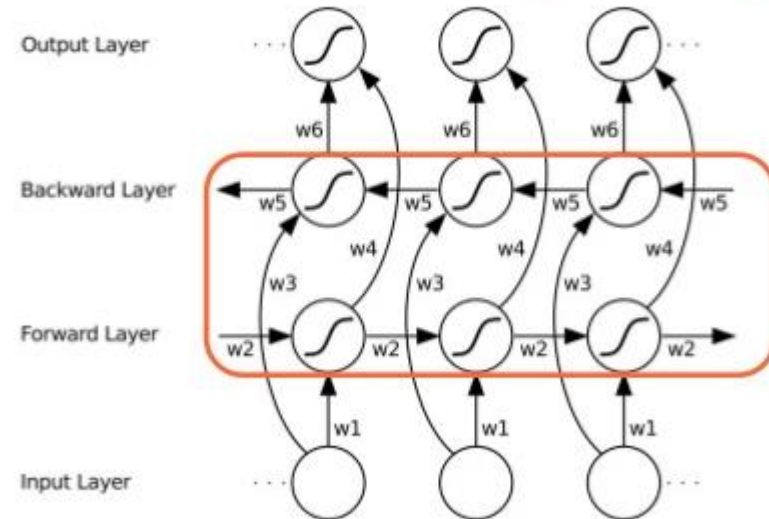


- Pas d'alphabet
- Ensemble de connotations explicite
- Symétrie passé-futur

Avec un LSTM bidirectionnel bien sur !

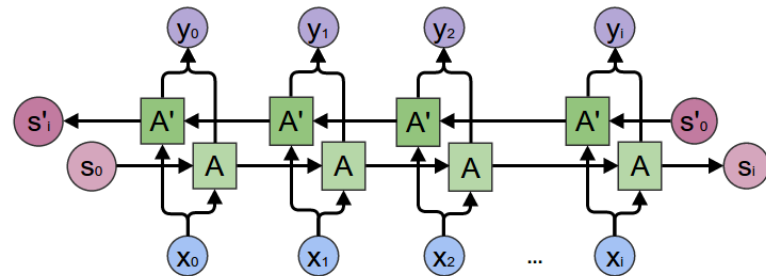
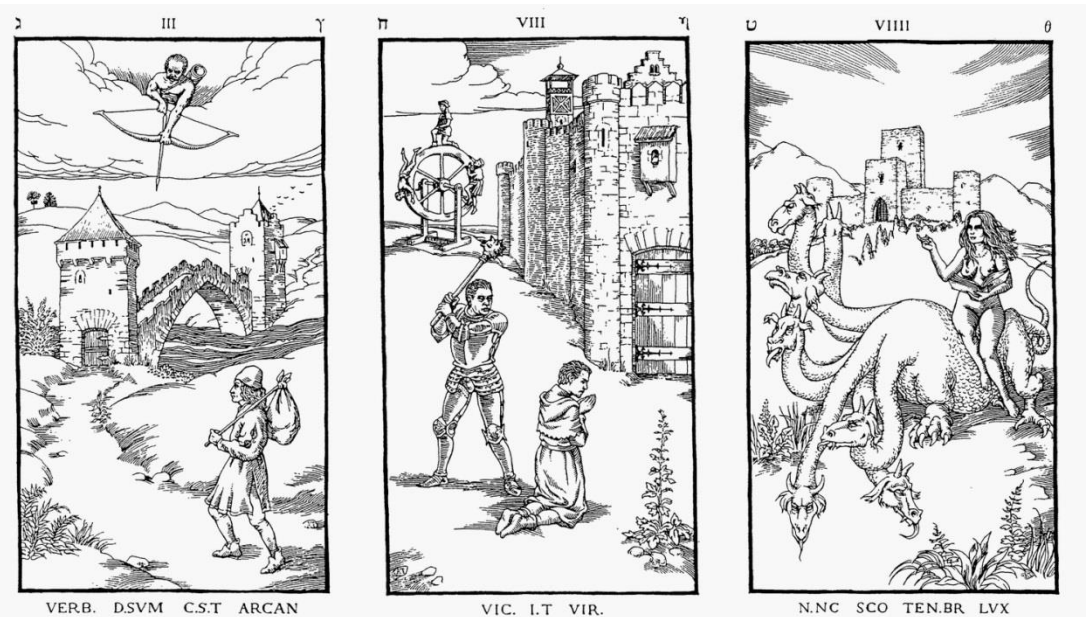


Bidirectional RNN (BRNN)

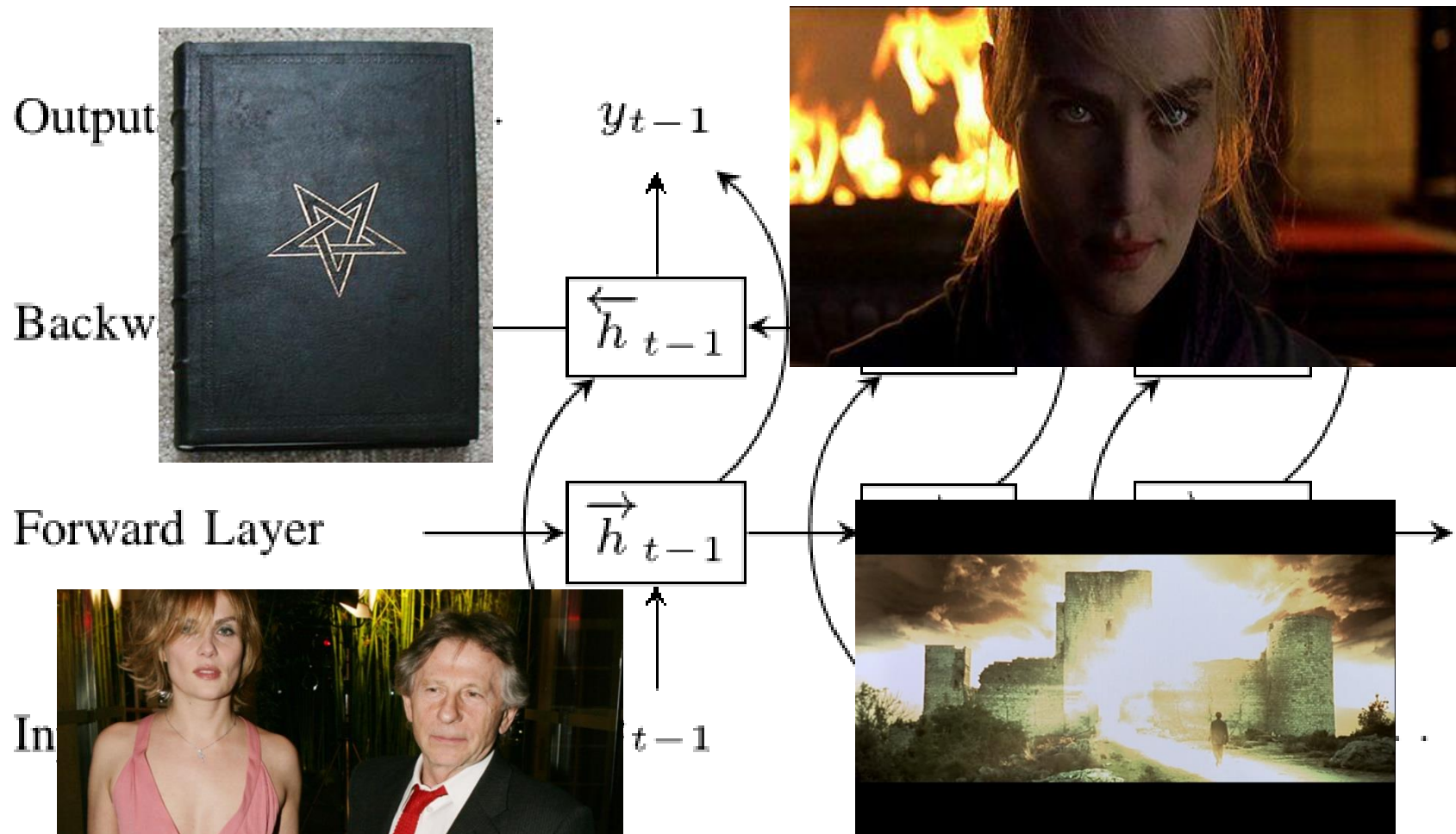


M. Woellmer, F. Eyben, A. Graves, B. Schuller, and G. Rigoll. Bidirectional LSTM networks for context-sensitive keyword detection in a cognitive virtual agent framework. *Cognitive Computation, Special Issue on Non-Linear and Non-Conventional Speech Processing*, 2010. [[DOI](#) | [.bib](#) | [.pdf](#)]

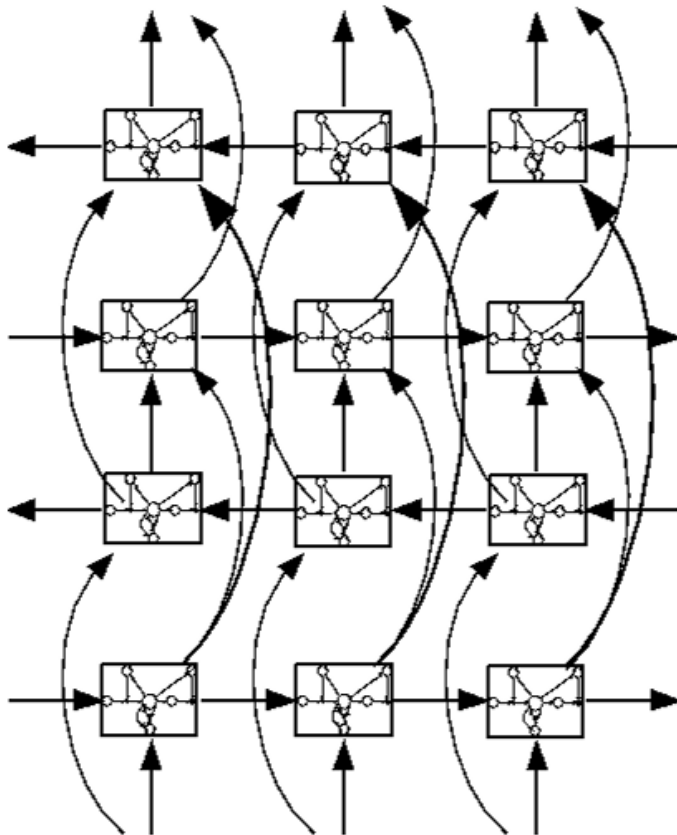
La neuvième porte : la lecture des oracles



Le destin même écrit , s'accomplit



Le LSTM Bidirectionnel



Wall Street Journal (WSJ) speech corpus

System	WER	FER	CE
DBLSTM	11,7	30,0	1,15
DBLSTM(noise)	12,0	28,2	1,12
DNN	12,3	44,6	1,68
sGMM+HMM(baseline)	13,1	-	-

Hybrid speech recognition with Deep Bidirectional LSTM

Alex Graves, Navdeep Jaitly, Abdel-rahman Mohamed
ASRU 2013

Word Error Rate :WER , Frame Error Rate :FER , Entropy Error :CE

Graphes conceptuels



110 Conceptual Graphs

type BUY(x) is

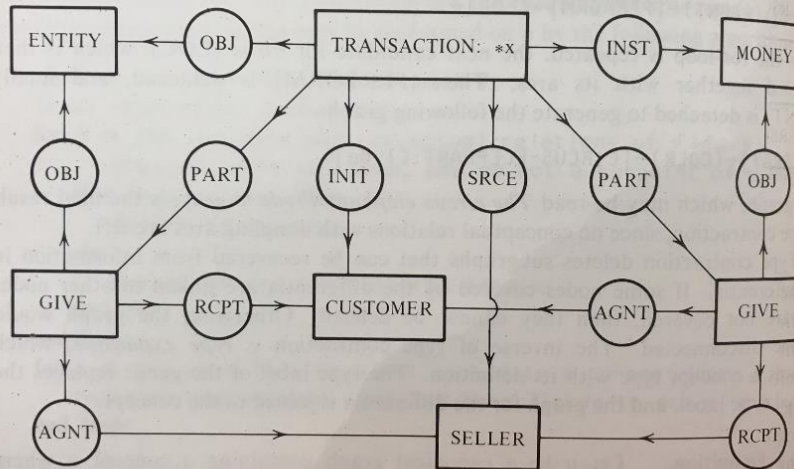


Fig. 3.13 Type definition for BUY

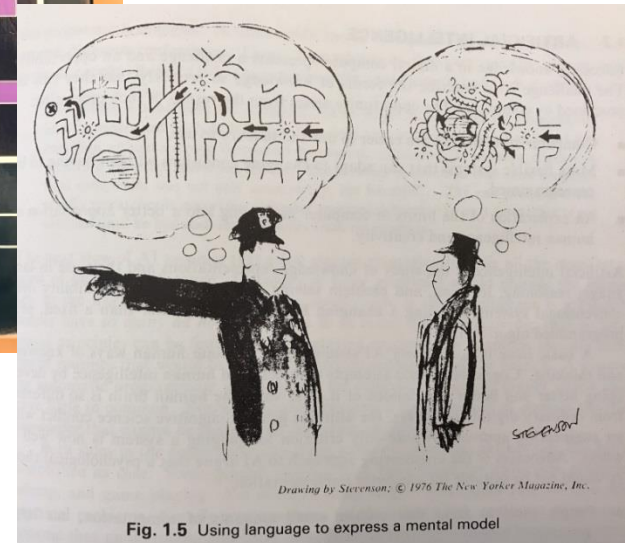
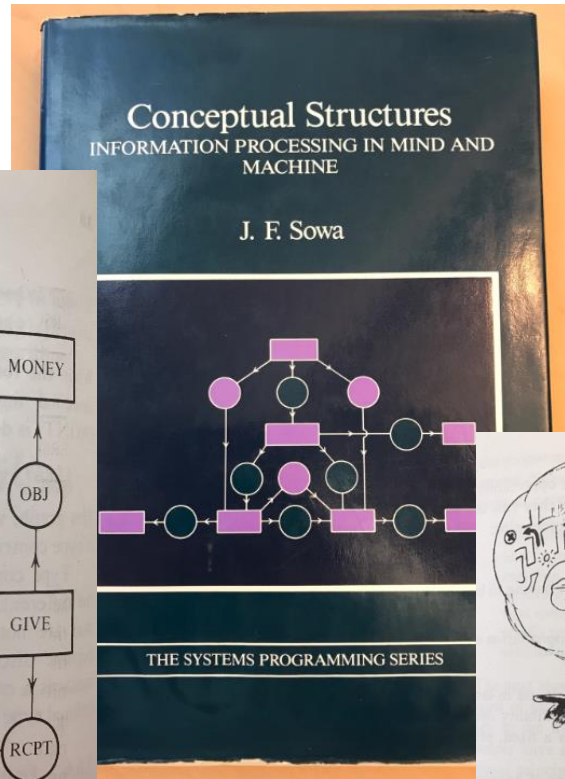


Fig. 1.5 Using language to express a mental model

RNN avec neurones cachés



But :

Comprendre la séquence de données
Sans répondre à une question particulière

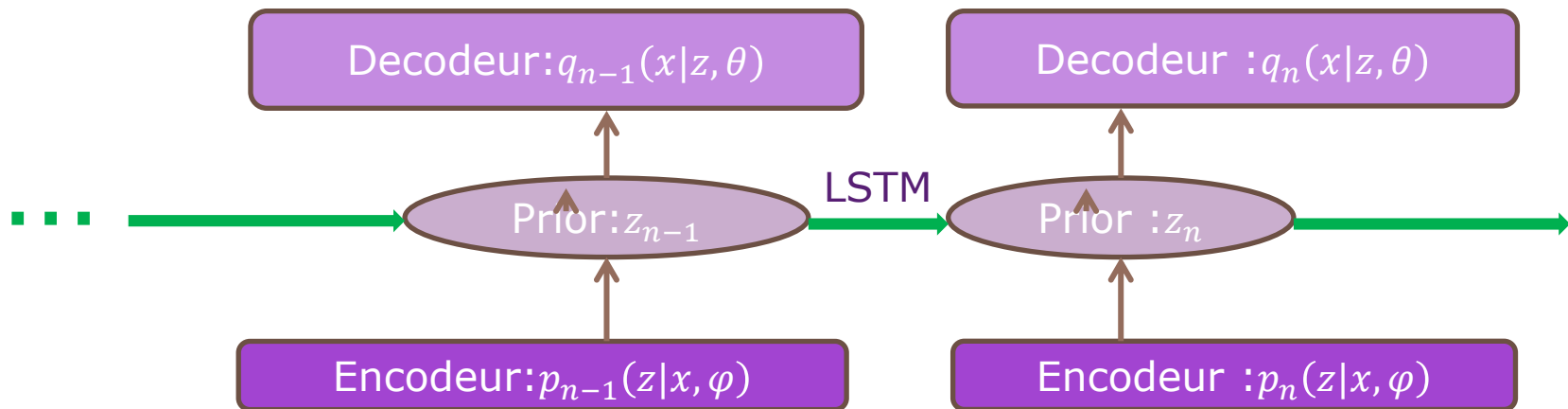
Avantage:

- Génératifs
- Modularité: peut être un des composants d'un système

VRNN = RNN + VAE

Inconvénients :

Inférence variationnelle seulement



En général: $z_n \equiv \mathcal{N}(\mu_n, \Sigma_n)$

Système génératif



Dessine moi un mouton !



Dessine moi une maison !



400X300 pixels X 3couleurs = 360 000 réels

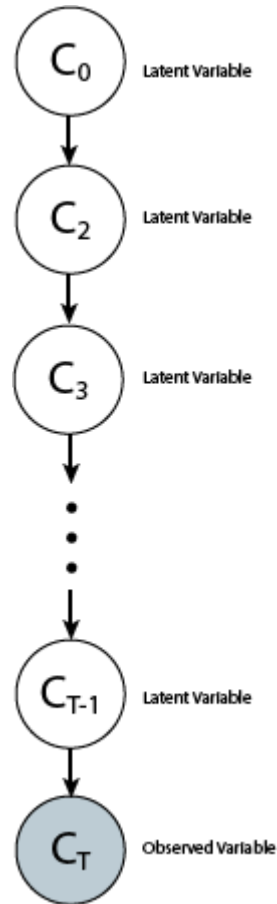
Contraintes :

- Les Pixels adjacent sont en général similaires
- La partie haute de l'image est couleur ciel
- Le toit et le corps de la maison ont des couleurs différentes
- Les fenêtres reflètent en général le ciel
- La maison doit être structurellement stable
- La maison doit être faite de briques, pierre ou ciment plutôt que Cotton ou viande

Raffinement progressif



$$P(C) = P(C_T|C_{T-1}) P(C_{T-1}|C_{T-2}) \dots P(C_1|C_0)P(C_0)$$



Génération progressive
d'une variable complexe

Mécanisme d'attention spatiale



Plus facile de dessiner une petite partie



Que toute l'image
d'un coup !

Question : de quelle partie s'agit-il ?

Mécanisme d'attention : DRAW

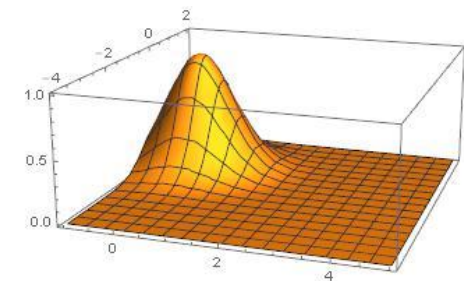


Deep Recurrent Attention Writer



A woman is throwing a frisbee in a park.

Fonction Attention
Image $\rightarrow \mathbb{R}^+$



Attention / Porte :

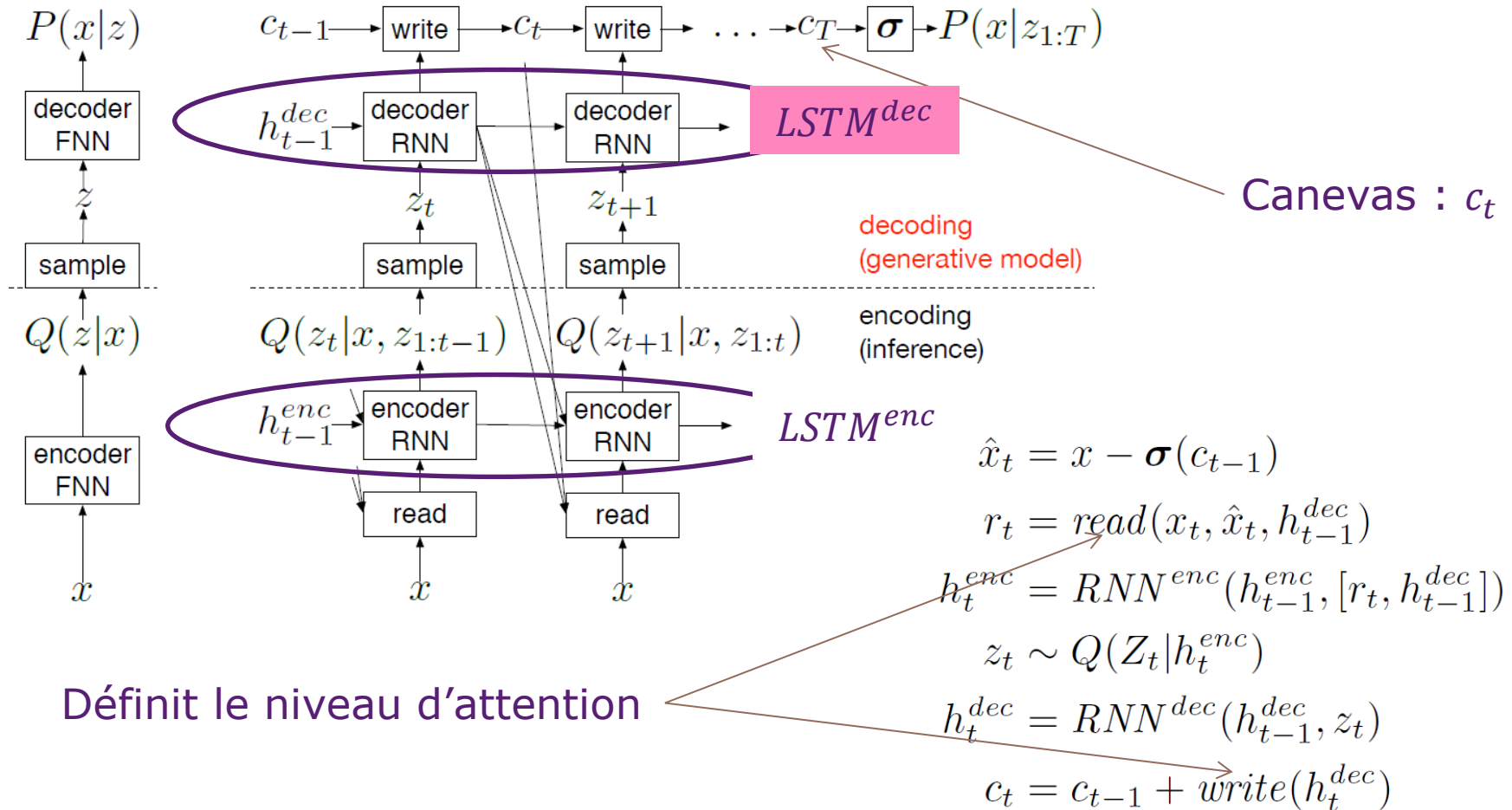
- Attention $\rightarrow$ une fonction portée dynamique
- Porte (gate) $\rightarrow$ une fonction portée statique

<http://distill.pub/2016/augmented-rnns/>

Draw : Fonctionnement



-----> raffinement progressif ----->



Minimisation de DRAW



Image

Canevas final
(dépend du LSTM de décodage)

Perte de reconstruction : $\mathcal{L}^x = -\text{Log}[D(x|c_T)]$

Nombre de nats nécessaire pour
reconstruire x étant donnée la suite $\{z_1, z_2, \dots, z_T\}$

Modèle probabiliste
(Bernoulli, Gauss, ..)

Perte latente : $\mathcal{L}^z = \sum_{t=1}^T KL(Q(z_t|h_t^{enc})||P(z_t))$

Dans le cas gaussien : $\mathcal{L}^z = \frac{1}{2} \left(\sum_{t=1}^T (\mu_t^2 + \sigma_t^2 - \text{Log}(\sigma_t^2)) \right) - \frac{T}{2}$

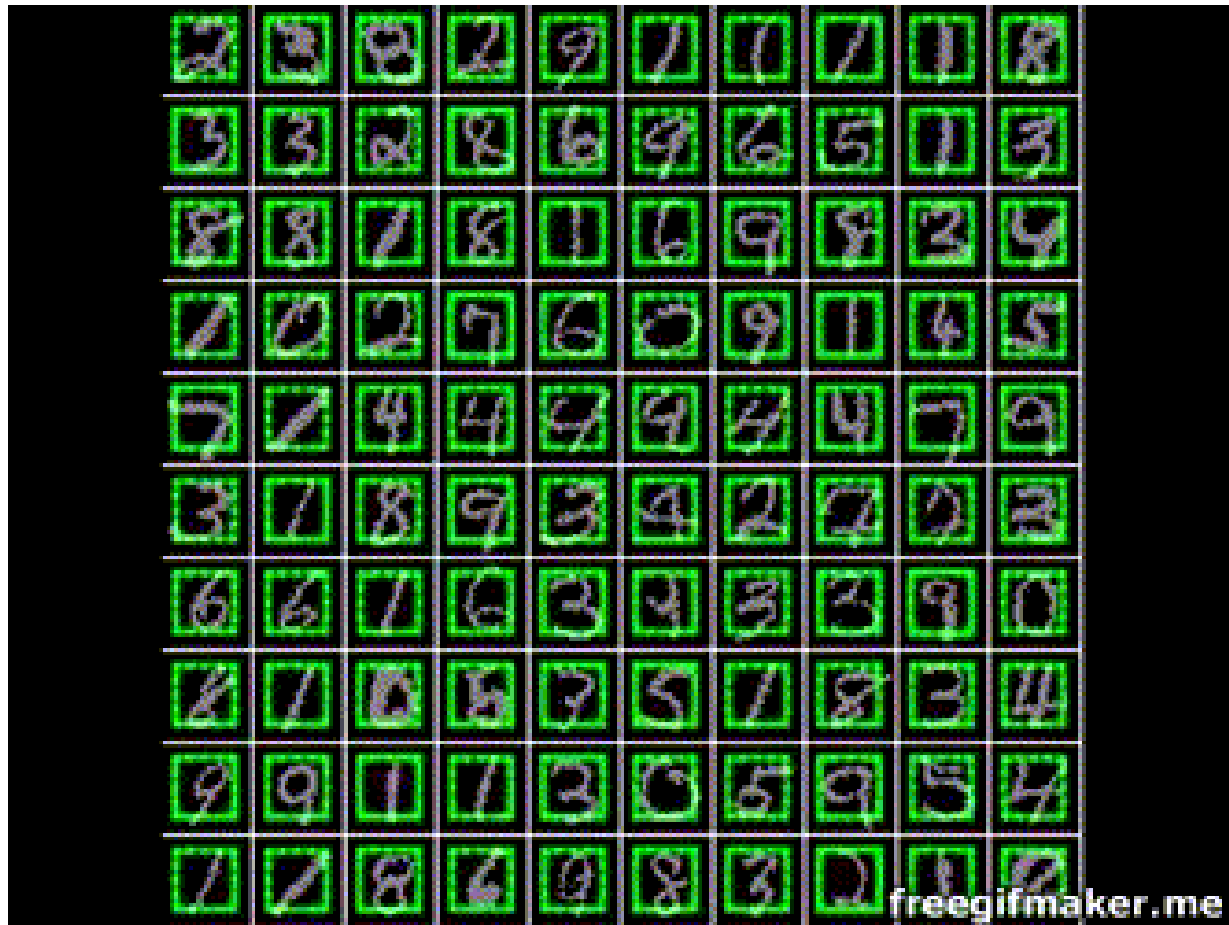
Produit du LSTM central

LSTM d'encodage

Minimum variationnel

→ Perte totale : $\mathcal{L}^{Total} = \mathbb{E}_{z \sim Q}[\mathcal{L}^x + \mathcal{L}^z]$

DRAW-MNIST



DRAW-2digits



Generating two digits

freegifmaker.me

Vers les AGI



Paradigme	Avantages
Mécanisme d'attention (DRAW)	Simplifier l'inférence par la localisation
LSTM (RNN)	Structurer l'inférence par un raisonnement par étape
Système génératif (VAE)	Généraliser l'inférence grâce aux capacités interpolatoires
Algorithmique différentiable (DNC)	Inférence hétérogène complexe
Mémoires dynamiques (GTMM)	Inférence avec mémoire long terme flexible
Multi-Interfaces neuronaux	Inférence transférable, coopérative et massive

An important distinction

- "Narrow" (classical) AI:
 - Systems explicitly designed to solve specific, reasonably well-defined problems
 - E.g. Deep Blue, Watson, etc.

- Artificial General Intelligence:
 - Systems designed to autonomously learn new tasks and adapt to changing environments

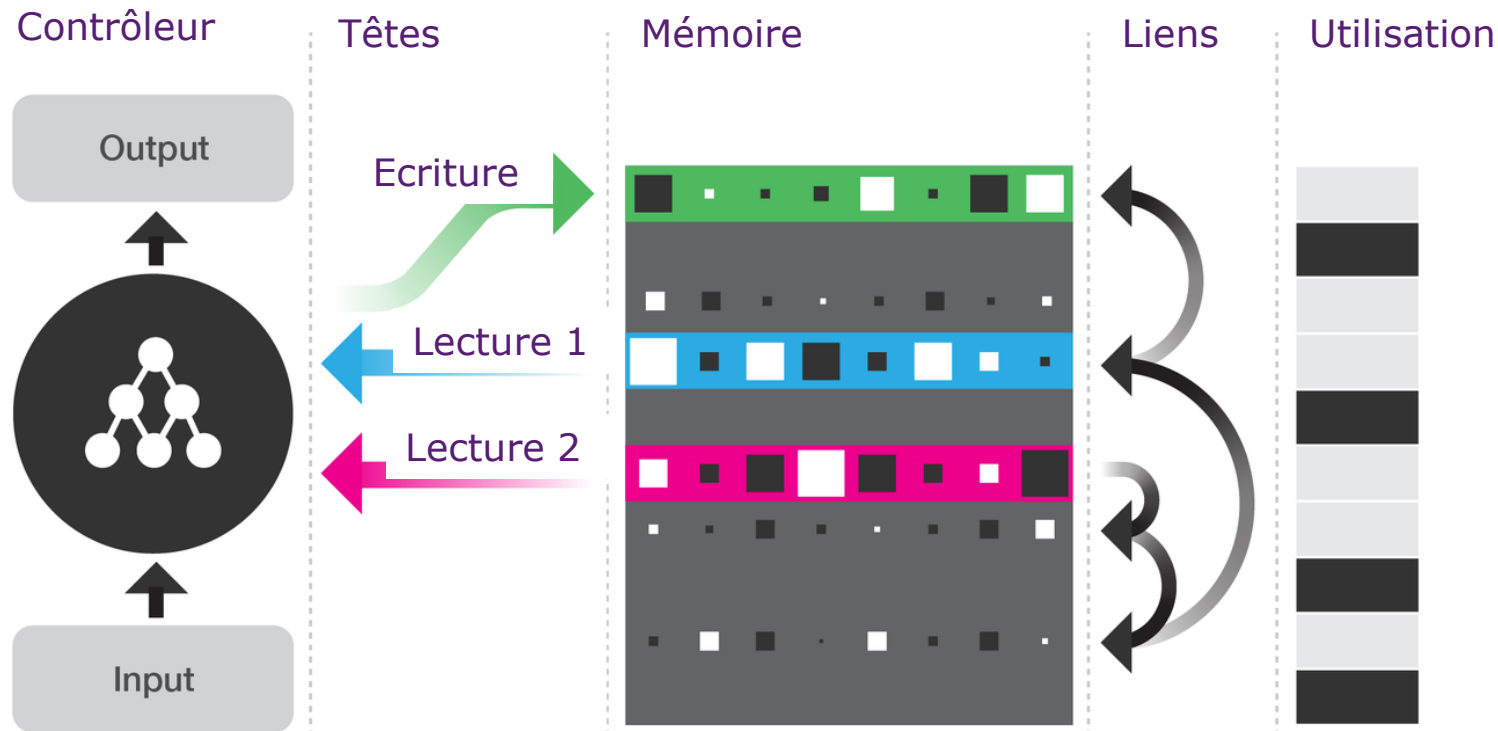
Intellifast 2012



DNC: algorithmique différentiable

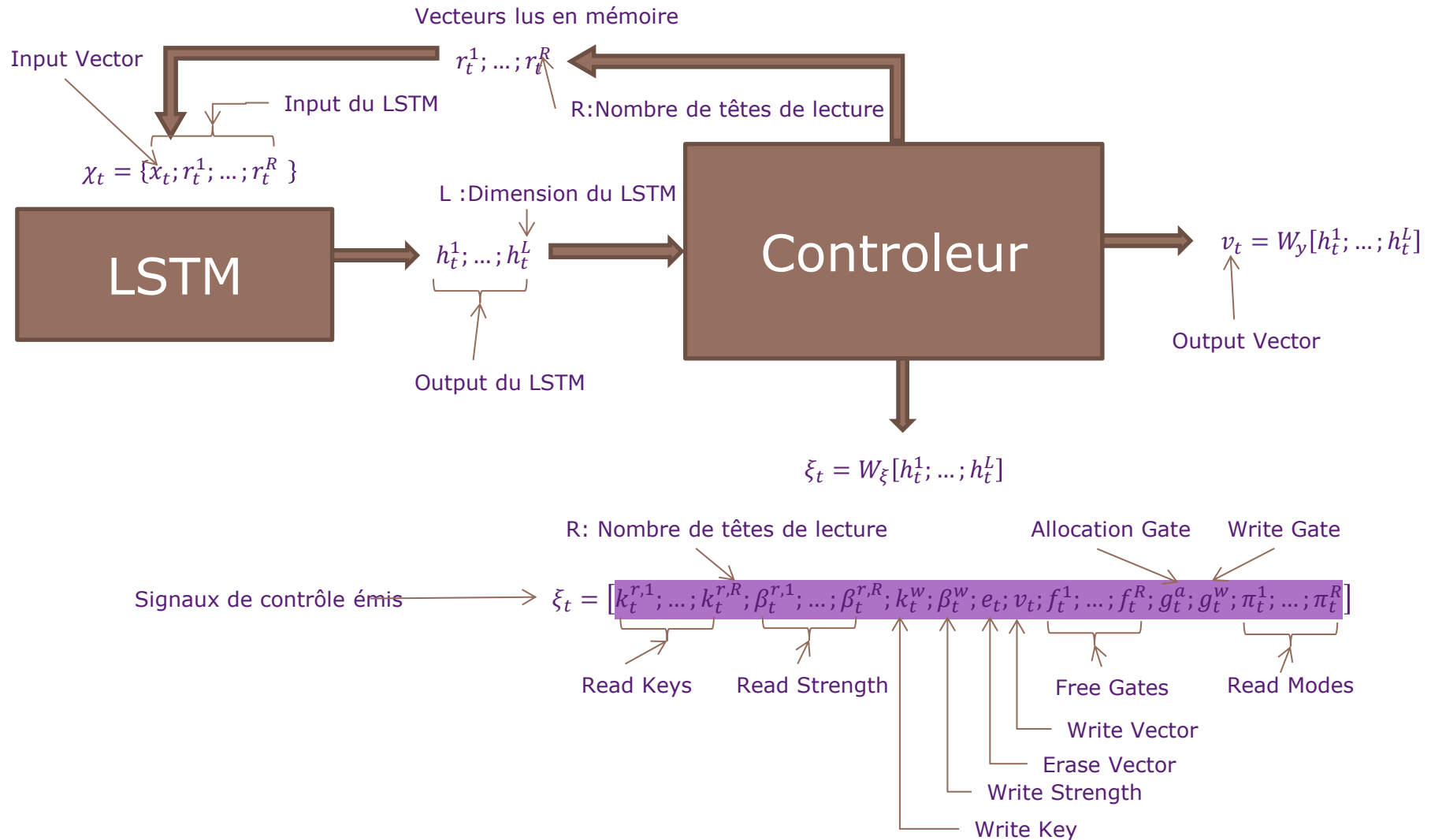


Architecture générale du réseau

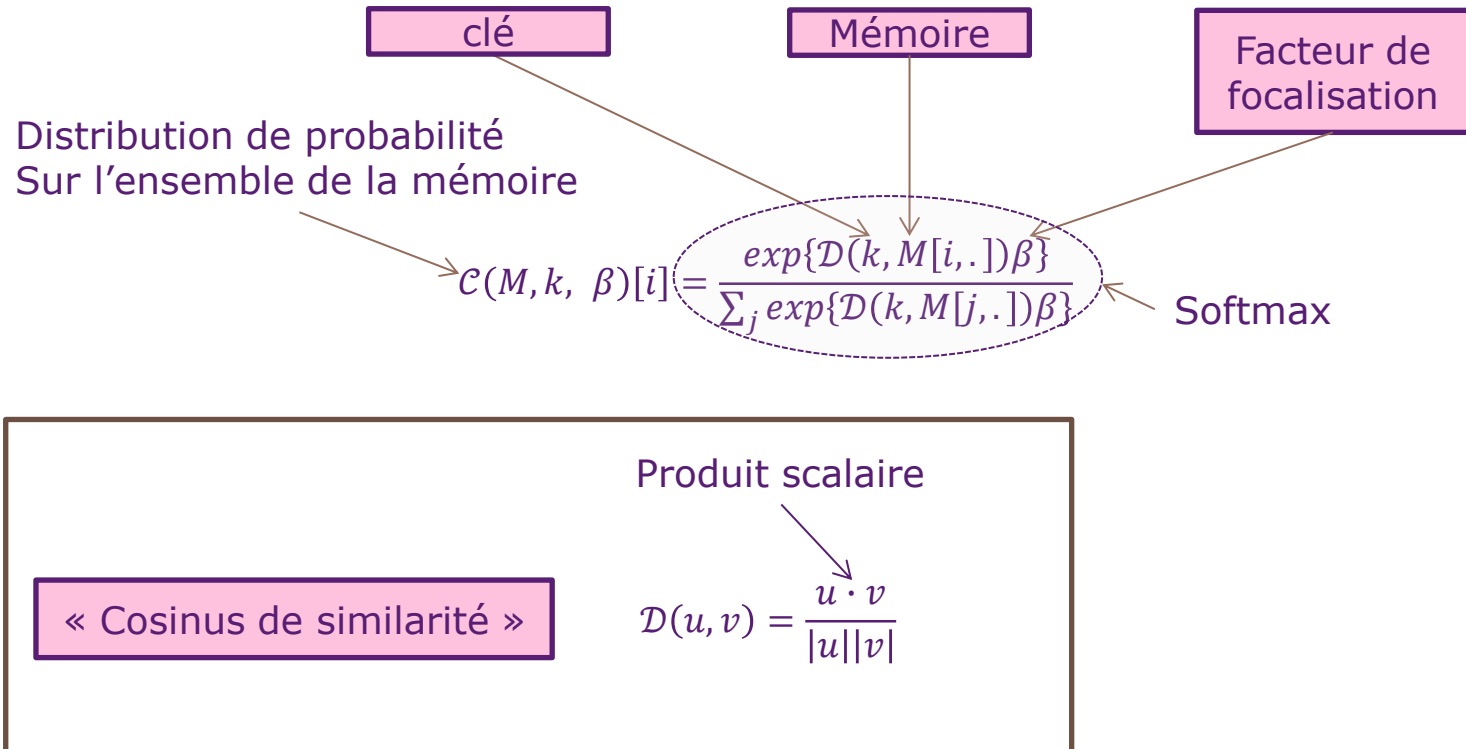


DNC : généralisation du mécanisme d'attention

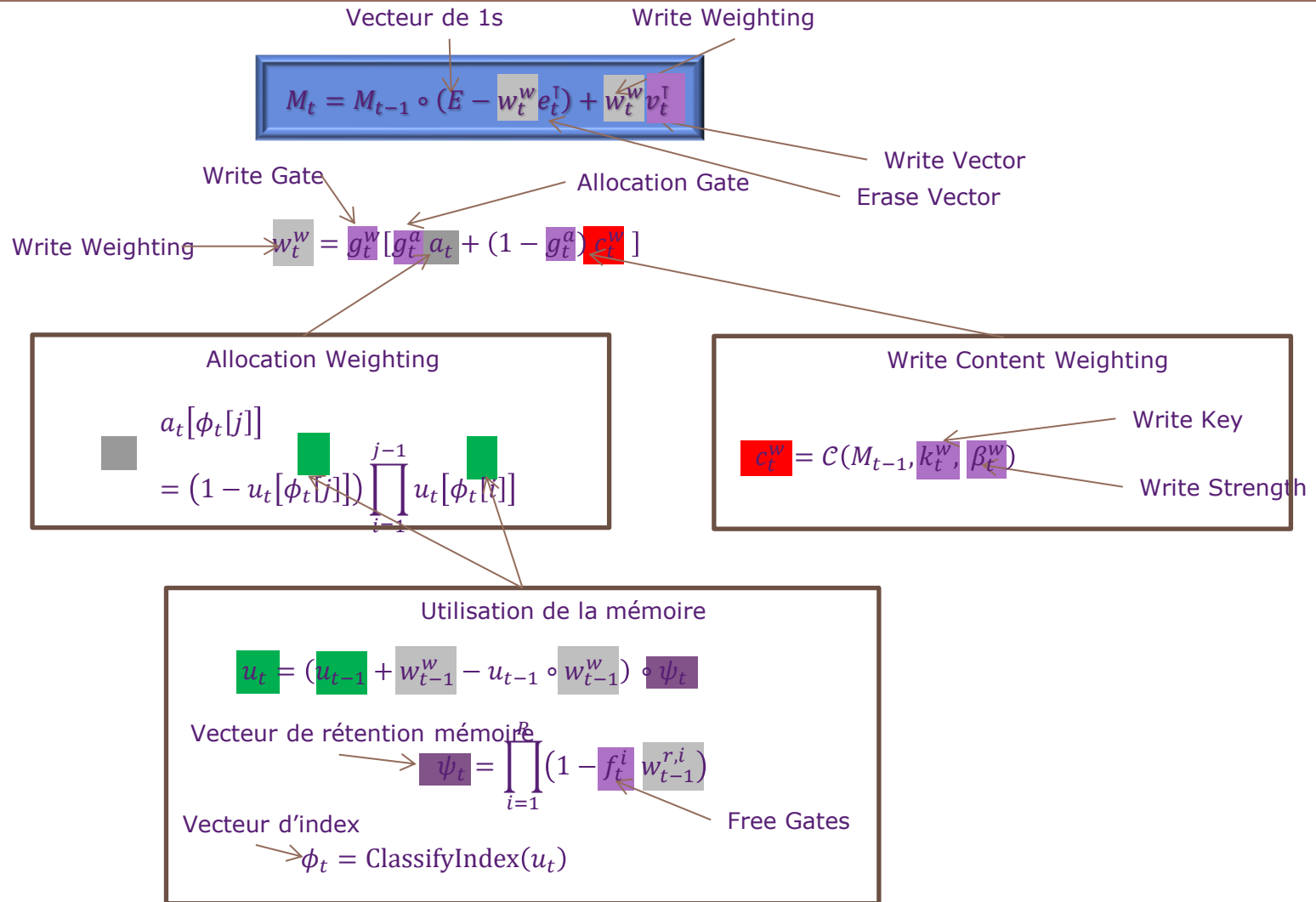
DNC : Contrôleur , Fonctions



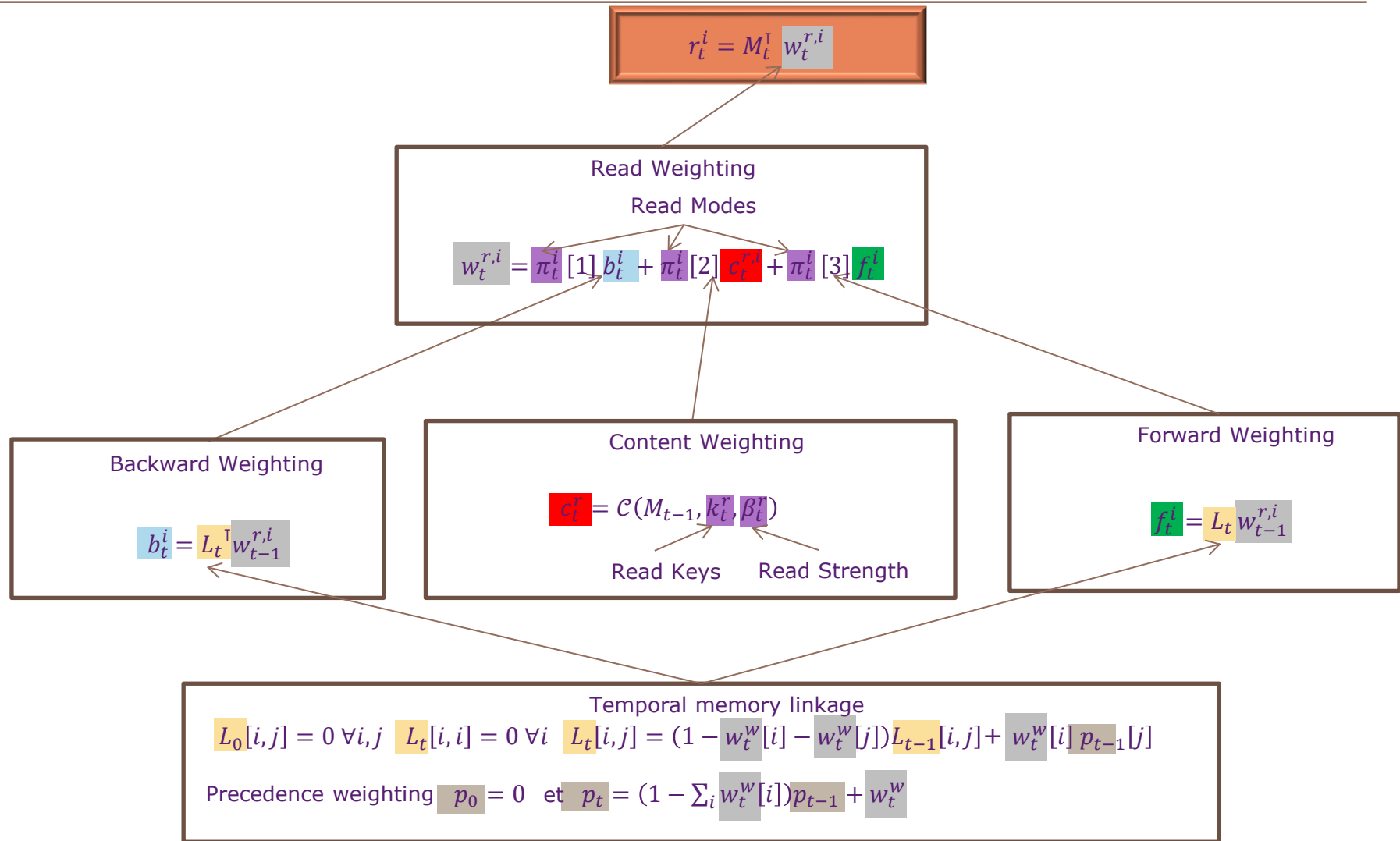
Adressage par contenu de la mémoire



DNC : Contrôleur : Ecriture mémoire



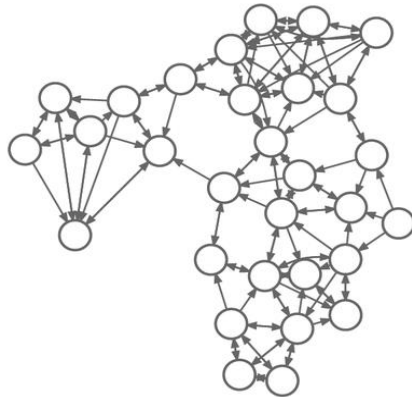
DNC : Contrôleur: lecture mémoire



Problème du voyageur de commerce



Random Training Graph



London Underground



Traversal

Shortest

Underground Input:

(OxfordCircus, TottenhamCtRd, Central)
 (TottenhamCtRd, OxfordCircus, Central)
 (BakerSt, Marylebone, Circle)
 (BakerSt, Marylebone, Bakerloo)
 (BakerSt, OxfordCircus, Bakerloo)
 ...
 (LeicesterSq, CharingCross, Northern)
 (TottenhamCtRd, LeicesterSq, Northern)
 (OxfordCircus, PiccadillyCircus, Bakerloo)
 (OxfordCircus, NottingHillGate, Central)
 (OxfordCircus, Euston, Victoria)

- 84 edges in total

Traversal Question:

(BondSt, _, Central),
 (_, _, Circle), (_, _, Circle),
 (_, _, Circle), (_, _, Circle),
 (_, _, Jubilee), (_, _, Jubilee),

Answer:

(BondSt, NottingHillGate, Central)
 (NottingHillGate, GloucesterRd, Circle)
 ...
 (Westminster, GreenPark, Jubilee)
 (GreenPark, BondSt, Jubilee)

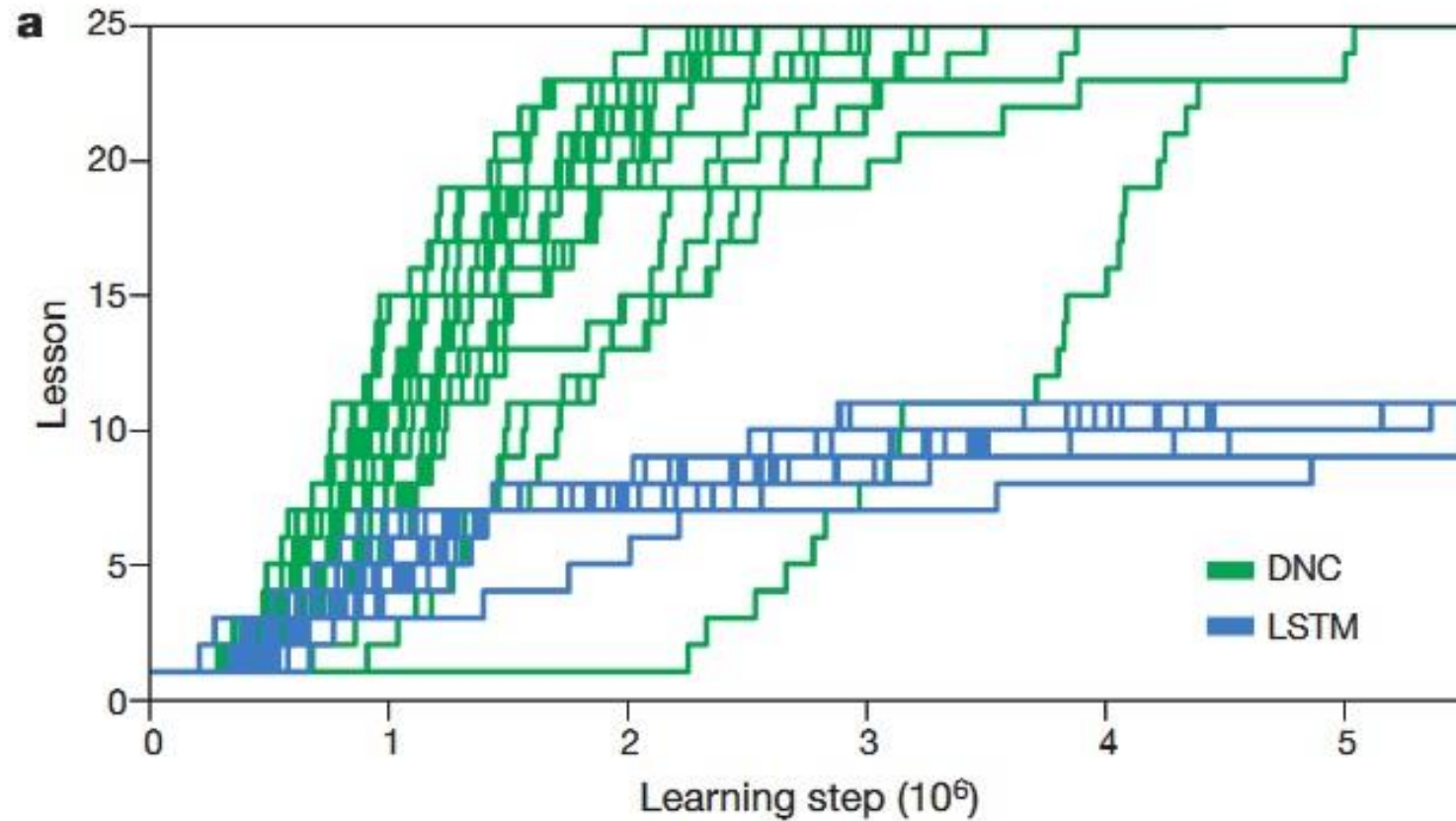
Shortest Path Question:

(Moorgate, PiccadillyCircus, _)

Answer:

(Moorgate, Bank, Northern)
 (Bank, Holborn, Central)
 (Holborn, LeicesterSq, Piccadilly)
 (LeicesterSq, PiccadillyCircus, Piccadilly)

Supériorité du DNC



Question ?



Est-il possible de construire une machine de Turing neurale universelle ?

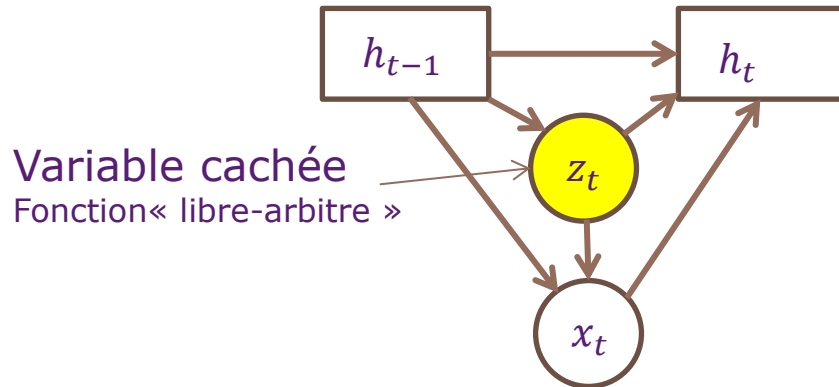
- Elle doit permettre de représenter tout programme
- Elle doit permettre de converger vers tout programme

Comparaison GTMM –Autres



	LSTM	DNC	VRNN	GTMM
Génératif	Non	Non	Oui	Oui
Mémorisation Longue	Oui	Oui	Non	Oui
Scalabilité	Faible	Forte	Faible	Forte
Inférence	Exacte	Exacte	Variationnelle	Variationnelle
Mémoire externe	Non	Oui	Non	Oui
Input Stream de grande dimension	Non	Non	Oui	Oui
Corrélation Capacité mémoire et capacité processing	Forte	Faible	Très forte	Faible

Le variationnel dans les GTM



Reseau	LSTM
Fonction de transition	$h_t = f_h(h_{t-1}, z_t, x_t)$
Fonction prior	$z_t = f_z(h_{t-1})$
Fonction d'observation	$x_t = f_x(h_{t-1}, z_t)$

$$\text{Log } p(x_{\leq T}) = \text{Log} \int p_{\theta}(x_{\leq T}, z_{\leq T}) dz_{\leq T} = \text{Log} \mathbb{E}_{q_{\phi}(z_{\leq T} | x_{\leq T})} \left[\frac{p_{\theta}(x_{\leq T} | z_{\leq T})}{q_{\phi}(z_{\leq T} | x_{\leq T})} \right]$$

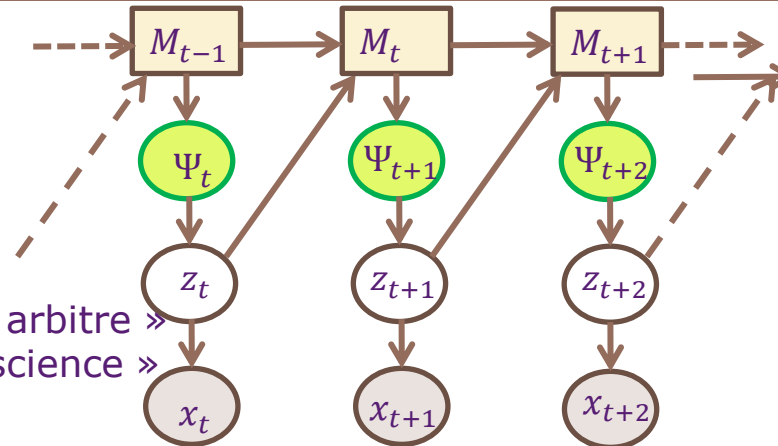
$$\geq \mathbb{E}_{q_{\phi}(z_{\leq T} | x_{\leq T})} [p_{\theta}(x_{\leq T} | z_{\leq T})] - \text{KL}[q_{\phi}(z_{\leq T} | x_{\leq T}) || p_{\theta}(z_{\leq T})] = \mathcal{F}(q; \theta)$$

On choisit un variationnel auto régressif $q_{\phi}(z_{\leq T} | x_{\leq T}) = \prod_{t=1}^T q_{\phi}(z_t | x_{\leq t})$

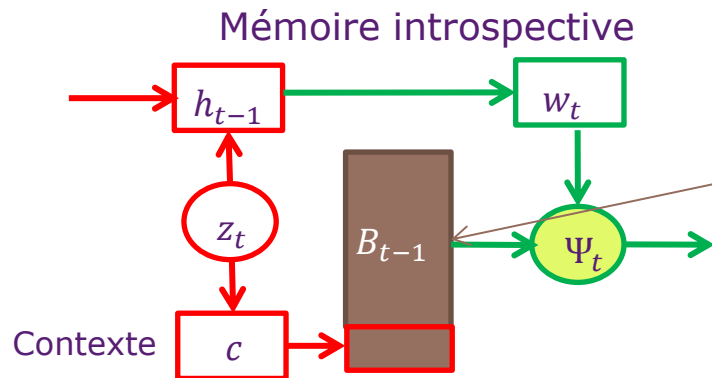
Cela permet d'écrire $\mathcal{F}(q; \theta) = \sum_{t=1}^T \mathcal{F}_t(q; \theta)$

$$\mathcal{F}_t(q; \theta) = \mathbb{E}_{q_{\phi}(z_t | z_{<t}, x_{\leq t})} [p_{\theta}(x_t | z_{\leq t}, x_{<t})] - \text{KL}[q_{\phi}(z_t | z_{<t}, x_{\leq t}) || p_{\theta}(z_t | z_{<t}, x_{<t})]$$

Différents accès mémoriels



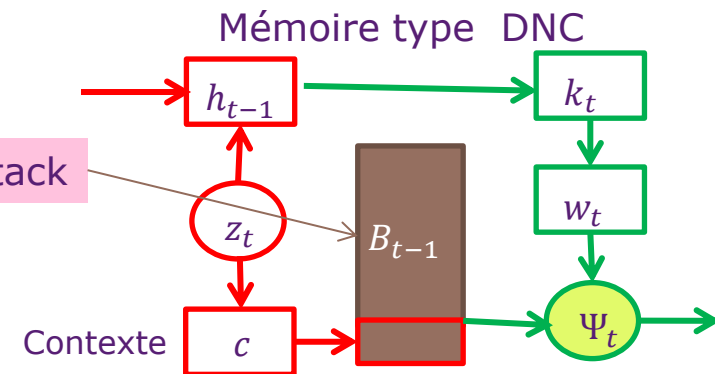
Mémorisation du « libre arbitre »
= « Phénomène de conscience »



Soft attention: $w_t^r = f_{att}(h_{t-1})$

Update de l'état: $h_t = f_{key}(h_{t-1}, z_{t-1}, c_t)$

Avantage : apprentissage rapide

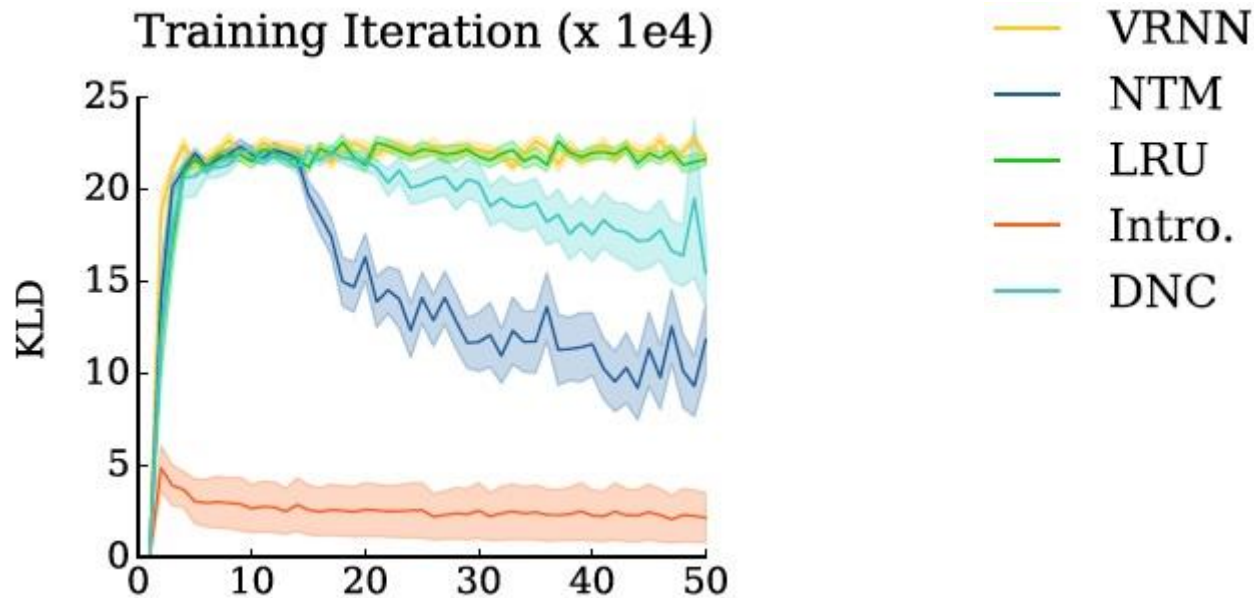


Lecture par contenu: $k_t = f_{key}(h_{t-1})$

$w_t^r = f_{att}(M_{t-1}, k_t)$
Update de l'état: $h_t = f_{key}(h_{t-1}, M_{t-1}, z_{t-1}, c_t)$

Avantage : apprentissage complexe

Supériorité du GTMM avec mémoire introspective

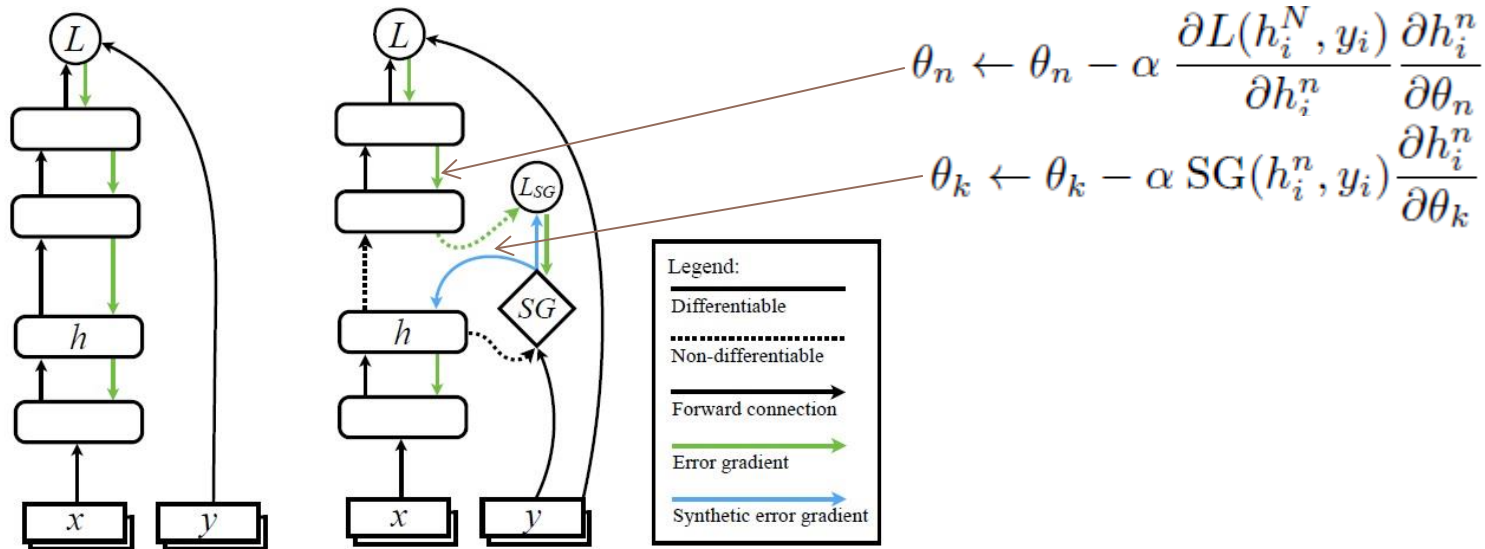


Tache: apprentissage de la répétition d'une séquence de digits

0	1	2	3	4	5	6	7	8	9	PRE-RECALL INTERVAL (l)										RECALL INTERVAL (k)				
3	4	3	8	0	2	4	4	9	4	9	7	0	9	8	1	8	1	0	1	3	4	3	8	0

Stimuli de distraction

Interfaces neuronales Découplés



Backpropagation usuelle Backpropagation avec DNI

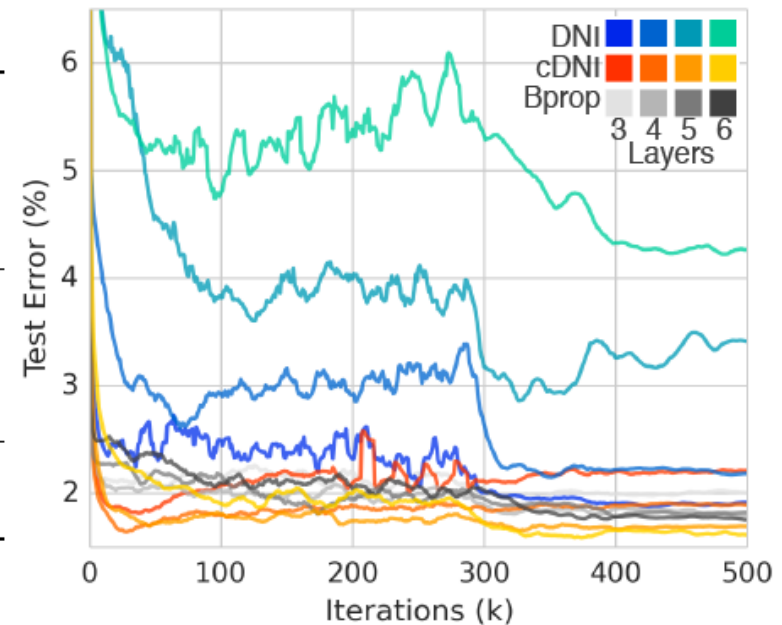
- Avantages du DNI:
- Désynchronisation de la backpropagation
 - Adaptation des updates de gradient à l'importance des niveaux
 - Une étape obligée vers l'apprentissage continu
 - Un parallèle biologique crédible pour la backpropagation

DNI – Exemple sur MNIST et CIFAR



Layers		MNIST (% Error)				CIFAR-10 (% Error)			
		No Bprop	Bprop	DNI	cDNI	No Bprop	Bprop	DNI	cDNI
FCN	3	9.3	2.0	1.9	2.2	54.9	43.5	42.5	48.5
	4	12.6	1.8	2.2	1.9	57.2	43.0	45.0	45.1
	5	16.2	1.8	3.4	1.7	59.6	41.7	46.9	43.5
	6	21.4	1.8	4.3	1.6	61.9	42.0	49.7	46.8
CNN	3	0.9	0.8	0.9	1.0	28.7	17.9	19.5	19.0
	4	2.8	0.6	0.7	0.8	38.1	15.7	19.5	16.4

Gagnants

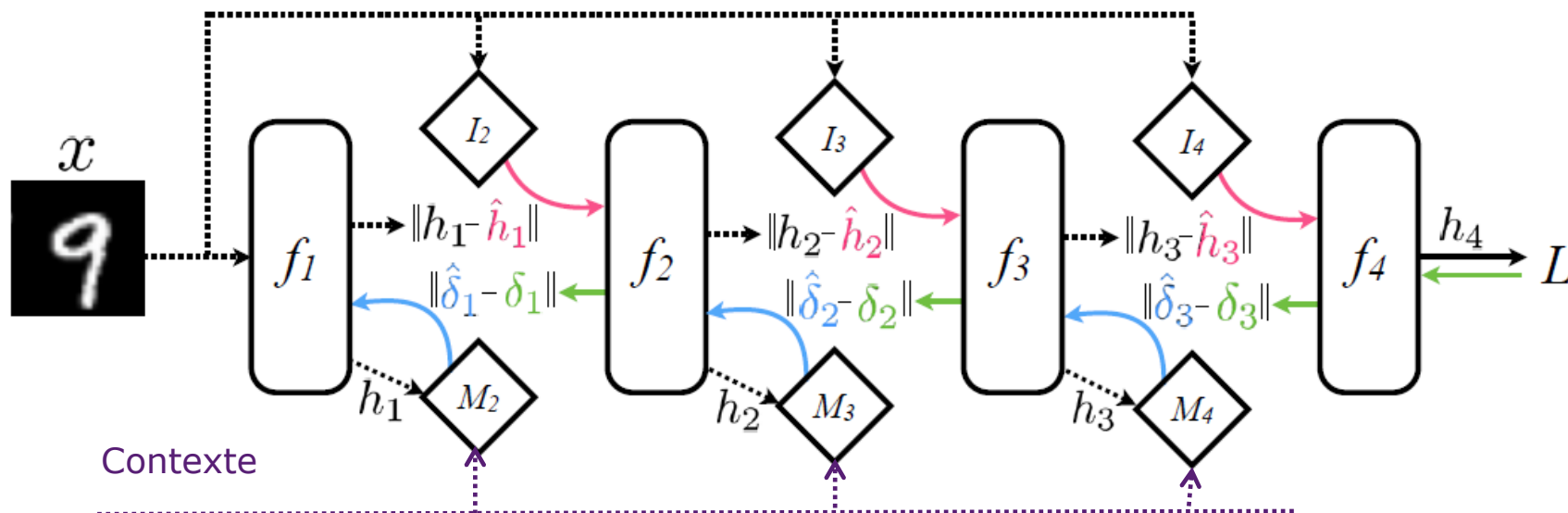


cDNI : Gradient synthétique conditionnel

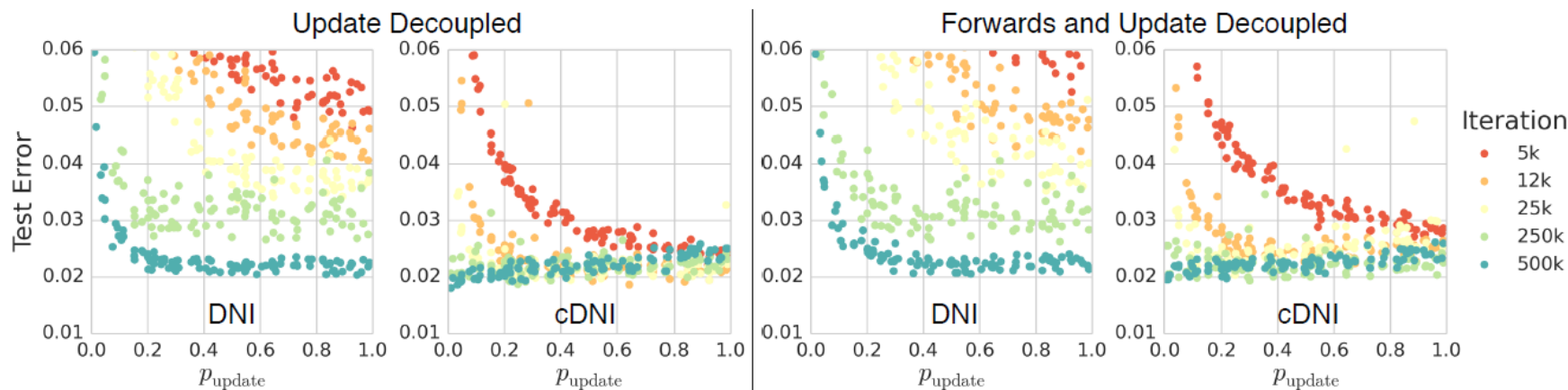
Leçon : Le gradient doit être considéré comme un nœud à part entière

Avantage du cDNI: Il augmente la précision du résultat en régularisant la convergence sans biais

DNI généralisé



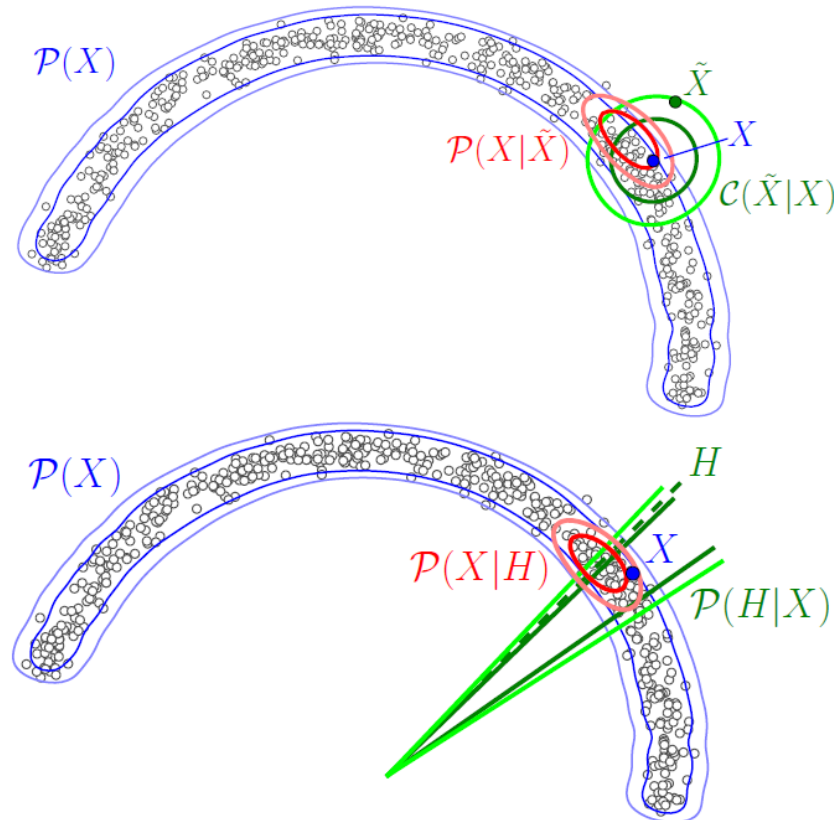
Tous les gradients et le forwards sont nodisés et conditionnels



L'hypothèse de variété non supervisée

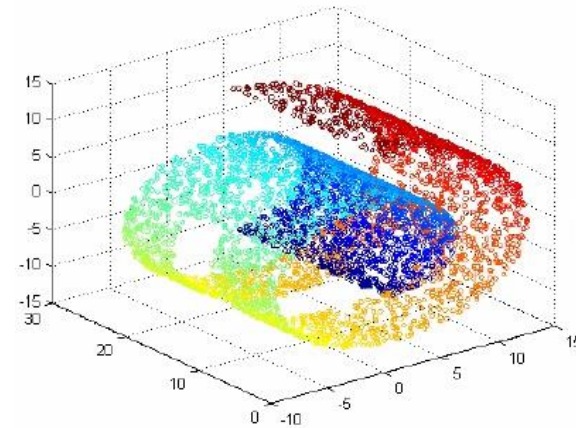
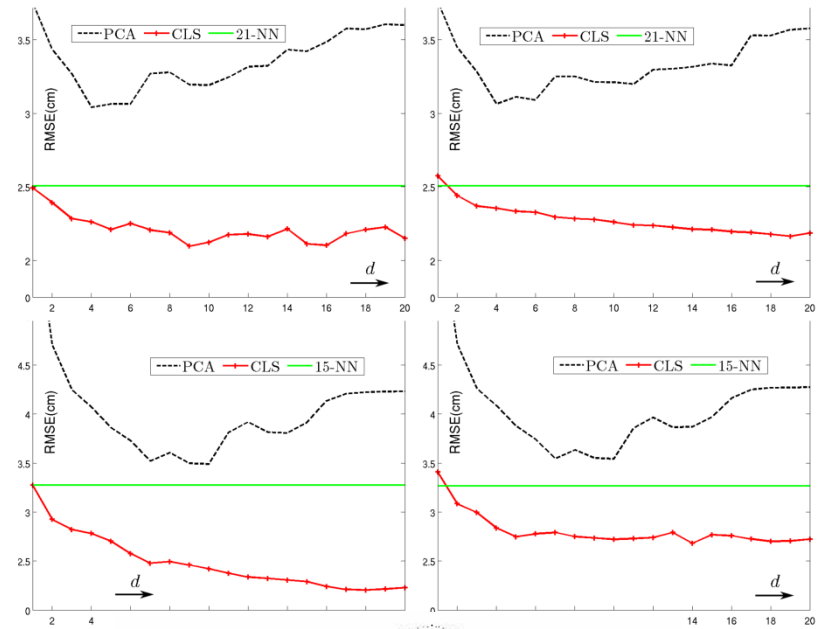
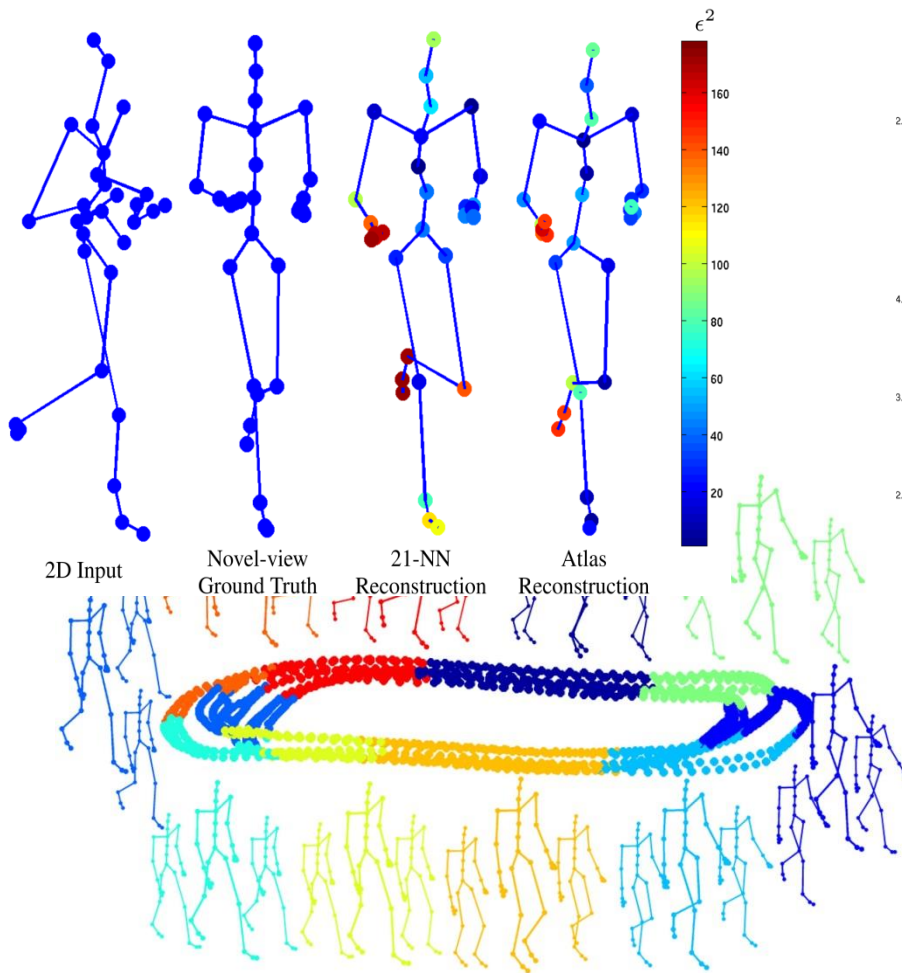


Hypothèse : Les données, quoique ensemble de points dans un espace de grande dimension, sont groupée selon une sous variété de faible dimension intrinsèque

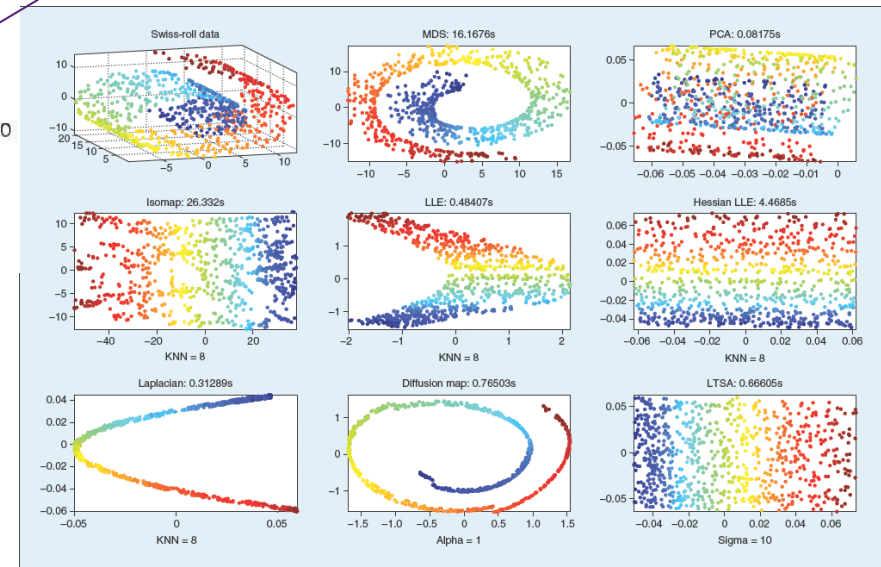
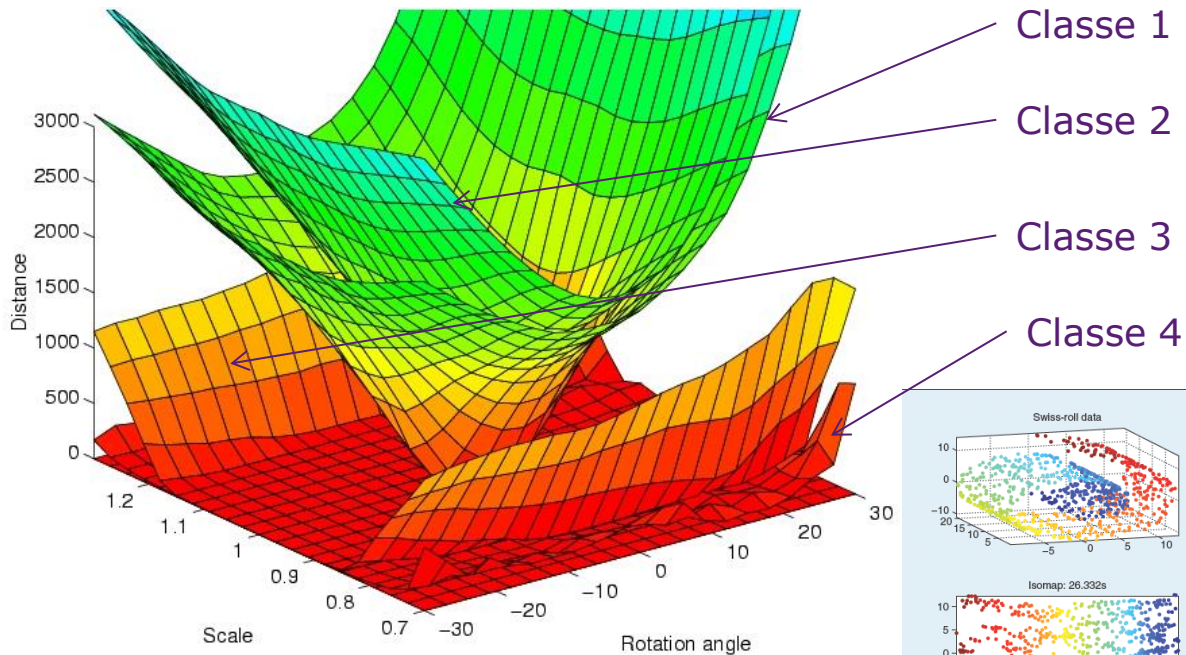


<http://colah.github.io/posts/2014-03-NN-Manifolds-Topology/>

Exemple de sous-variété



Le classifieur de sous-varieté tangente



Minimisation de l'autoencodeur associé



$$r = \text{Enc} \circ \text{Dec} \qquad J(x) = \frac{\partial \text{enc}(x)}{\partial x}$$

$SVD_i[M]$ est le i ème plus important vecteur de la décomposition singulière de M

1^{ier} ordre Contrastive $\mathcal{L}_{CAE} = (\|r(x) - x\|^2 + \lambda \|J(x)\|^2)$

2^{ieme} ordre stochastique Contrastive

$$\mathcal{L}_{CAE} = \left(\|r(x) - x\|^2 + \lambda \|J(x)\|^2 + \gamma \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \sigma^2 I)} [\underbrace{\|J(x) - J(x + \epsilon)\|}_{\text{Emule un hessien}}] \right)$$

1^{ier} ordre stochastique MTC

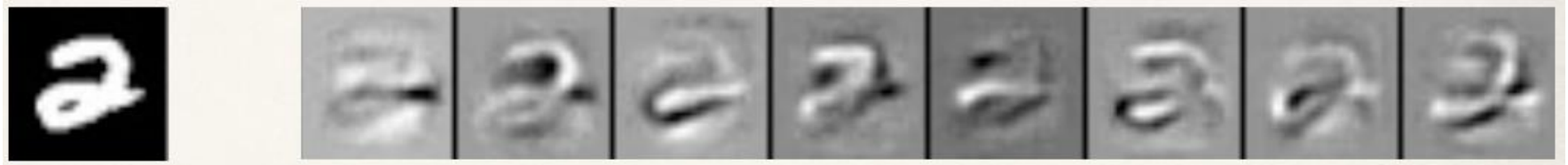
$$\mathcal{L}_{CAE} = \left(\|r(x) - x\|^2 + \lambda \left\| \sum_{i=1}^n SVD_i[J(x)] \right\|^2 \right)$$

MTC sur MNIST



input

Les différents vecteurs tangents



MTC sur CIFAR10



input

Les différents vecteurs tangents



↑
PCA



↑
Autoencodeur contractif

MTC: performances



K-NN	NN	SVM	DBN	CAE	DBM	CNN	MTC
3.09%	1.60%	1.40%	1.17%	1.04%	0.95%	0.95%	0.81%

MNIST : Modèles invariants par permutation

Prévision financière

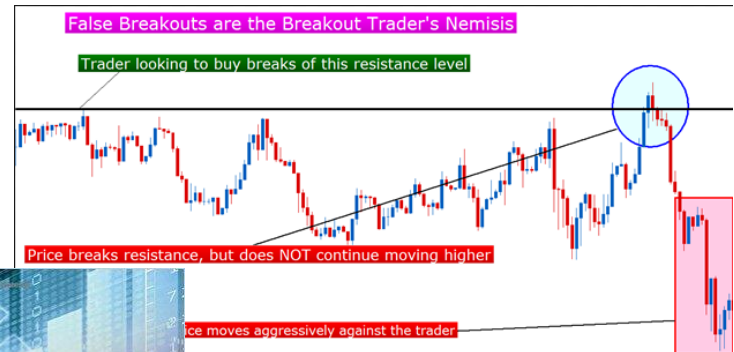
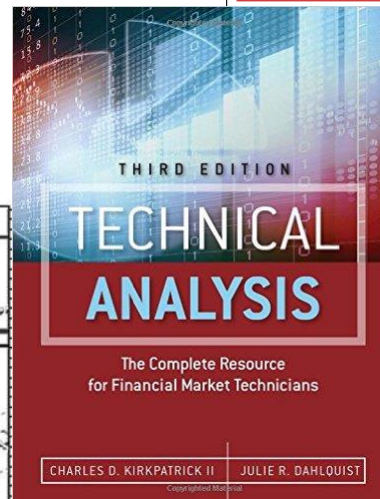
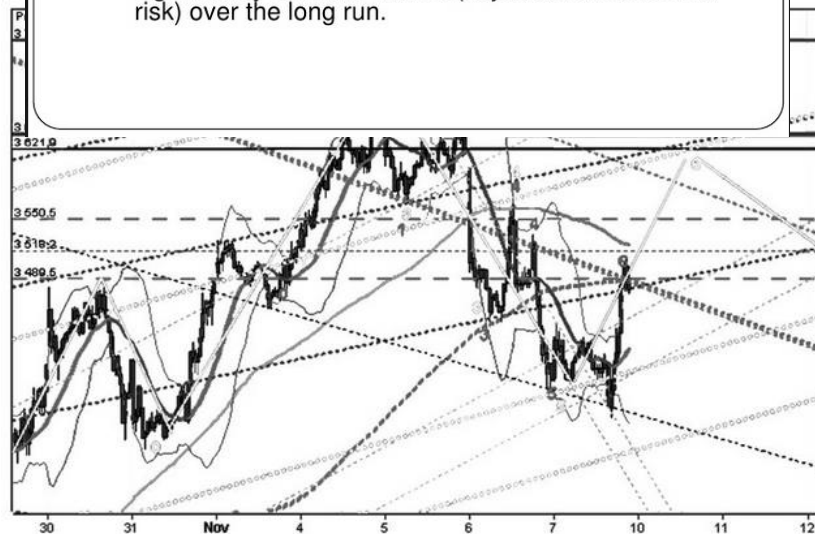


Prévision : l'Age de la magie

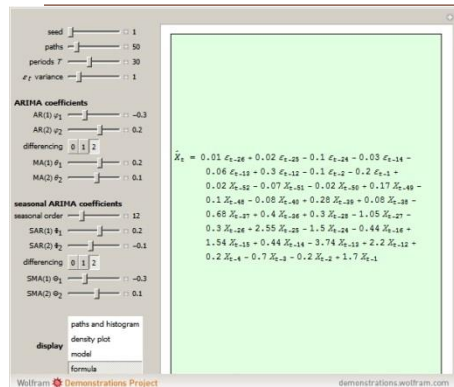


Summary of Tests of the EMH

- Weak form is supported, so technical analysis cannot consistently outperform the market.
- Semi-strong form is mostly supported, so fundamental analysis cannot consistently outperform the market.
- Strong form is generally not supported. If you have secret ("insider") information, you CAN use it to earn excess returns on a consistent basis.
- Ultimately, most believe that the market is very efficient, though not perfectly efficient. It is unlikely that any system of analysis could consistently and significantly beat the market (adjusted for costs and risk) over the long run.



Prévision : Vision mathématique pure



$$(1 - \phi_1 B) (1 - \Phi_1 B^4) (1 - B) (1 - B^4) y_t = (1 + \theta_1 B) (1 + \Theta_1 B^4) e_t.$$

(Non-seasonal AR(1))
(Non-seasonal difference)
(Non-seasonal MA(1))
(Seasonal MA(1))



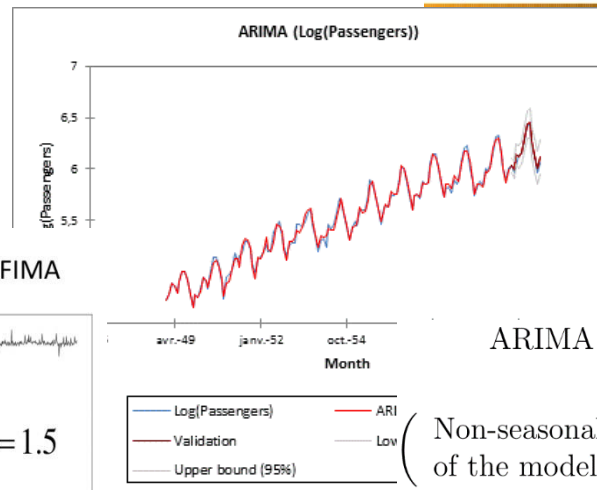
ARMA(p,q) Models

- A model with autoregressive terms can be combined with a model having moving average terms to get an ARMA(p,q) model:

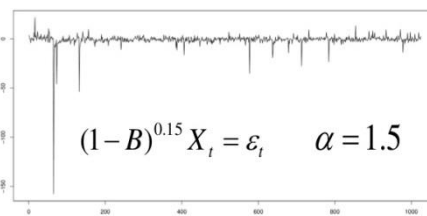
$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \dots - \theta_q \varepsilon_{t-q}$$
- ARMA(p,q) models can describe a wide variety of behaviors for stationary time series.
- Theoretical behavior of the ACF and PACF for autoregressive-moving average processes:

	ACF	PACF
AR(p)	Die out	Cut off after the order p of the process
MA(q)	Cut off after the order q of the process	Die out
ARMA(p,q)	Die out	Die out

Note that:
- ARMA(p,0) = AR(p)
- ARMA(0,q) = MA(q)
In practice, the values of p and q each rarely exceed 2.
In this context...
- "Die out" means "tend to zero gradually"
- "Cut off" means "disappear" or "is zero"



Pure symmetric α -stable ARFIMA



$$\text{ARIMA} \underbrace{(p, d, q)}_{\text{Non-seasonal part of the model}} \underbrace{(P, D, Q)_m}_{\text{Seasonal part of the model}}$$

Prévision : martingalité orthodoxe



TESTING THE MARTINGALE HYPOTHESIS*

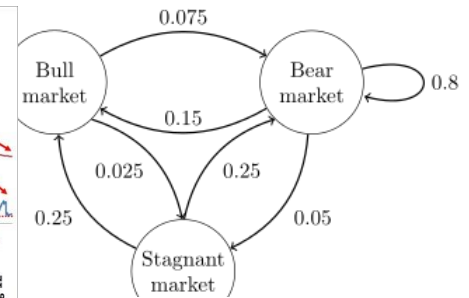
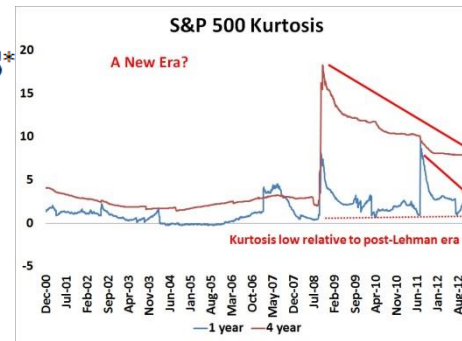
J. CARLOS ESCANCIANO

Indiana University, Bloomington, USA

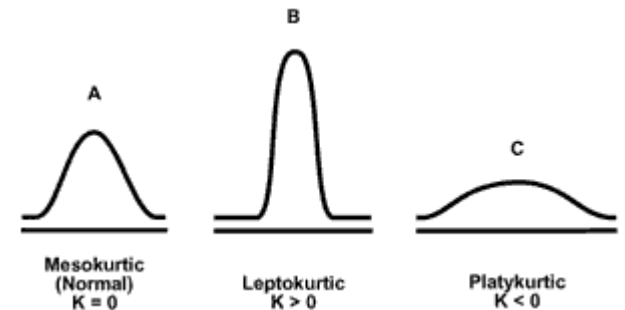
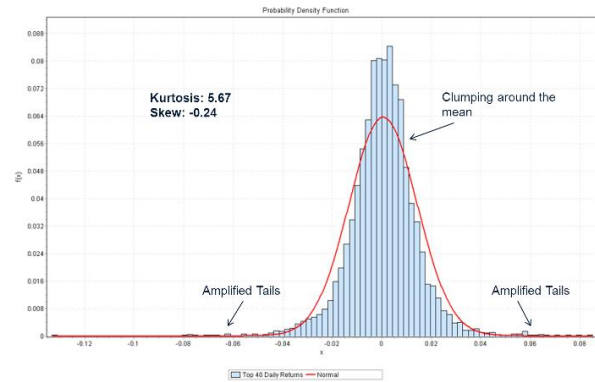
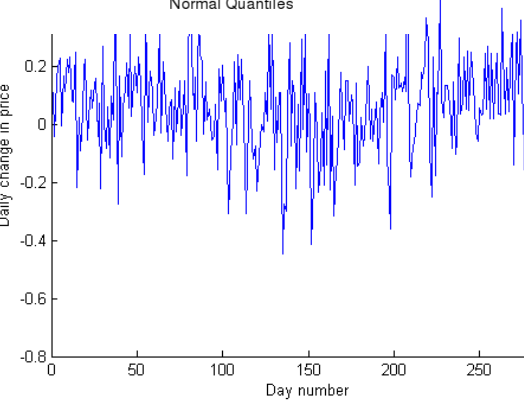
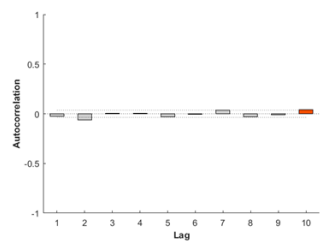
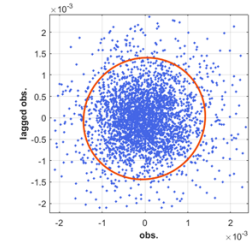
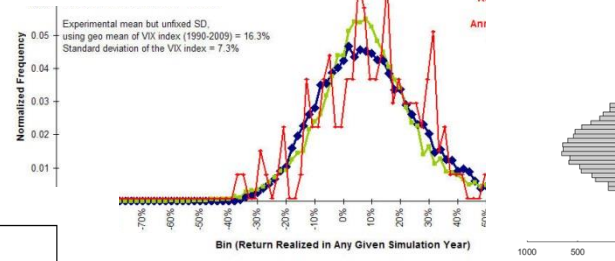
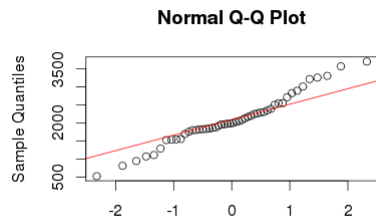
IGNACIO N. LOBATO

Instituto Tecnológico Autónomo de México, México D.F., Mexico

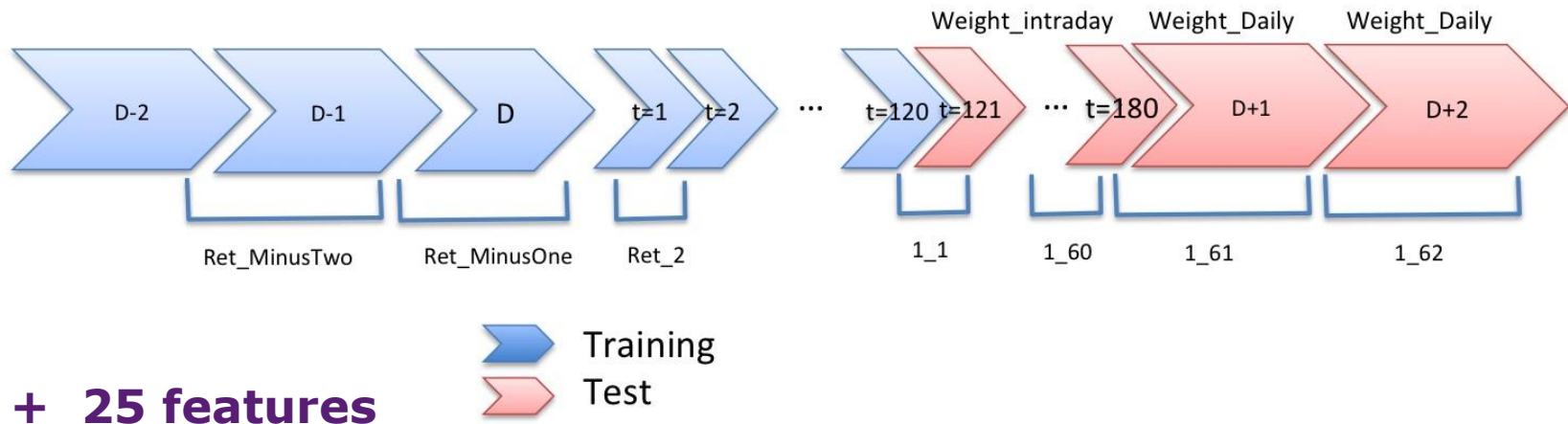
October 25, 2007



Invariance test on the daily changes in the 10-year yield to maturity



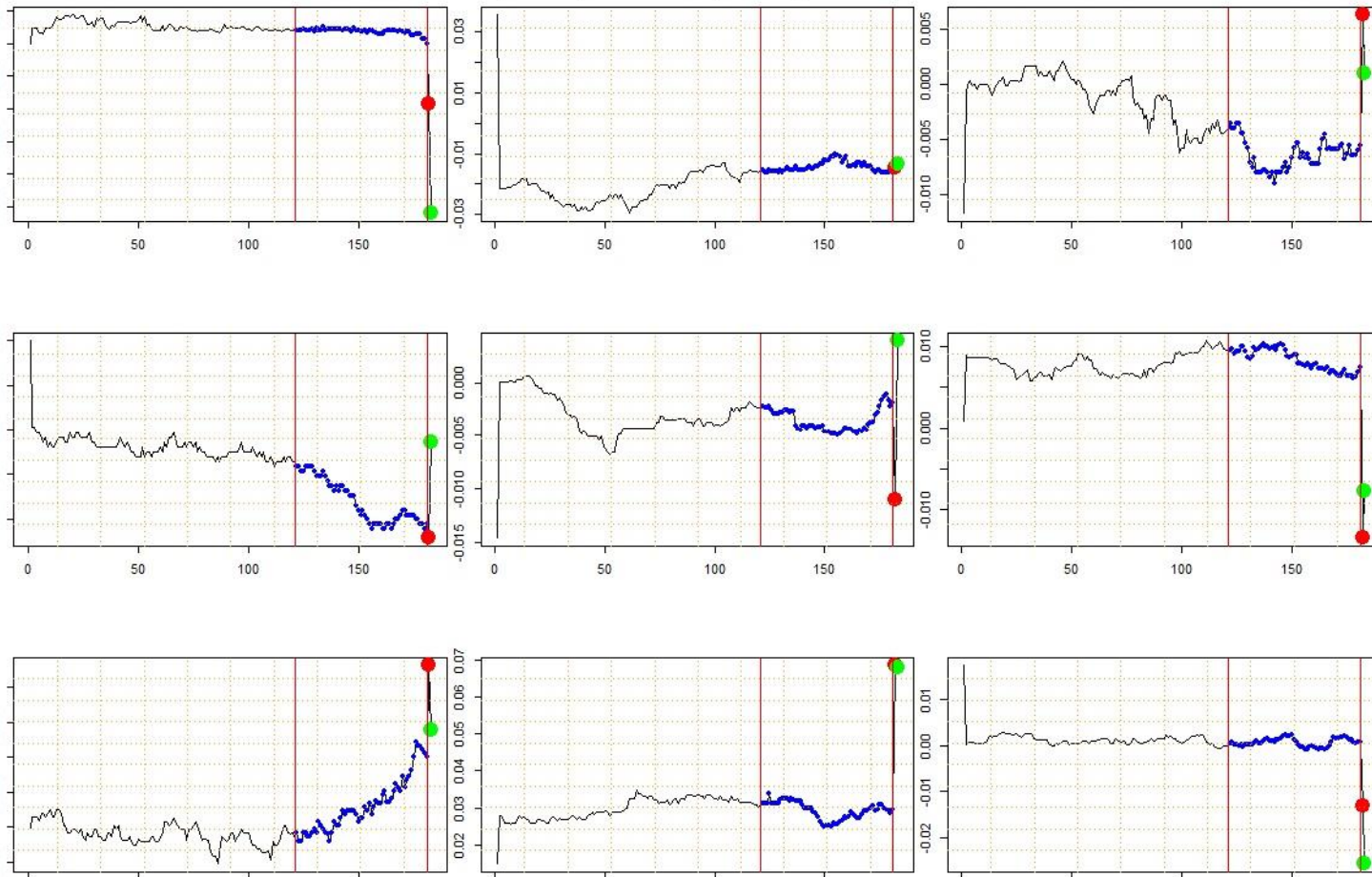
Winton Stock Market Challenge



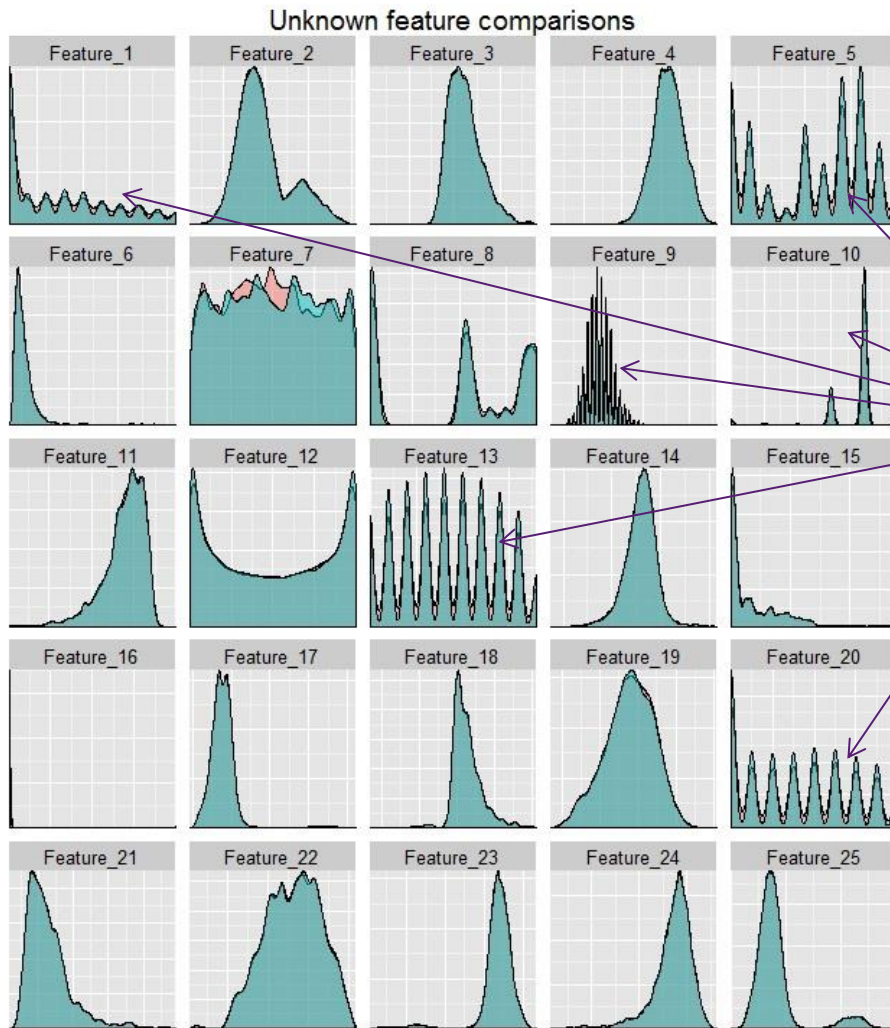
Training : 40000 séries

Notation : 100000 séries

Winton Stock Market Challenge (2)

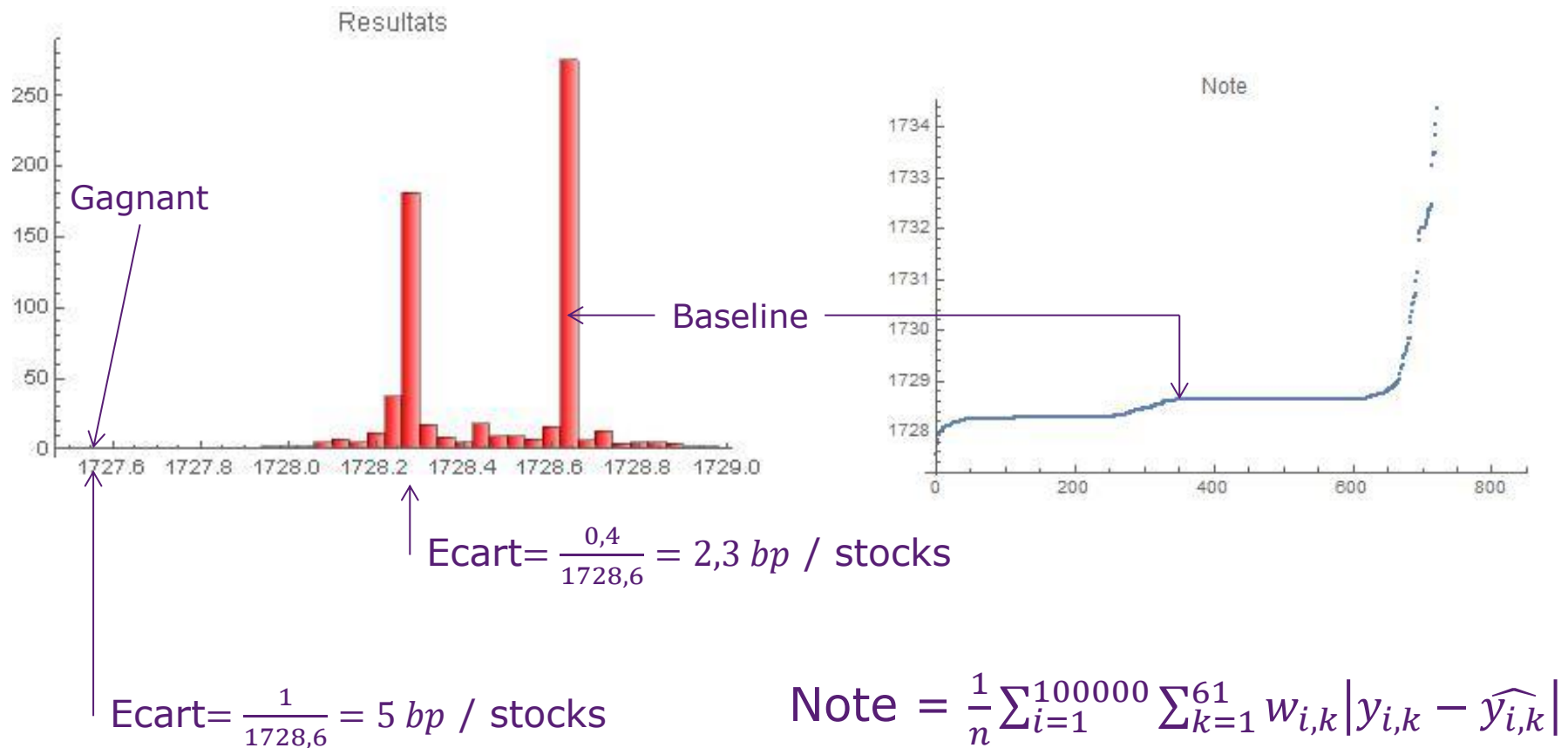


Winton Stock Market Challenge (3)



Certaines features
sont évidemment catégorielles

Winton Stock Market Challenge: résultats



Stock Market Challenge : ma solution



Une série boursière est comme un discours dont il faut apprendre la langue

```
model:add(SliceList)
if dropoutFlag then model:add(nn.Sequencer(nn.Dropout(dropoutProb))) end
model:add(nn.Sequencer(nn.FastLSTM(embeddingSize+nbfeatures, embeddingSize, rhoOutput)))
model:add(nn.Sequencer(nn.Linear(embeddingSize, dictionarySize))
if not(CrossEntropyFlag) then model:add(nn.Sequencer(nn.LogSoftMax())) end
return model
```

```
taille du dataset de training :36
error goal 0.2 and max iteration = 20000
error for iteration 500 is 1.8108169855281
error for iteration 1000 is 1.5390473429868
error for iteration 1500 is 1.3987408592645
error for iteration 2000 is 1.2643873510552
error for iteration 2500 is 1.190822624829
error for iteration 3000 is 1.0666057409108
error for iteration 3500 is 0.92792516314329
error for iteration 4000 is 0.80726509513093
error for iteration 4500 is 0.76559975886313
error for iteration 5000 is 0.77901771165743
error for iteration 5500 is 0.66204038208886
error for iteration 6000 is 0.57159945965616
error for iteration 6500 is 0.64896469790744
error for iteration 7000 is 0.76506822398161
```

```
Module[{j0 = 1, patternSet},
  patternSet = Union[
    Table[With[{j1 = 7, k = k0}, Function[{set, item},
      (set[[item, 1]] = j1) && set[[item, 13]] = k]], {k0, 1, 3}];
  BuildAndWriteReducedSets[patternSet]
]
_1_ train,test written = {36, 133}
_2_ train,test written = {36, 124}
_3_ train,test written = {44, 122}
```

```
error for iteration 11500 is 0.33068015678106
error for iteration 12000 is 0.28179228332878
error for iteration 12500 is 0.23297878925448
global error = 0.19949678173875
time for learning :5504.9666700363 sec
```

LSTM

$$1728.62346 - 1728.62301 = \frac{100\,000}{36}$$

1.25

$$1728.62301 - \frac{1728.62346 - 1728.62301}{36} \cdot 100\,000$$

1727.37

Mon score pour 36 stocks

Mon score théorique pour les 100000 stocks

1 — * Just Pay your Bills 1727.53576

2 — * Humberto Brandão 1727.73786



Zero prediction

1728.62346



Prédire des cours de bourse avec des news



Table 2: Weekly Summary Statistics by Market Capitalization

Decile	Log market cap	News stories per week	Proportion of firms without news	Return	Return w/news	Return w/o news	Difference	t-statistics
Smallest 1	9.42	0.16	0.95	-0.14%	2.00%	-0.24%	2.24%	3.47
2	10.56	0.59	0.94	-0.12%	1.51%	-0.23%	1.75%	3.59
3	11.22	1.33	0.92	-0.14%	-0.02%	-0.15%	0.12%	0.40
4	11.80	2.49	0.89	-0.06%	-0.04%	-0.06%	0.01%	0.04
5	12.35	4.01	0.87	-0.05%	-0.28%	-0.01%	-0.26%	-0.86
6	12.88	5.81	0.85	-0.01%	-0.26%	0.03%	-0.29%	-0.99
7	13.43	7.63	0.82	0.04%	-0.07%	0.06%	-0.13%	-0.48
8	14.04	10.19	0.76	0.05%	0.08%	0.04%	0.04%	0.15
9	14.83	13.30	0.66	0.09%	0.12%	0.07%	0.05%	0.25
Largest 10	16.48	22.42	0.34	0.06%	0.07%	0.04%	0.04%	0.18

Notes: This table presents weekly statistics for decile portfolios grouped on market capitalization over calendar years 2003 to 2010 (417 weeks). We divide the firms into deciles based on the market capitalization at the beginning of the month. Each week we note the number of news articles with relevance of at least 0.35, proportion of firms without news, average return, average return for firms with news, average return for firms without news, difference of return between firms with and without news and the *t*-statistics for the difference.

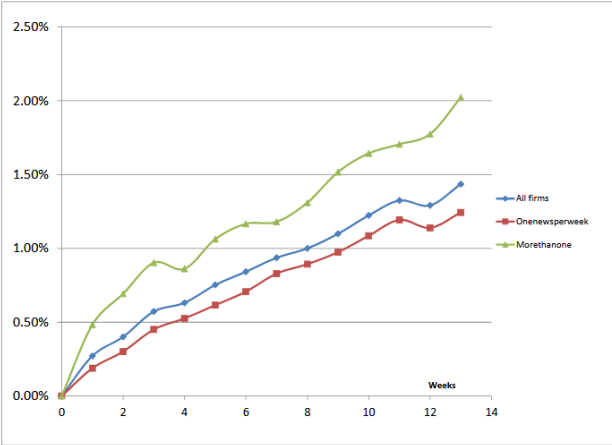


Figure 4: Cumulative Weekly Post-News Long-Short Quintile Returns

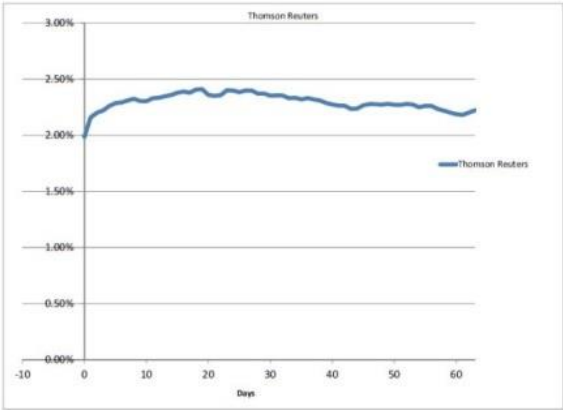


Figure 1: Cumulative Daily News and Post-News Long-Short Quintile Returns

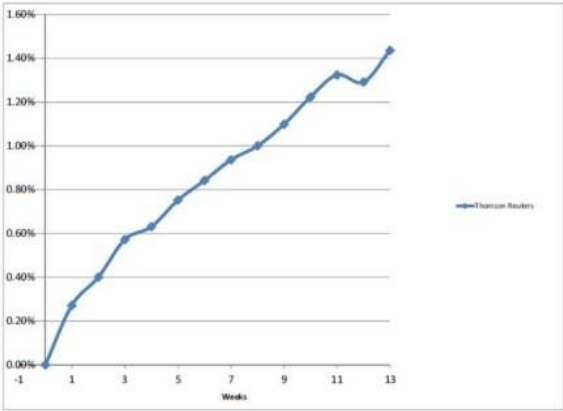
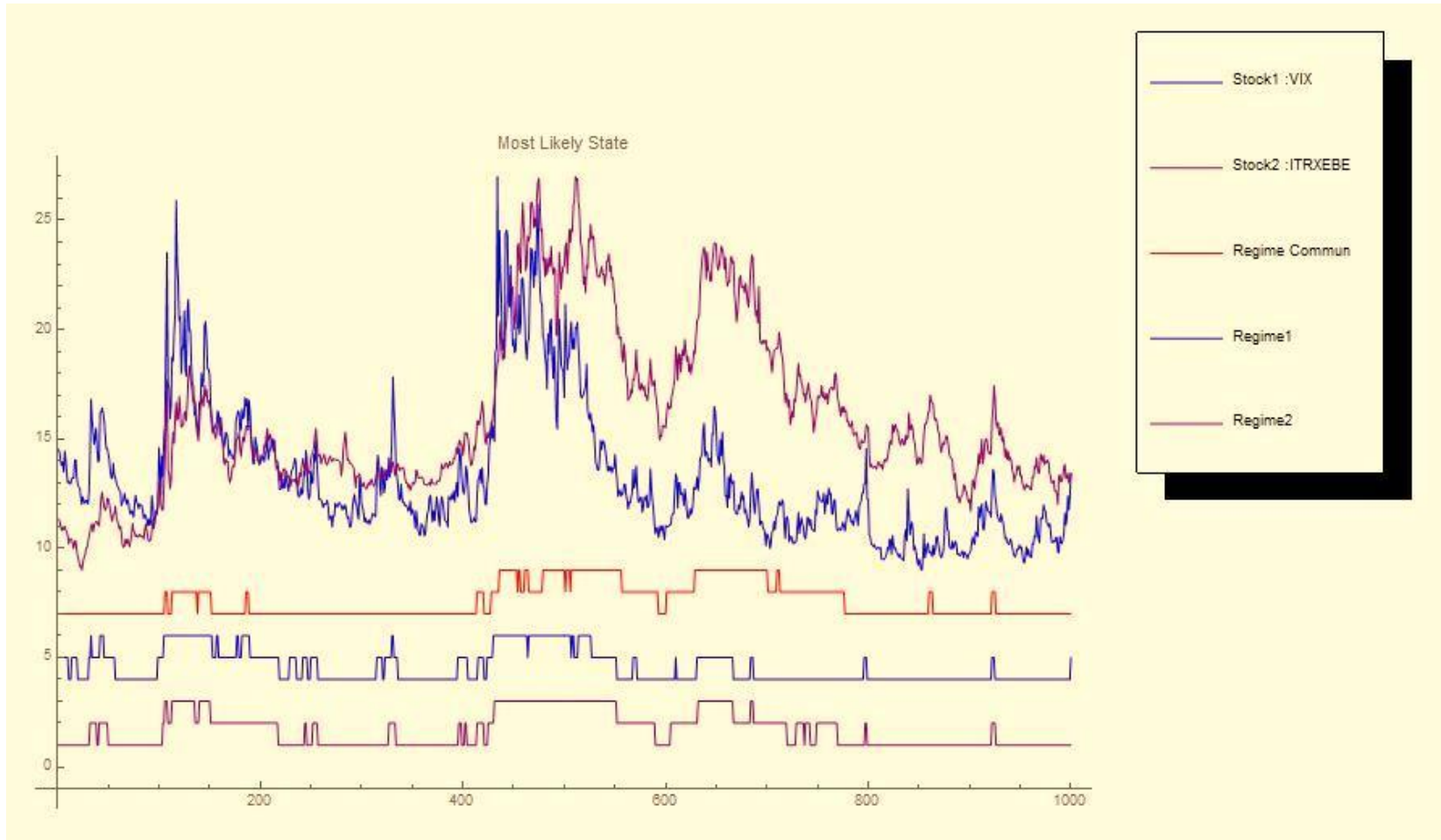
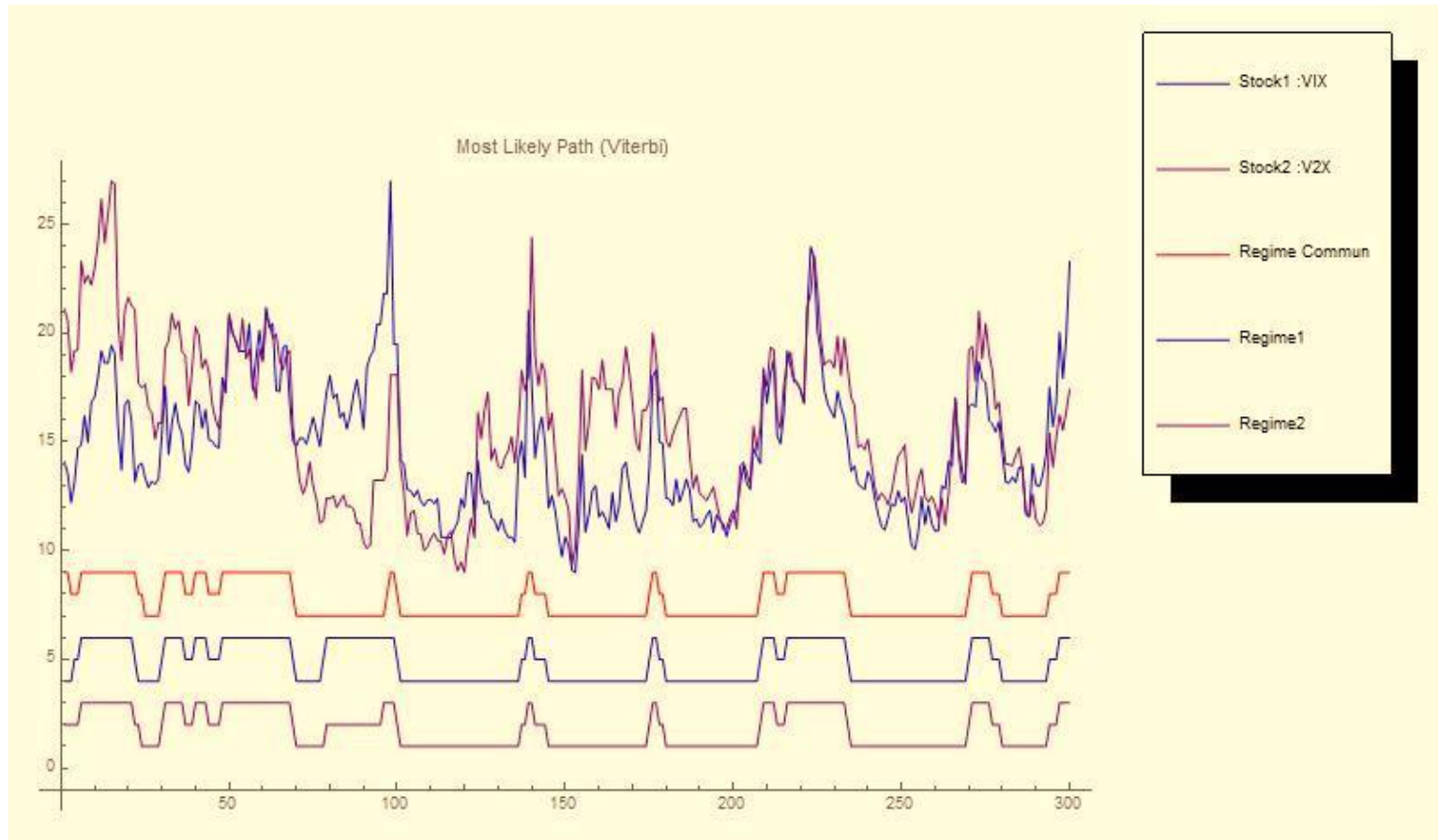


Figure 2: Cumulative Weekly Post-News Long-Short Decile Returns

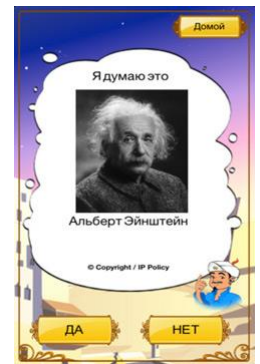
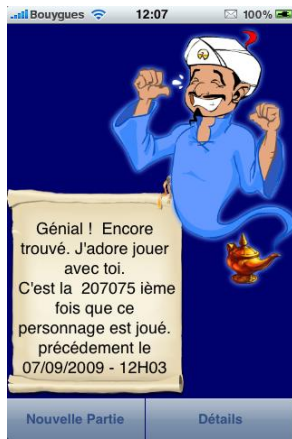
HMM multi-dimensionnel (VIX/ITERXEBE)



HMM multi-dimensionnel (VIX/V2X)



Akinator : un succes planétaire



Plus de 100000 personnages !
14 langues !

Akinator : le principe



Bayes :

$$P(S = S_k | G = G_n) = \frac{P(G = G_n | S = S_k) P(S = S_k)}{P(G = G_n)}$$

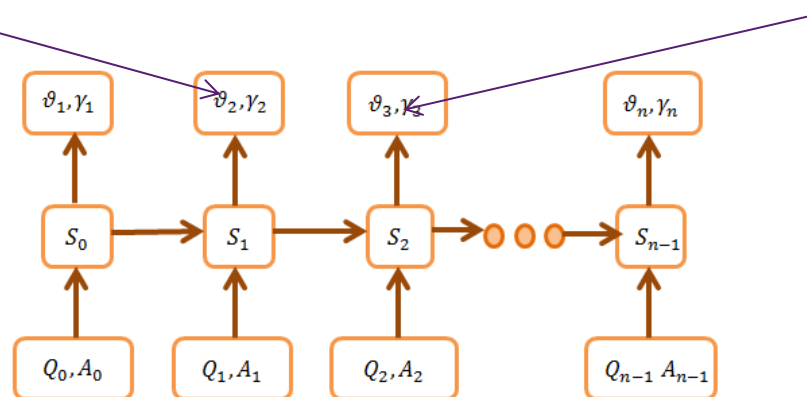
Maximiser l'entropie croisée

$$H(d_{before}, d_{after}^{q_n}) = \int_S \left(d_{before}(S) \log \left[d_{after}^{q_n}(S) \right] \right) dS$$

$$\{P(g_n = \{q, a\} | G_i)\}_{r \in R}$$

$$\{P(g_i | S = S_k, G = G_{i-1})\}_{k \in K}$$

LSTM :





Interpolateur / extrapolateur pour série financière à réalité augmentée

- **Tirant parti de l'existence**
 - d'une grande quantité de séries synchrones
 - de séries de données non numériques
- **Capable de tirer parti de l'existence de corrélations complexes (spatiales et temporelles) non simplement exprimables par des concepts mathématique usuels**
- **Tirant parti de la mise a disposition de grandes puissances de calculs (GPU, clusters ..)**
- **Utilisant les concepts de réseau de neurones précédemment exposés**

Cœur de l'architecture du réseau idéal



Choix architecturaux

Génératif

Réactivité a la complexité

Mémoire Differentiable

Grande sensibilité au passé

Manifold Tang. Cl.

Profondeur de complexité explorée

Interface Neur. Déc.

Modulariser et distribuer l'apprentissage du réseau



- **Architecture du réseau neuronal est un facteur clé**
 1. Réseau profond
 2. Génératif / discriminatif
 3. Inférence variationnelle
 4. Mécanisme d'attention
 5. Mémoire différentiable
 6. Découplage des interfaces neuronaux
 7. Contrôle des sous-variétés du gradient
- **Les performances restituées par réseau neuronal correctement pensé sont souvent suprahumaines**
- **La prévision de séries financières en réalité augmentée est possible, à condition qu'elle soit**
 1. Probabiliste
 2. A input large.

Thank you for your time



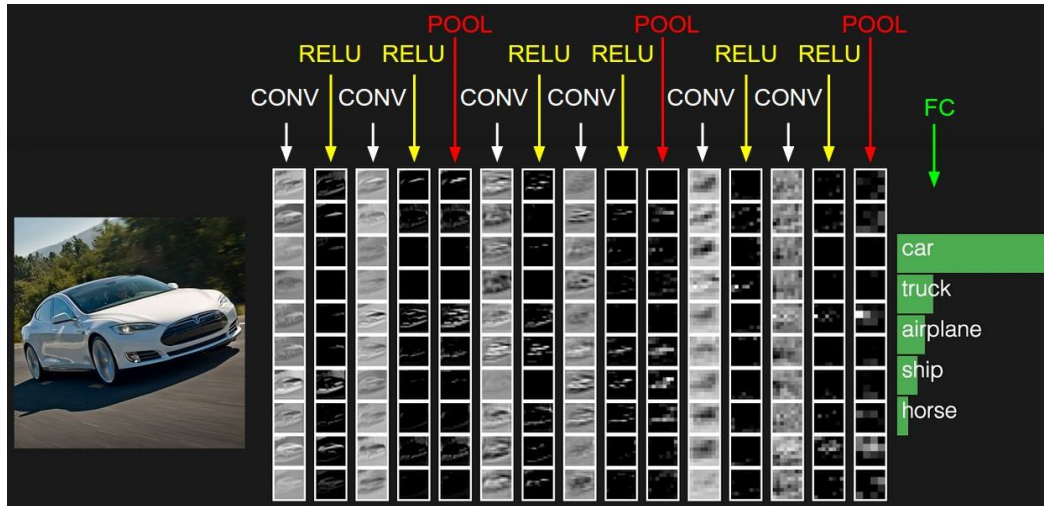
Let's try it !





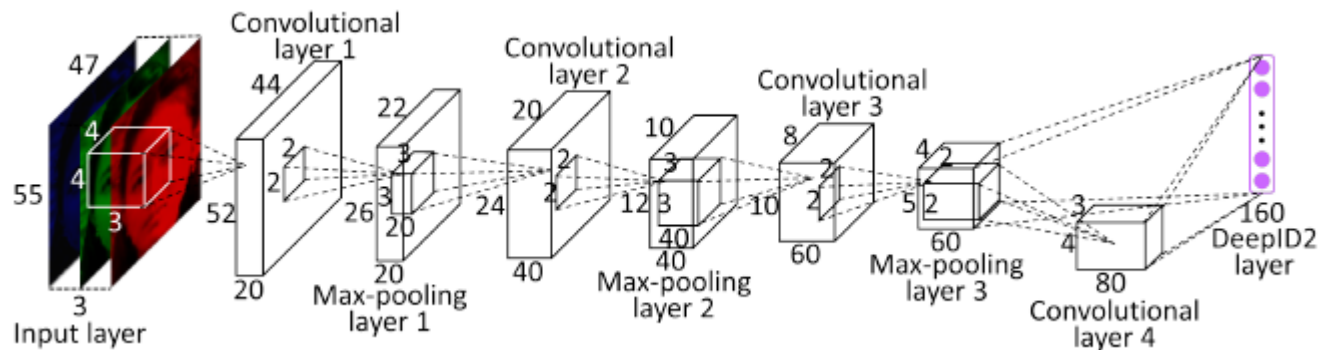
- 1. Fonctionnement de la vision humaine**
- 2. Réseaux convolutionnels**
- 3. Performance des réseaux convolutionnels en vision**
- 4. Chatbots**
- 5. Reinforcement learning**

Réseaux convolutionnels



Deux types de cellule:

1. Convolution
2. Pooling

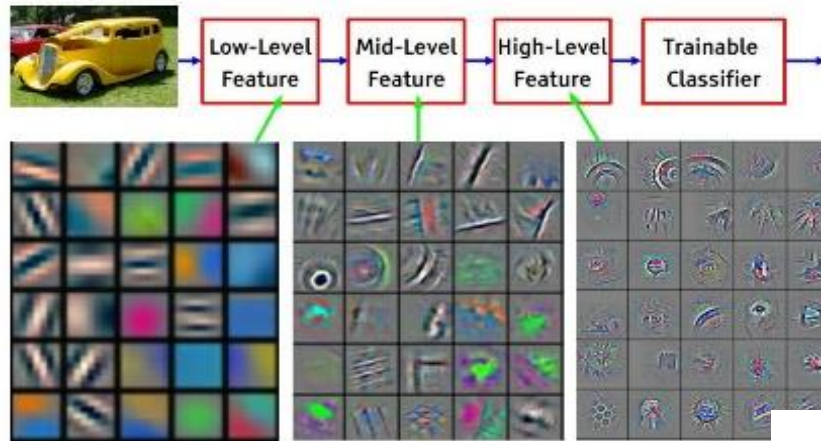


Rerseau convolutionnel : traditionnels

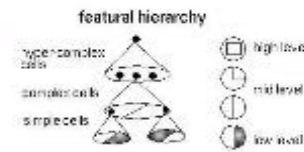


ConvNet : Interpretation

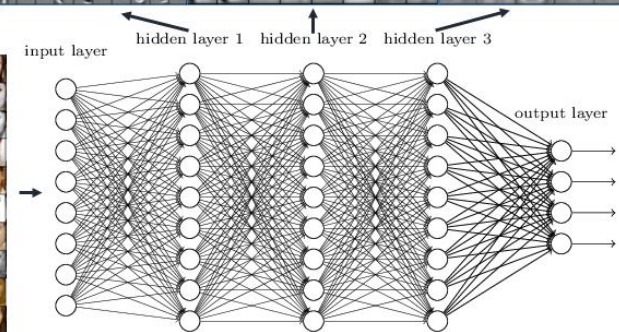
Chaque couche analyse un niveau d'agrégation



Feature visualization of convolutional net trained on ImageNet from [Zeller & Fergu



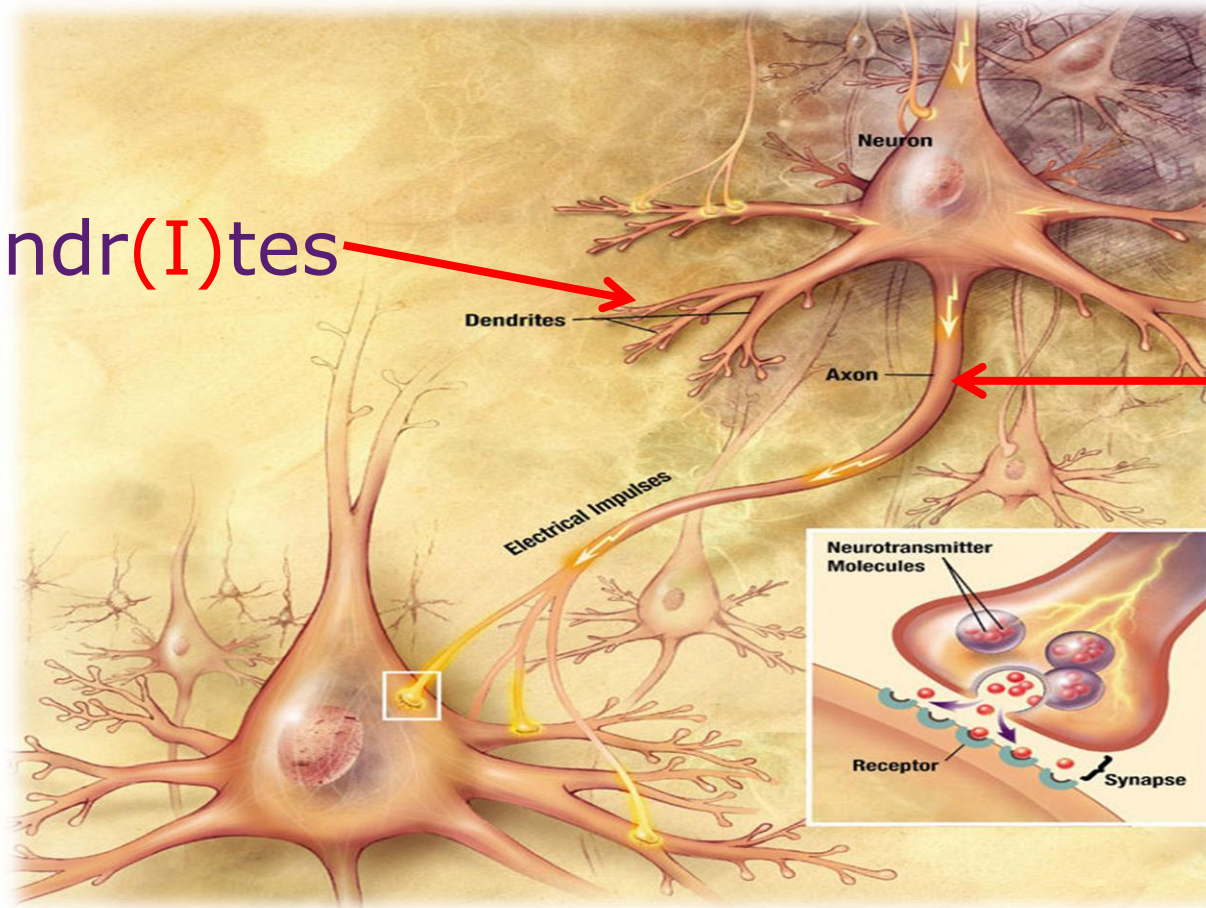
Deep neural networks learn hierarchical feature representations



Comment cela se passe dans le cerveau

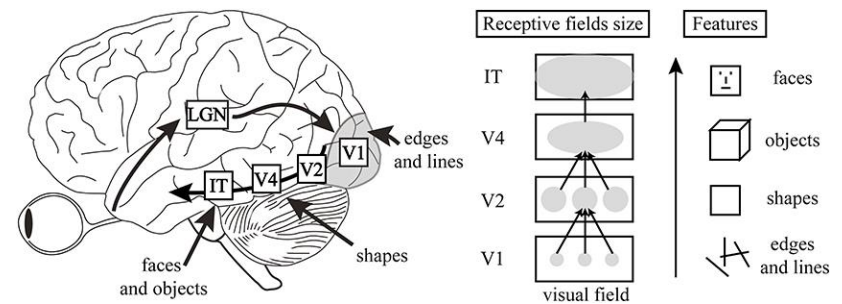
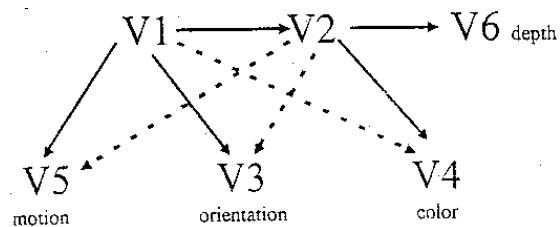
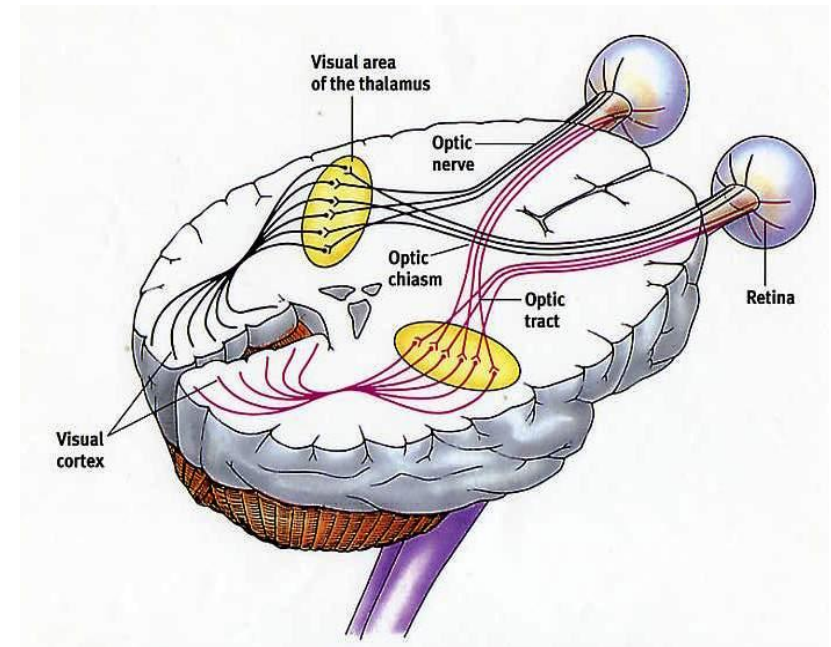
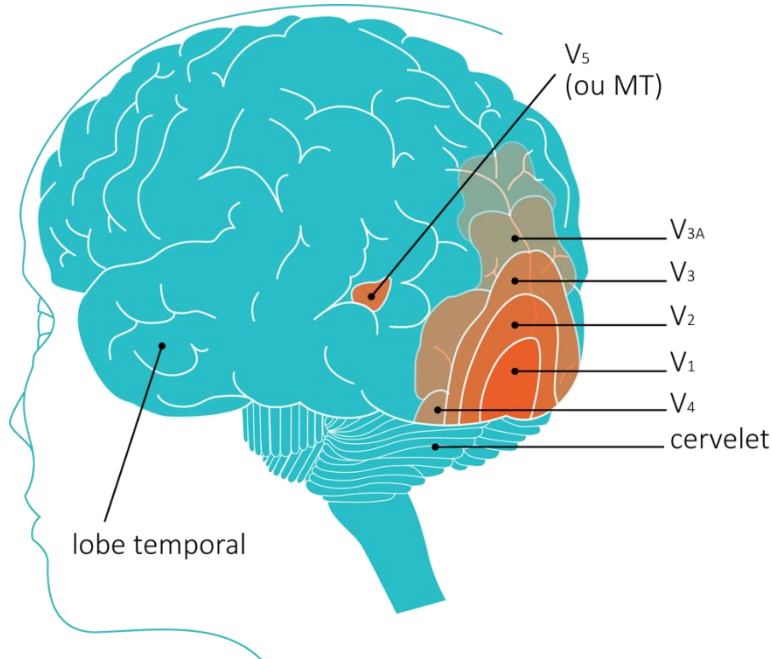


Dendr(I)tes

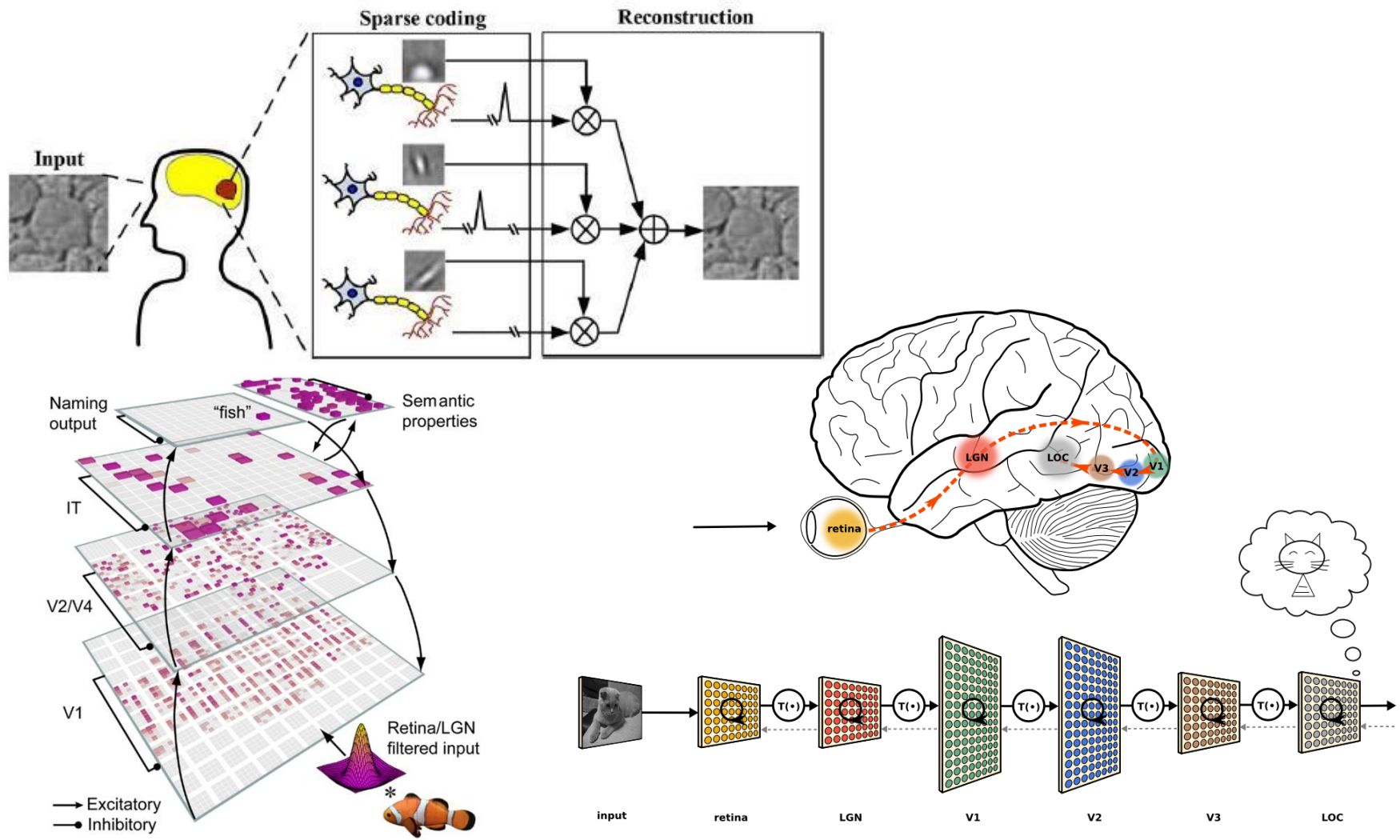


Ax(O)ne

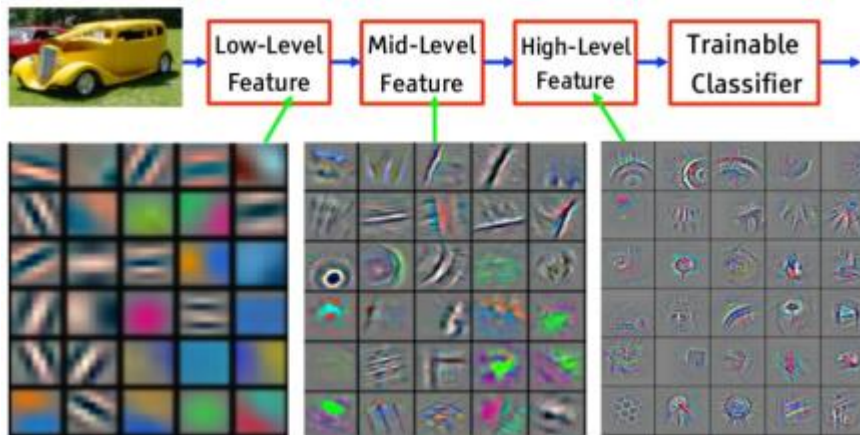
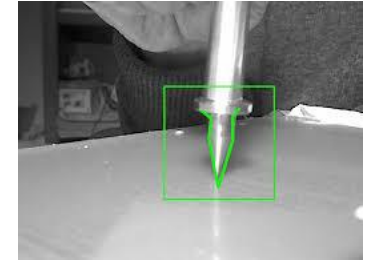
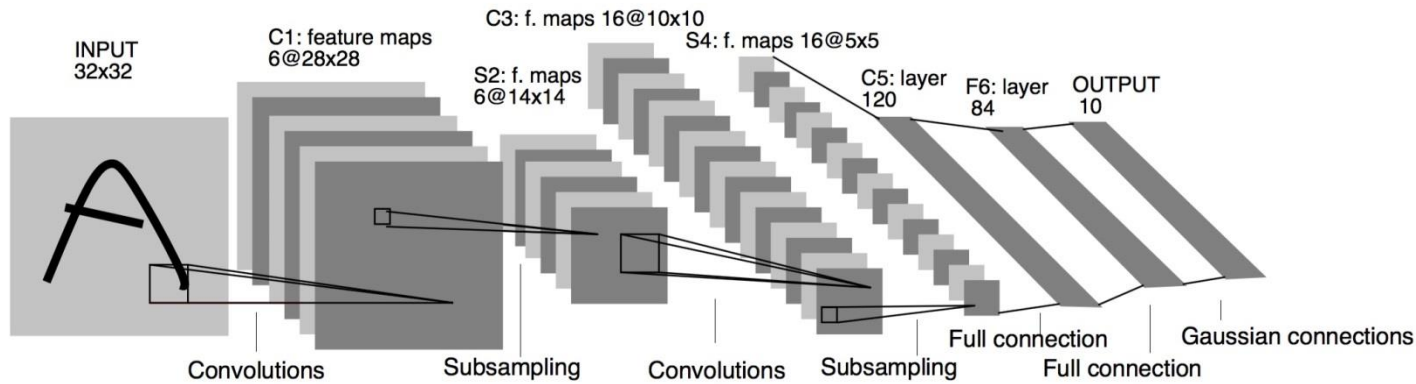
Traitement neuronal de la vision



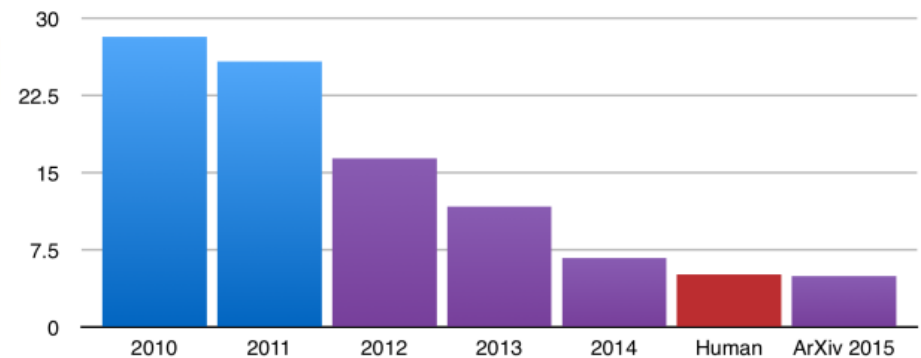
Fonctionnement de la vision



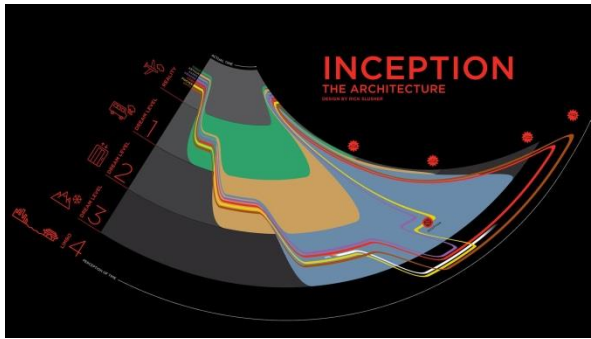
Lenet



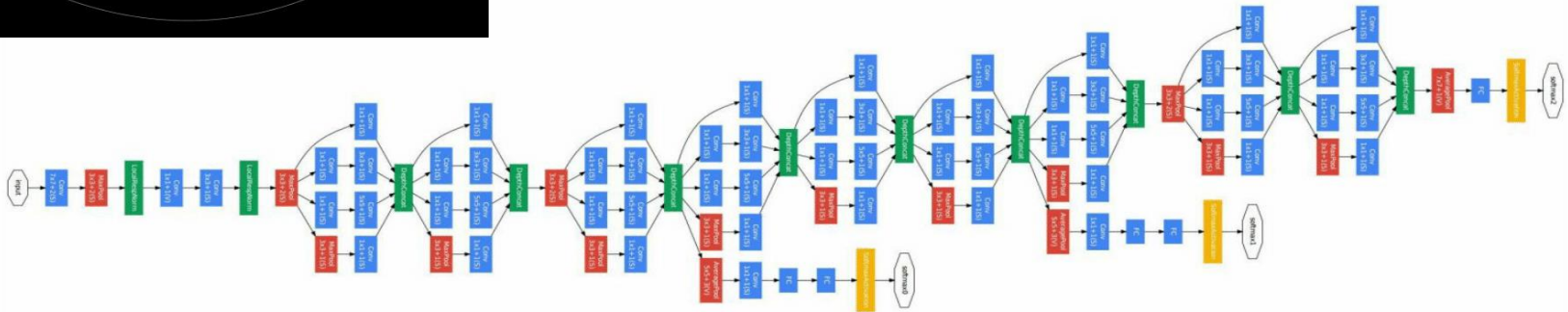
ILSVRC top-5 error on ImageNet



Inception (2015)



22 niveaux!



The Inception Architecture (GoogLeNet, 2015)

Lenet: Génération de représentant

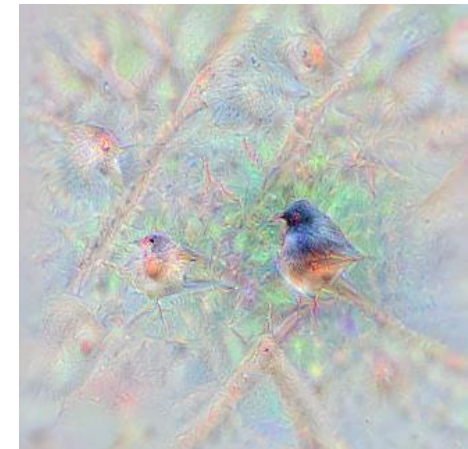
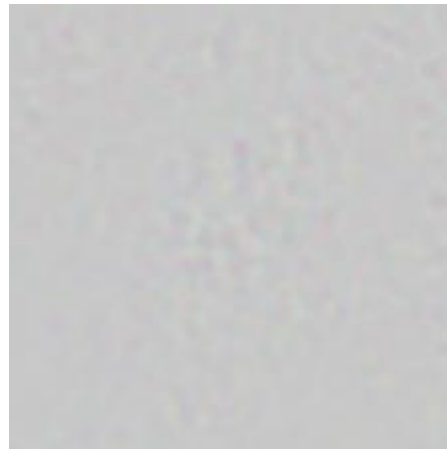


Going Deeper with Convolutions 2014

Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, Andrew Rabinovich

Understanding Neural Networks Through Deep Visualization 2015

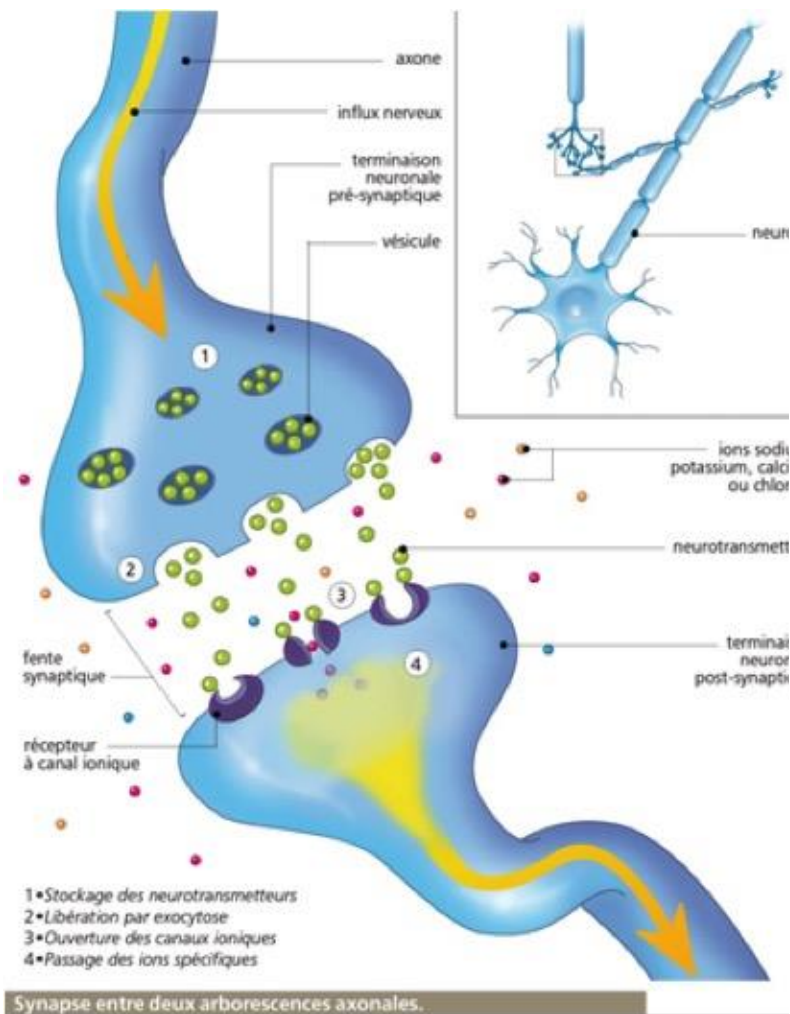
Jason Yosinski, Jeff Clune, Anh Nguyen, Thomas Fuchs, Hod Lipson



GoogLeNet classes : <http://www.auduno.com/2015/07/29/visualizing-googlenet-classes/>



Une liaison synaptique en fonctionnement



Nature chimique	Nom du neurotransmetteur	Quelques sites de libération synaptique	Principaux rôles biologiques
Amines	Acétylcholine	Synapse neuromusculaire	Contraction des muscles striés Ralentissement cardiaque Motricité digestive Sécrétion
	Dopamine	Cerveau	Contrôle de la motricité (maladie de Parkinson) Contrôle de l'éveil comportemental
	Adrénaline	Bulbe rachidien	Augmentation rythme cardiaque, augmentation pression artérielle (stress, effort physique, danger...)
	Noradrénaline	Système orthosympathique Cortex cérébral Hypothalamus Bulbe rachidien	Stimulation cardiaque
	Sérotonine	Cortex cérébral Hypothalamus Cervelet Bulbe rachidien	Sommeil Thermorégulation Chronobiologie
Acides aminés	GABA (Acide Gamma-amino-butyrique)	Motoneurones de la moelle épinière Cerveau Moelle épinière	Neuromédiateur inhibiteur
	Glutamate et aspartate	Système nerveux central	Neuromédiateur exciteur
Peptides	Substance P	Cerveau Neuromédiateur des neurones sensitifs (peau vers moelle épinière)	Perceptions douloureuses : nociception
	Enképhalines	Cortex cérébral et moelle épinière	Analgésie : diminution de la sensibilité à la douleur

Chatbot contre chatboot



Robot parlant: SpeechToText + chatbot + TextToSpeech



Chatbot a la Watson



Reinforcement

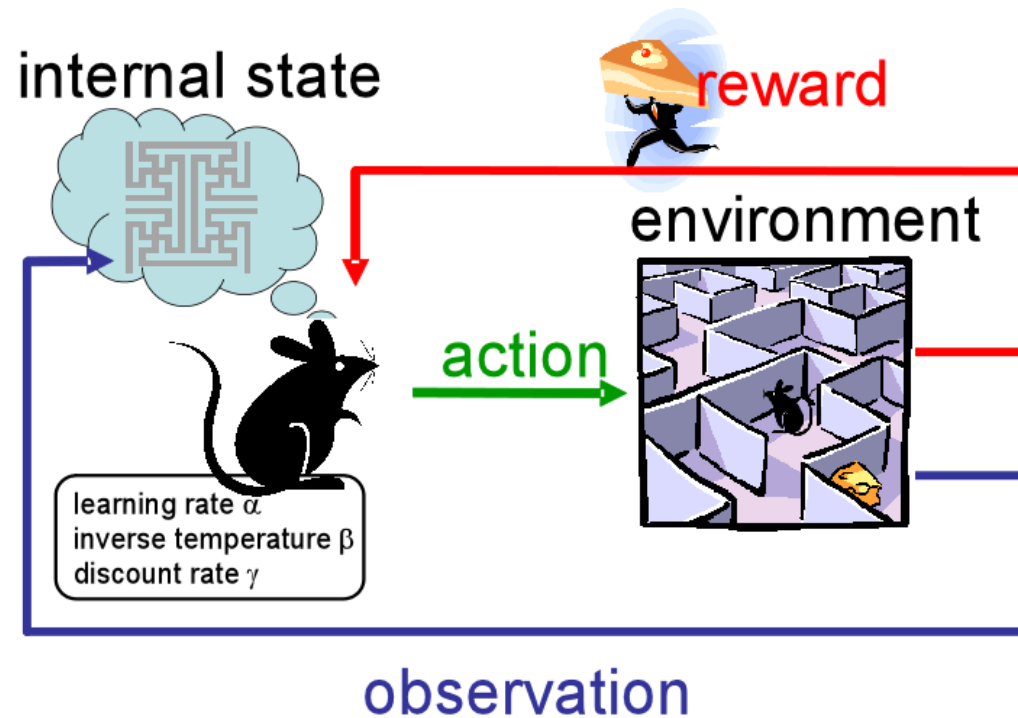


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“It’s a special hearing aid. It filters out criticism and amplifies compliments.”

Reinforcement learning





Approaches To Reinforcement Learning

Value-based RL

- ▶ Estimate the **optimal value function** $Q^*(s, a)$
- ▶ This is the maximum value achievable under any policy

Policy-based RL

- ▶ Search directly for the **optimal policy** π^*
- ▶ This is the policy achieving maximum future reward

Model-based RL

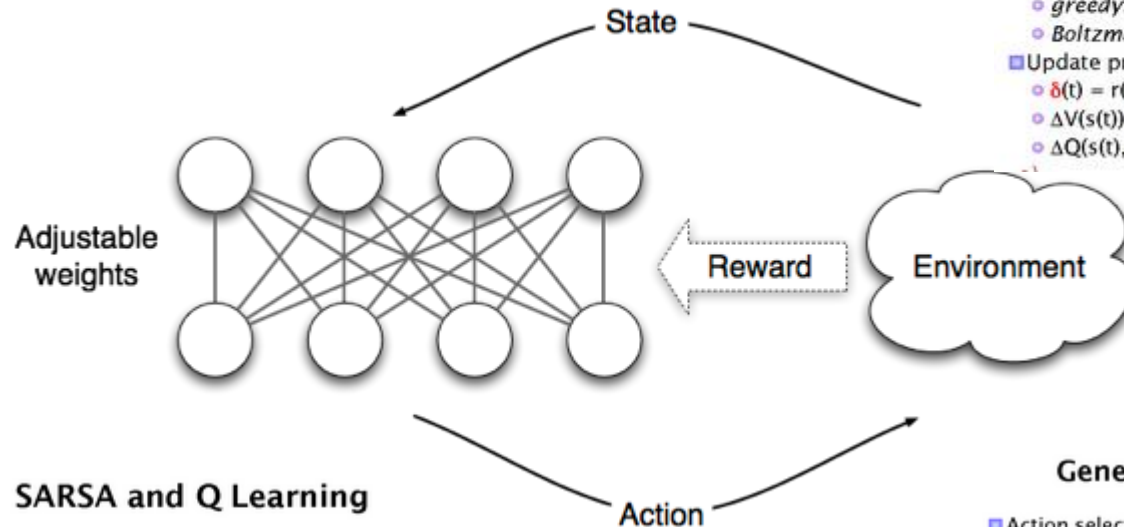
- ▶ Build a model of the environment
- ▶ Plan (e.g. by lookahead) using model

Reinforcement



Reinforcement Learning

- Predict reward: *value function*
 - $V(s) = E[r(t) + \gamma r(t+1) + \gamma^2 r(t+2) \dots | s(t)=s]$
 - $Q(s,a) = E[r(t) + \gamma r(t+1) + \gamma^2 r(t+2) \dots | s(t)=s, a(t)=a]$
- Select action *How to implement these steps?*
 - *greedy*: $a = \operatorname{argmax}_a Q(s,a)$
 - *Boltzmann*: $P(a|s) \propto \exp[\beta Q(s,a)]$
- Update prediction: *TD error*
 - $\delta(t) = r(t) + \gamma V(s(t+1)) - V(s(t))$
 - $\Delta V(s(t)) = \alpha \delta(t)$ *How to tune these parameters?*
 - $\Delta Q(s(t),a(t)) = \alpha \delta(t)$



SARSA and Q Learning

- Action value function
 - $Q(s,a) = E[r(t) + \gamma r(t+1) + \gamma^2 r(t+2) \dots | s(t)=s, a(t)=a]$
- Action selection
 - ϵ -greedy: $a = \operatorname{argmax}_a Q(s,a)$ with prob $1-\epsilon$
 - Boltzman: $P(a_i|s) = \exp[\beta Q(s,a_i)] / \sum_j \exp[\beta Q(s,a_j)]$
- SARSA: on-policy update
 - $\Delta Q(s(t),a(t)) = \alpha \{ r(t) + \gamma Q(s(t+1),a(t+1)) - Q(s(t),a(t)) \}$
- Q learning: off-policy update
 - $\Delta Q(s(t),a(t)) = \alpha \{ r(t) + \gamma \max_{a'} Q(s(t+1),a') - Q(s(t),a(t)) \}$

Generalized Q-learning Model

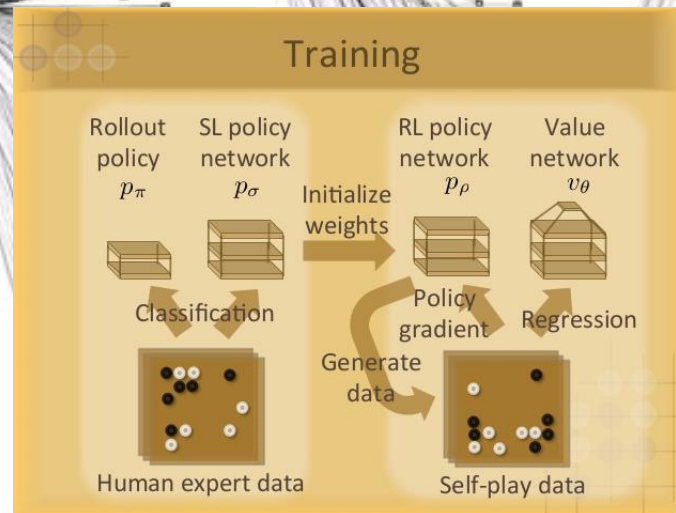
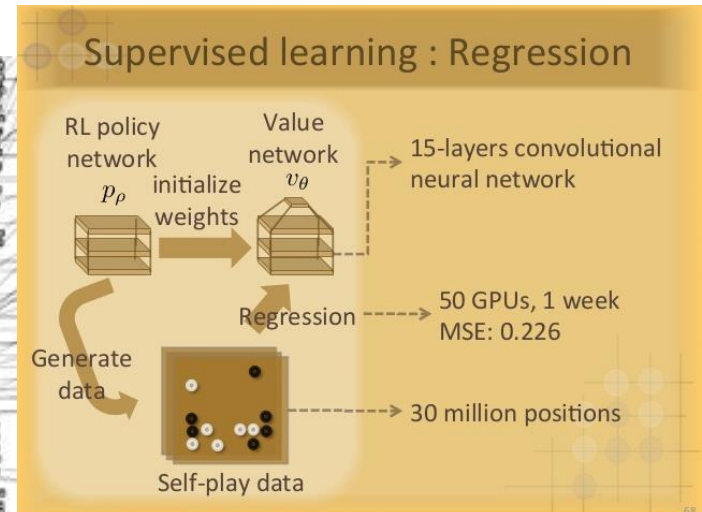
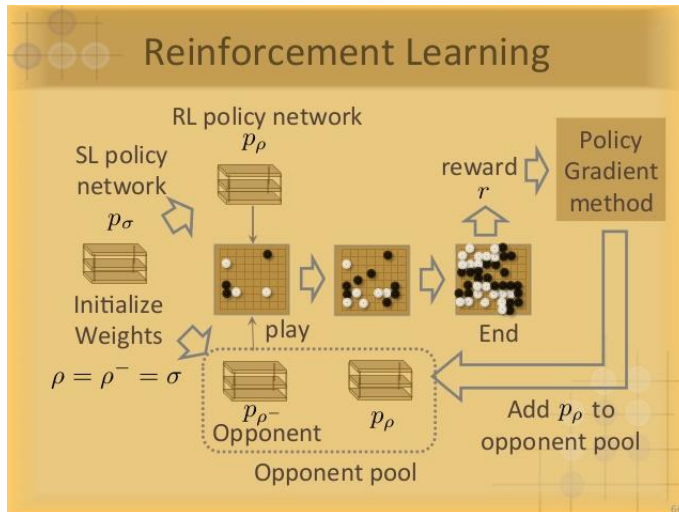
(Ito & Doya, 2009)

- Action selection

$$P(a(t)=L) = \exp Q_L(t) / (\exp Q_L(t) + \exp Q_R(t))$$
- Action value update: $i \in \{L, R\}$

$$Q_i(t+1) = \begin{cases} (1-\alpha_1)Q_i(t) + \alpha_1 \kappa_1 & \text{if } a(t)=i, r(t)=1 \\ (1-\alpha_1)Q_i(t) + \alpha_1 \kappa_2 & \text{if } a(t)=i, r(t)=0 \\ (1-\alpha_2)Q_i(t) & \text{if } a(t) \neq i, r(t)=1 \\ (1-\alpha_2)Q_i(t) & \text{if } a(t) \neq i, r(t)=0 \end{cases}$$
- Parameters
 - α_1 : learning rate
 - α_2 : forgetting rate
 - κ_1 : reward reinforcement
 - κ_2 : no-reward aversion

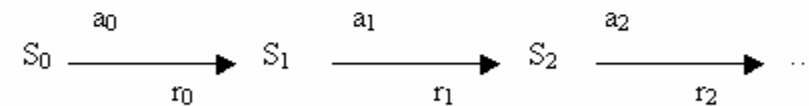
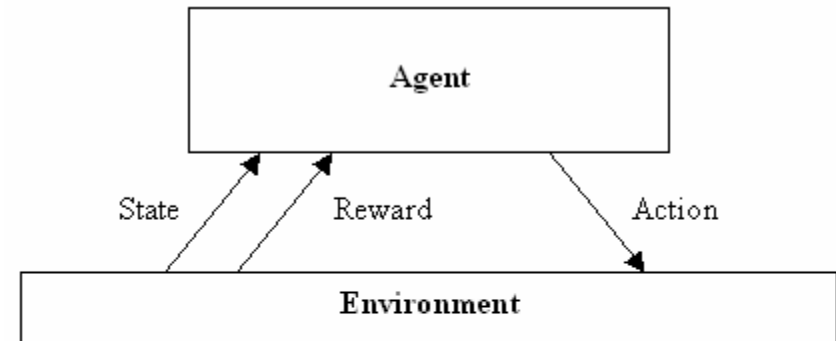
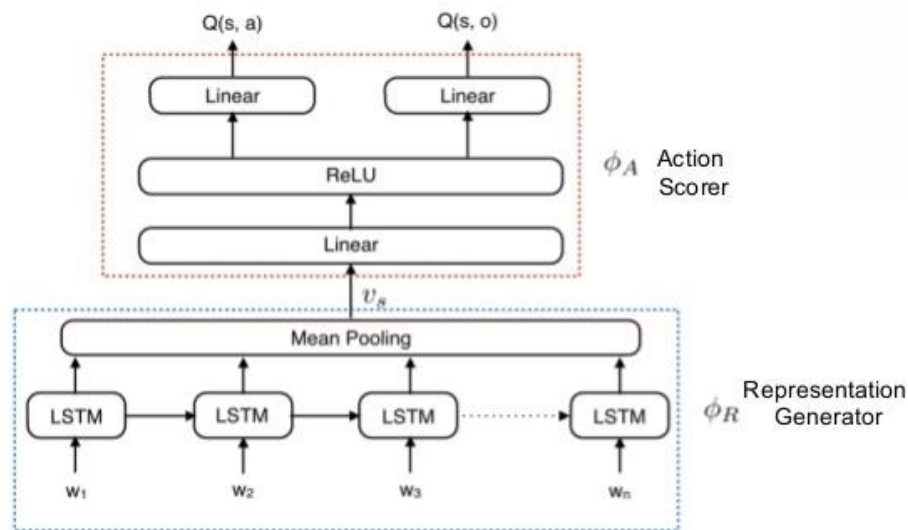
Reinforcement dans Alphago



Reinforcement profond



Deep Reinforcement Learning Framework



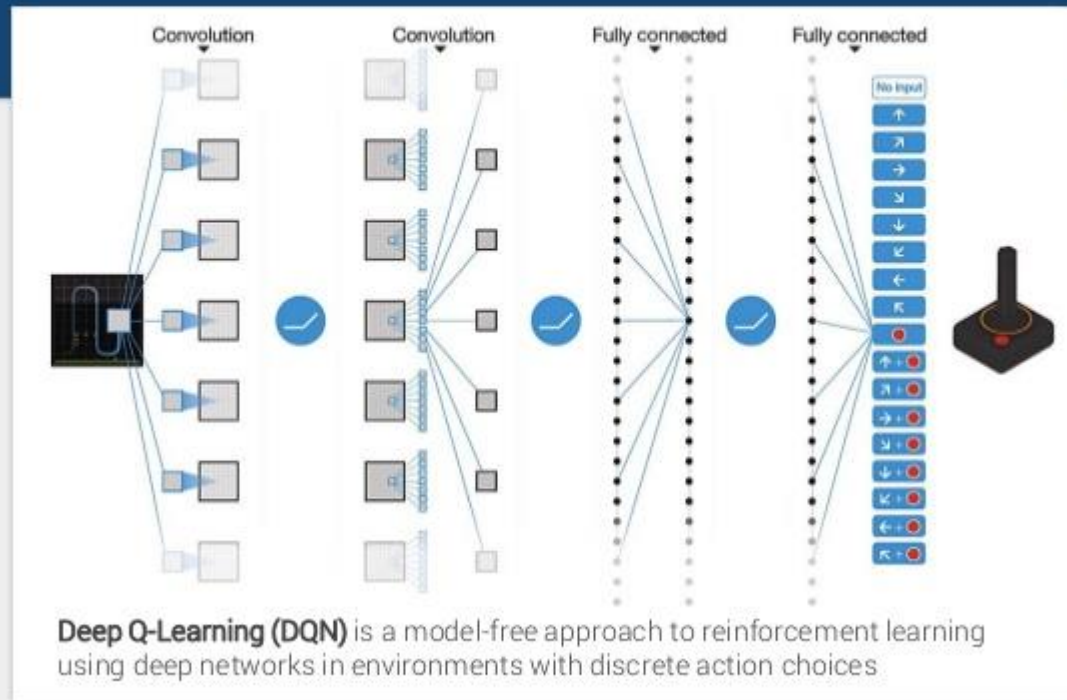
Goal: learn to choose actions that maximize:
 $r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$, where $0 \leq \gamma < 1$

Deep Q-Learning



Deep Learning in Google Research

DeepMind Deep Q-Learning





Toy video QA problem



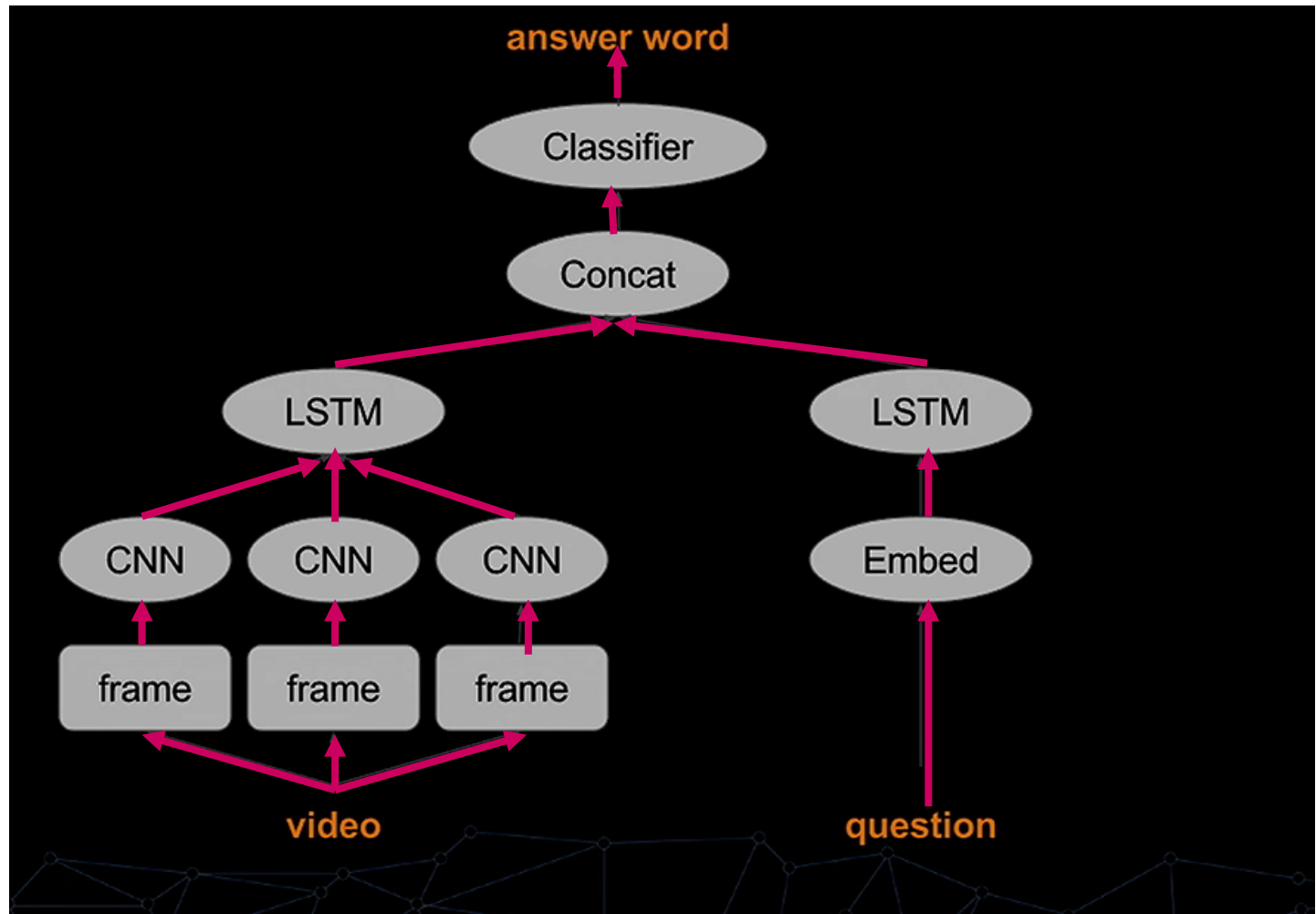
- > What is the woman doing?
- > **packing**
- > What is the color of her shirt?
- > **black**



La solution



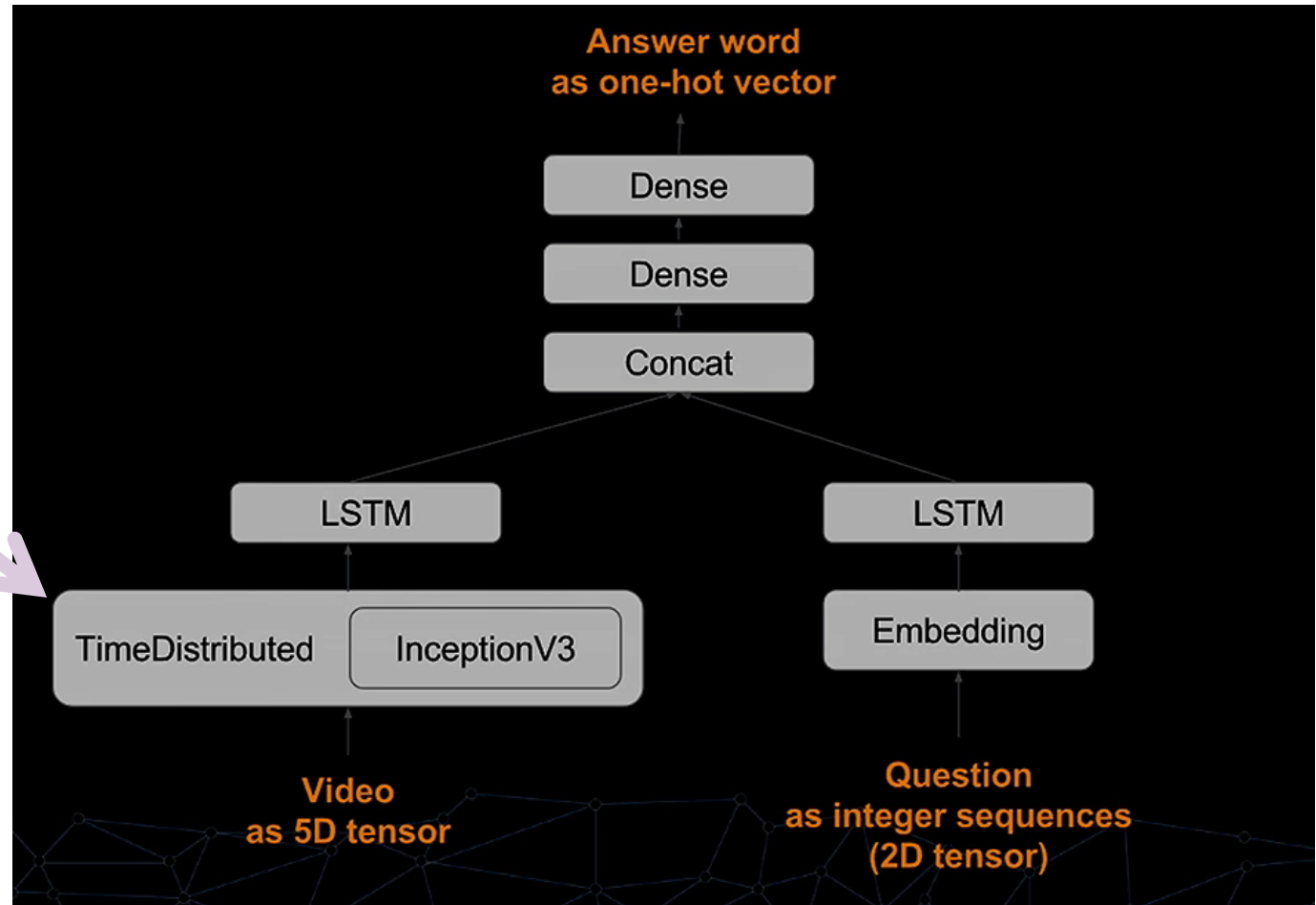
↑
Sémantique
Transportée
par un vecteur
↑



Traduite en Keras



Pre-trained



Le code du modèle (1 : partie video)



Pre-trained
Et
Non-trainable

```
video = tf.keras.layers.Input(shape=(None, 150, 150, 3))
cnn = tf.keras.applications.InceptionV3(weights='imagenet',
                                         include_top=False,
                                         pool='avg')
cnn.trainable = False
encoded_frames = tf.keras.layers.TimeDistributed(cnn)(video)
encoded_vid = tf.layers.LSTM(256)(encoded_frames)
```

Le modèle (2 : partie textuelle)



Transformation des mots en
vecteurs : vocabulaire = 10000

```
question = tf.keras.layers.Input(shape=(100), dtype='int32')  
x = tf.keras.layers.Embedding(10000, 256, mask_zero=True)(question)  
encoded_q = tf.keras.layers.LSTM(128)(x)
```

Remplir les 256 mots
Si nécessaire avec des blancs

Le modèle (3 : formation de la réponse)



```
x = tf.keras.layers.concat([encoded_vid, encoded_q])  
x = tf.keras.layers.Dense(128, activation=tf.nn.relu)(x)  
outputs = tf.keras.layers.Dense(1000)(x)
```

2 couches Linéaires
séparées
par une activation

Le modèle (conditions d'apprentissage)



Base de learning

```
model = tf.keras.models.Model([video, question], outputs)
model.compile(optimizer=tf.AdamOptimizer(),
              loss=tf.softmax_crossentropy_with_logits)
```

Méthode
d'optimisation

Spécification de la dernière couche d'activation

Le code en entier



```
video = tf.keras.layers.Input(shape=(None, 150, 150, 3))
cnn = tf.keras.applications.InceptionV3(weights='imagenet',
                                         include_top=False,
                                         pool='avg')

cnn.trainable = False
encoded_frames = tf.keras.layers.TimeDistributed(cnn)(video)
encoded_vid = tf.layers.LSTM(256)(encoded_frames)

question = tf.keras.layers.Input(shape=(100), dtype='int32')
x = tf.keras.layers.Embedding(10000, 256, mask_zero=True)(question)
encoded_q = tf.keras.layers.LSTM(128)(x)

x = tf.keras.layers.concat([encoded_vid, encoded_q])
x = tf.keras.layers.Dense(128, activation=tf.nn.relu)(x)
outputs = tf.keras.layers.Dense(1000)(x)

model = tf.keras.models.Model([video, question], outputs)
model.compile(optimizer=tf.AdamOptimizer(),
              loss=tf.softmax_crossentropy_with_logits)
```

Intégration de TensorFlow et de Keras



```
import tensorflow as tf
from keras import backend as K
from keras.layers import Dense
from keras.objectives import categorical_crossentropy

# this placeholder will contain our input digits, as flat vectors
img = tf.placeholder(tf.float32, shape=(None, 784))
# Keras layers can be called on TensorFlow tensors:
# fully-connected layer with 128 units and ReLU activation
x = Dense(128, activation='relu')(img)
x = Dense(128, activation='relu')(x)
```

Keras TensorFlow



KETE Combo rocks

Img : defini dans tensorflow

Comment on lance un learning



On change de méthode d'optimisation (si on veut)

```
# for a multi-class classification problem
model.compile(optimizer='rmsprop',
              loss='categorical_crossentropy',
              metrics=['accuracy'])
model.fit(X_train, Y_train, nb_epoch=5, batch_size=32)
```

C'est parti !

Utiliser un modèle calibré



Evaluate your performance in one line:

```
loss_and_metrics = model.evaluate(X_test, Y_test, batch_size=32)
```

Or generate predictions on new data:

```
classes = model.predict_classes(X_test, batch_size=32)  
proba = model.predict_proba(X_test, batch_size=32)
```

Utiliser des modèles pre-trained



Extract features with VGG16

```
from keras.applications.vgg16 import VGG16
from keras.preprocessing import image
from keras.applications.vgg16 import preprocess_input
import numpy as np

model = VGG16(weights='imagenet', include_top=False)

img_path = 'elephant.jpg'
img = image.load_img(img_path, target_size=(224, 224))
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
x = preprocess_input(x)

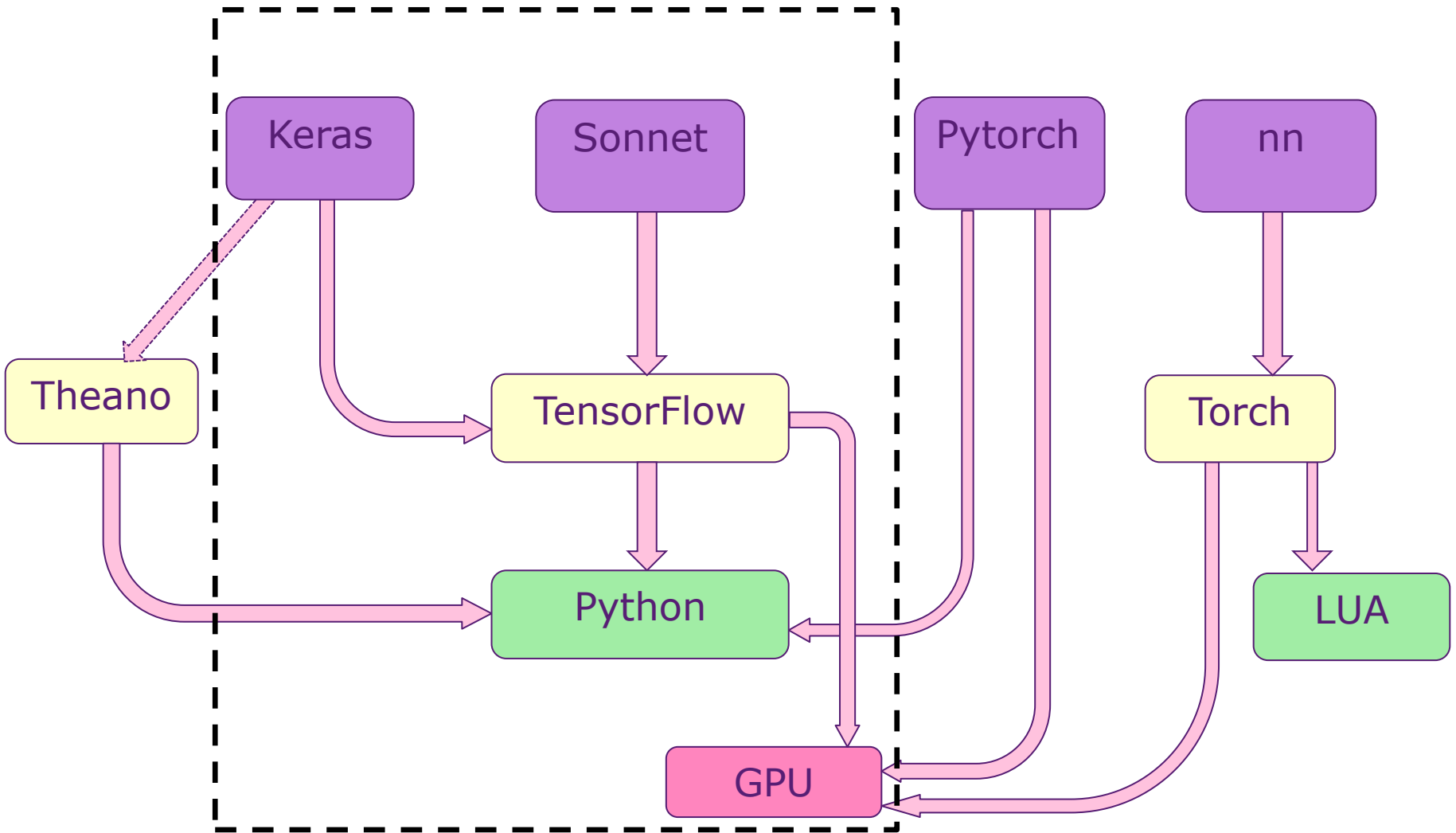
features = model.predict(x)
```

On charge
un modèle

On extrait
Les
features

On ne charge
que le ConvNet

Environnement de dev



Keras (approche fonctionnelle)



```
from keras.layers import Input, Dense
from keras.models import Model

# This returns a tensor
inputs = Input(shape=(784,))

# a layer instance is callable on a tensor, and returns a tensor
x = Dense(64, activation='relu')(inputs)
x = Dense(64, activation='relu')(x)
predictions = Dense(10, activation='softmax')(x)

# This creates a model that includes
# the Input layer and three Dense layers
model = Model(inputs=inputs, outputs=predictions)
model.compile(optimizer='rmsprop',
              loss='categorical_crossentropy',
              metrics=['accuracy'])
model.fit(data, labels) # starts training
```

Activation déclarée dans la couche

S'appuie sur

- Ou bien Tensorflow
- Ou bien Theano

Sonnet (Deepmind 2017)



Approche fonctionnelle

```
def custom_build(inputs, is_training, keep_prob):  
    """A custom build method to wrap into a sonnet Module."""  
    outputs = snt.Conv2D(output_channels=32, kernel_shape=4, stride=2)(inputs)  
    outputs = snt.BatchNorm()(outputs, is_training=is_training)  
    outputs = tf.nn.relu(outputs)  
    outputs = snt.Conv2D(output_channels=64, kernel_shape=4, stride=2)(outputs)  
    outputs = snt.BatchNorm()(outputs, is_training=is_training)  
    outputs = tf.nn.relu(outputs)  
    outputs = snt.BatchFlatten()(outputs)  
    outputs = tf.nn.dropout(outputs, keep_prob=keep_prob)  
    outputs = snt.Linear(output_size=10)(outputs)  
    return outputs
```

Package nn de Torch

Learning to Learn in TensorFlow

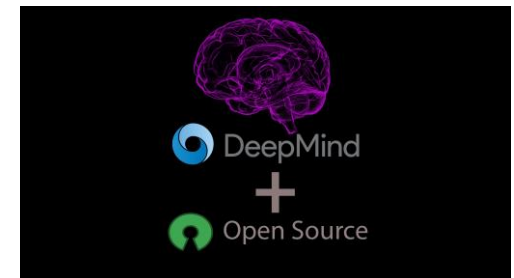
Dependencies

- TensorFlow >=1.0
- Sonnet >=1.0

```
class MetaOptimizer(object):
```

```
    """Learning to learn (meta) optimizer.
```

```
    Optimizer which has an internal RNN which takes as input, at each iteration,  
    the gradient of the function being minimized and returns a step direction.  
    This optimizer can then itself be optimized to learn optimization on a set of  
    tasks.
```





Deux premières idées d'applications

Akinator Financier:

- Sélectionner un produit financier sur la base d'un questionnaire dynamique, avec enrichissement automatique de la base de connaissance

Pricer ultra rapide a base de Réseau neuronal :

- Pricer en situation (sur une tablette en face du client)
- Comparer un ensemble de structures